KR Network Cloud Technologies Pvt. Ltd

1) What is Linux?

Linux is an operating system based on UNIX, and was first introduced by Linus Torvalds. It is based on the Linux Kernel, and can run on different hardware platforms manufactured by Intel, MIPS, HP, IBM, SPARC and Motorola. Another popular element in Linux is its mascot, a penguin figure named Tux.

2) What is the difference between UNIX and LINUX?

Unix originally began as a propriety operating system from Bell Laboratories, which later on spawned into different commercial versions. On the other hand, Linux is free, open source and intended as a non-propriety operating system for the masses.

4) What is Linux Kernel?

Linux Kernel is a low-level systems software whose main role is to manage hardware resources for the user. It is also used to provide an interface for user-level interaction.

5) What is LILO?

LILO is a boot loader for Linux. It is used mainly to load the Linux operating system into main memory so that it can begin its operations.

6) What is a swap space?

A swap space is a certain amount of space used by Linux to temporarily hold some programs that are running concurrently. This happens when RAM does not have enough memory to hold all programs that are executing.

7) What is the advantage of open source?

Open source allows you to distribute your software, including source codes freely to anyone who is interested. People would then be able to add features and even debug and correct errors that are in the source code. They can even make it run better, and then redistribute these enhanced source code freely again. This eventually benefits everyone in the community.

8) What are the basic components of Linux?

Just like any other typical operating system, Linux has all of these components: kernel, shells and GUIs, system utilities, and application program. What makes Linux advantageous over other operating system is that every aspect comes with additional features and all codes for these are downloadable for free.

9) Does it help for a Linux system to have multiple desktop environments installed?

In general, one desktop environment, like KDE or Gnome, is good enough to operate without issues. It's all a matter of preference for the user, although the system allows switching from one environment to another. Some programs will work on one environment and not work on the other, so it could also be considered a factor in selecting which environment to use.

10) What is the basic difference between BASH and DOS?

The key differences between the BASH and DOS console lies in 3 areas:
– BASH commands are case sensitive while DOS commands are not;
– under BASH, / character is a directory separator and acts as an escape character. Under DOS, / serves as a command argument delimiter and is the directory separator
– DOS follows a convention in naming files, which is 8 character file name followed by a dot and 3 character for the extension. BASH follows no such convention.

11) What is the importance of the GNU project?

This so-called Free software movement allows several advantages, such as the freedom to run programs for any purpose and freedom to study and modify a program to your needs. It also

allows you to redistribute copies of a software to other people, as well as freedom to improve software and have it released to the public.

12) Describe the root account.

The root account is like a systems administrator account, and allows you full control of the system. Here you can create and maintain user accounts, assigning different permissions for each account. It is the default account every time you install Linux.

13) What is CLI?

CLI is short for Command Line Interface. This interface allows user to type declarative commands to instruct the computer to perform operations. CLI offers an advantage in that there is greater flexibility. However, other users who are already accustom with using GUI find it difficult to remember commands including attributes that come with it.

14) What is GUI?

GUI, or Graphical User Interface, makes use of images and icons that users click and manipulate as a way of communicating with the computer. Instead of having to remember and type commands, the use of graphical elements makes it easier to interact with the system, as well as adding more attraction through images, icons and colors.

15) How do you open a command prompt when issuing a command?

To open the default shell (which is where the command prompt can be found), press Ctrl-Alt-F1. This will provide a command line interface (CLI) from which you can run commands as needed.

16) How can you find out how much memory Linux is using?

From a command shell, use the “concatenate” command: cat /proc/meminfo for memory usage information. You should see a line starting something like: Mem: 64655360, etc. This is the total memory Linux thinks it has available to use.

17) What is typical size for a swap partition under a Linux system?

The preferred size for a swap partition is twice the amount of physical memory available on the system. If this is not possible, then the minimum size should be the same as the amount of memory installed.

18) What are symbolic links?

Symbolic links act similarly to shortcuts in Windows. Such links point to programs, files or directories. It also allows you instant access to it without having to go directly to the entire pathname.

19) Does the Ctrl+Alt+Del key combination work on Linux?

Yes, it does. Just like Windows, you can use this key combination to perform a system restart. One difference is that you won't be getting any confirmation message and therefore, reboot is immediate.

20) How do you refer to the parallel port where devices such as printers are connected?

Whereas under Windows you refer to the parallel port as the LPT port, under Linux you refer to it as /dev/lp . LPT1, LPT2 and LPT3 would therefore be referred to as /dev/lp0, /dev/lp1, or /dev/lp2 under Linux.

21) Are drives such as harddrive and floppy drives represented with drive letters?

No. In Linux, each drive and device has different designations. For example, floppy drives are referred to as /dev/fd0 and /dev/fd1. IDE/EIDE hard drives are referred to as /dev/hda, /dev/hdb, /dev/hdc, and so forth.

22) How do you change permissions under Linux?

Assuming you are the system administrator or the owner of a file or directory, you can grant permission using the chmod command. Use + symbol to add permission or – symbol to deny permission, along with any of the following letters: u (user), g (group), o (others), a (all), r (read), w (write) and x (execute). For example the command chmod go+rw FILE1.TXT grants read and write access to the file FILE1.TXT, which is assigned to groups and others.

23) In Linux, what names are assigned to the different serial ports?

Serial ports are identified as /dev/ttyS0 to /dev/ttyS7. These are the equivalent names of COM1 to COM8 in Windows.

24) How do you access partitions under Linux?

Linux assigns numbers at the end of the drive identifier. For example, if the first IDE hard drive had three primary partitions, they would be named/numbered, /dev/hda1, /dev/hda2 and /dev/hda3.

25) What are hard links?

Hard links point directly to the physical file on disk, and not on the path name. This means that if you rename or move the original file, the link will not break, since the link is for the file itself, not the path where the file is located.

26) What is the maximum length for a filename under Linux?

Any filename can have a maximum of 255 characters. This limit does not include the path name, so therefore the entire pathname and filename could well exceed 255 characters.

27)What are filenames that are preceded by a dot?

In general, filenames that are preceded by a dot are hidden files. These files can be configuration files that hold important data or setup info. Setting these files as hidden makes it less likely to be accidentally deleted.

28) Explain virtual desktop.

This serves as an alternative to minimizing and maximizing different windows on the current desktop. Using virtual desktops, each desktop is a clean slate where you can open one or more programs. Rather than minimizing/restoring all those programs as needed, you can simply shuffle between virtual desktops with programs intact in each one.

29) How do you share a program across different virtual desktops under Linux?

To share a program across different virtual desktops, in the upper left-hand corner of a program window look for an icon that looks like a pushpin. Pressing this button will “pin” that application in place, making it appear in all virtual desktops, in the same position onscreen.

30) What does a nameless (empty) directory represent?

This empty directory name serves as the nameless base of the Linux file system. This serves as an attachment for all other directories, files, drives and devices.

31) What is the pwd command?

The pwd command is short for print working directory command. It's counterpart in DOS is the cd command, and is used to display the current location in the directory tree.

32) What are daemons?

Daemons are services that provide several functions that may not be available under the base operating system. Its main task is to listen for service request and at the same time to act on these requests. After the service is done, it is then disconnected and waits for further requests.

33) How do you switch from one desktop environment to another, such as switching from KDE to Gnome?

Assuming you have these two environments installed, just log out from the graphical interface. Then at the Log in screen, type your login ID and password and choose which session type you wish to load. This choice will remain your default until you change it to something else.

34) What are the kinds of permissions under Linux?

There are 3 kinds of permissions under Linux:
– Read: users may read the files or list the directory
– Write: users may write to the file of new files to the directory
– Execute: users may run the file or lookup a specific file within a directory

35) How does case sensitivity affect the way you use commands?

When we talk about case sensitivity, commands are considered identical only if every character is encoded as is, including lowercase and uppercase letters. This means that CD, cd and Cd are three different commands. Entering a command using uppercase letters, where it should be in lowercase, will produce different outputs.

36) What are environmental variables?

Environmental variables are global settings that control the shell's function as well as that of other Linux programs. Another common term for environmental variables is global shell variables.

37) What are the different modes when using vi editor?

There are 3 modes under vi:
– Command mode – this is the mode where you start in
– Edit mode – this is the mode that allows you to do text editing
– Ex mode – this is the mode wherein you interact with vi with instructions to process a file

38) Is it possible to use shortcut for a long pathname?

Yes, there is. A feature known as filename expansion allows you do this using the TAB key. For example, if you have a path named /home/iceman/assignments directory, you would type as follows: /ho[tab]/ice[tab]/assi[tab] . This, however, assumes that the path is unique, and that the shell you're using supports this feature.

39) What is redirection?

Redirection is the process of directing data from one output to another. It can also be used to direct an output as an input to another process.

40) What is grep command?

grep a search command that makes use of pattern-based searching. It makes use of options and parameters that is specified along the command line and applies this pattern into searching the required file output.

41) What could possibly be the problem when a command that was issued gave a different result from the last time it was used?

One highly possible reason for getting different results from what seems to be the same command has something to do with case sensitivity issues. Since Linux is case sensitive, a command that was previously used might have been entered in a different format from the present one. For example, to lists all files in the directory, you should type the command ls, and not LS. Typing LS would either result in an error message if there is no program by that exact name exist, or may produce a different output if there is a program named LS that performs another function.

42) What are the contents in /usr/local?

It contains locally installed files. This directory actually matters in environments where files are stored on the network. Specifically, locally-installed files go to /usr/local/bin, /usr/local/lib, etc.). Another application of this directory is that it is used for software packages installed from source, or software not officially shipped with the distribution.

43) How do you terminate an ongoing process?

Every process in the system is identified by a unique process id or pid. Use the kill command followed by the pid in order to terminate that process. To terminate all process at once, use kill 0.

44) How do you insert comments in the command line prompt?

Comments are created by typing the # symbol before the actual comment text. This tells the shell to completely ignore what follows. For example: "# This is just a comment that the shell will ignore."

45) What is command grouping and how does it work?

You can use parentheses to group commands. For example, if you want to send the current date and time along with the contents of a file named OUTPUT to a second file named MYDATES, you can apply command grouping as follows: (date cat OUTPUT) > MYDATES

46) How do you execute more than one command or program from a single command line entry?

You can combine several commands by separating each command or program using a semicolon symbol. For example, you can issue such a series of commands in a single entry:

Answer:

1 ls –l cd .. ls –a MYWORK which is equivalent to 3 commands: ls -l cd.. ls -a MYWORK

47) Write a command that will look for files with an extension “c”, and has the occurrence of the string “apple” in it.

Answer:

1 Find ./ -name “*.c” | xargs grep –i “apple”

48) Write a command that will display all .txt files, including its individual permission.

Answer:

1 ls -a -l *.txt

49) Write a command that will do the following:
-look for all files in the current and subsequent directories with an extension c,v
-strip the,v from the result (you can use sed command)
-use the result and use a grep command to search for all occurrences of the word ORANGE in the files.

Answer:

1 Find ./ -name “*.c,v” | sed ‘s/,v//g’ | xargs grep “ORANGE”

50) What, if anything, is wrong with each of the following commands?
a) ls -l-s
b) cat file1, file2
c) ls – s Factdir

Answers:
a) there should be space between the 2 options: ls -l -s
b) do not use commas to separate arguments: cat file1 file2
c) there should be no space between hyphen and option label: ls –s Factdir

51. What is initrd image and what is its function in the linux booting process ?

The initial RAM disk (initrd) is an initial root file system that is mounted prior to when the real root file system is available.The initrd is bound to the kernel and loaded as part of the kernel boot procedure. The kernel then mounts this initrd as part of the two-stage boot process to load the modules to make the real file systems available and get at the real root file system. Thus initrd image plays a vital role in linux booting process.

52. Explain the terms suid, sgid and sticky bit ?

In addition to the basic file permissions in Linux, there are few special permissions that are available for executable files and directories.

SUID : If setuid bit is set, when the file is executed by a user, the process will have the same rights as the owner of the file being executed.

SGID : Same as above, but inherits group previleges of the file on execution, not user previleges. Similar way when you create a file within directory,it will inherit the group ownership of the directories.

Sticky bit : Sticky bit was used on executables in linux so that they would remain in the memory more time after the initial execution, hoping they would be needed in the near future. But mainly it is on folders, to imply that a file or folder created inside a stickybit enabled folder could only be deleted by the owner. A very good implementation of sticky bit is /tmp ,where every user has write permission but only users who own a file can delete them.

53. List out few of the differences between Softlink and Hardlink ?

a) Hardlink cannot be created for directories. Hard link can only be created for a file.

b) Symbolic links or symlinks can link to a directory.

c) Removing the original file that your hard link points to does not remove the hardlink itself; the hardlink still provides the content of the underlying file.

d) If you remove the hard link or the symlink itself, the original file will stay intact.

e) Removing the original file does not remove the attached symbolic link or symlink, but without the original file, the symlink is useless

54. How do you sent a mail attachment via bash console ?

"mutt" is an opensource tool for sending emails with attachments from the linux bash command line. We can install "mutt" from the binary rpm or via packagemanager.

For Ubuntu / Debian based destros.

apt-get install mutt

For Redhat / Fedor based destros,

yum install mutt

Usage :

mutt -s "Subject of Mail" -a "path of attachment file" "email address of recipient" < "message text containing body of the message"

Eg : mutt -s "Backup Data" -a /home/backup.tar.gz admin@mywebsite.com < /tmp/message.txt

55. What is the difference between umask and ulimit ?

umask stands for ‘User file creation mask’, which determines the settings of a mask that controls which file permissions are set for files and directories when they are created. While ulimit is a linux built in command which provides control over the resources available to the shell and/or to processes started by it.

You can limit user to specific range by editing /etc/security/limits.conf at the same time system wide settings can be updated in /etc/sysctl.conf

56. What are the run levels in linux and how to change them ?

A run level is a state of init and the whole system that defines what system services are operating and they are identified by numbers.There are 7 different run levels present (run level 0-6) in Linux system for different purpose. The descriptions are given below.

0: Halt System (To shutdown the system)
1: Single user mode
2: Basic multi user mode without NFS
3: Full multi user mode (text based)
4: unused

5: Multi user mode with Graphical User Interface
6: Reboot System

To change the run level, edit the file “/etc/inittab” and change initdefault entry (id:5:initdefault:). If we want to change the run level on the fly, it can be done using ‘init’ command.

For example, when we type ‘init 3' in the commandline , this will move the system from current runlevel to runlevl 3. Current level can be listed by typing the command 'who -r'

57. What is the functionality of a Puppet Server ?

Puppet is an open-source and enterprise application for configuration management toll in UNIX like operating system. Puppet is an IT automation software used to push configuration to its clients (puppet agents) using code. Puppet code can do a variety of tasks from installing new software, to check file permissions, or updating user accounts and lots of other tasks.

58. What is SeLinux?

SELinux is an acronym for Security-enhanced Linux. It is an access control implementation and security feature for the Linux kernel. It is designed to protect the server against misconfigurations and/or compromised daemons. It put limits and instructs server daemons or programs what files they can access and what actions they can take by defining a security policy.

59. What is crontab and explain the fields in a crontab ?

The cron is a deamon that executes commands at specific dates and times in linux. You can use this to schedule activities, either as one-time events or as recurring tasks. Crontab is the program used to install, deinstall or list the tables used to drive the cron daemon in a server. Each user can have their own crontab, and though these are files in /var/spool/cron/crontabs, they are not intended to be edited directly. Here are few of the command line options for crontab.

crontab -e Edit your crontab file.
crontab -l Show your crontab file.
crontab -r Remove your crontab file.

Traditional cron format consists of six fields separated by white spaces:

<Minute> <Hour> <Day_of_the_Month> <Month_of_the_Year> <Day_of_the_Week> <command/program to execute>

The format is explained in detail below.

```
* * * * * *
| | | | | |
| | | | | +-- Year (range: 1900-3000)
| | | | +---- Day of the Week (range: 1-7, 1 standing for Monday)
| | | +------ Month of the Year (range: 1-12)
| | +-------- Day of the Month (range: 1-31)
| +---------- Hour (range: 0-23)
+------------ Minute (range: 0-59)
```

60. What is initrd image and what is its function in the linux booting process ?

The initial RAM disk (initrd) is an initial root file system that is mounted prior to when the real root file system is available.The initrd is bound to the kernel and loaded as part of the kernel boot procedure. The kernel then mounts this initrd as part of the two-stage boot process to load the

modules to make the real file systems available and get at the real root file system. Thus initrd image plays a vital role in linux booting process.

61. Explain the terms suid, sgid and sticky bit ?

In addition to the basic file permissions in Linux, there are few special permissions that are available for executable files and directories.

SUID : If setuid bit is set, when the file is executed by a user, the process will have the same rights as the owner of the file being executed.

SGID : Same as above, but inherits group previleges of the file on execution, not user previleges. Similar way when you create a file within directory,it will inherit the group ownership of the directories.

Sticky bit : Sticky bit was used on executables in linux so that they would remain in the memory more time after the initial execution, hoping they would be needed in the near future. But mainly it is on folders, to imply that a file or folder created inside a stickybit enabled folder could only be deleted by the owner. A very good implementation of sticky bit is /tmp ,where every user has write permission but only users who own a file can delete them.

62. List out few of the differences between Softlink and Hardlink ?

a) Hardlink cannot be created for directories. Hard link can only be created for a file.

b) Symbolic links or symlinks can link to a directory.

c) Removing the original file that your hard link points to does not remove the hardlink itself; the hardlink still provides the content of the underlying file.

d) If you remove the hard link or the symlink itself, the original file will stay intact.

e) Removing the original file does not remove the attached symbolic link or symlink, but without the original file, the symlink is useless

63. How do you sent a mail attachment via bash console ?

"mutt" is an opensource tool for sending emails with attachments from the linux bash command line. We can install "mutt" from the binary rpm or via packagemanager.

For Ubuntu / Debian based destros.

apt-get install mutt

For Redhat / Fedor based destros,

yum install mutt

Usage :

mutt -s "Subject of Mail" -a "path of attachment file" "email address of recipient" < "message text containing body of the message"

Eg : mutt -s "Backup Data" -a /home/backup.tar.gz admin@mywebsite.com < /tmp/message.txt

64. What is the difference between umask and ulimit ?

umask stands for 'User file creation mask', which determines the settings of a mask that controls which file permissions are set for files and directories when they are created. While ulimit is a linux built in command which provides control over the resources available to the shell and/or to processes started by it.

You can limit user to specific range by editing /etc/security/limits.conf at the same time system wide settings can be updated in /etc/sysctl.conf

65. What are the run levels in linux and how to change them ?

A run level is a state of init and the whole system that defines what system services are operating and they are identified by numbers.There are 7 different run levels present (run level 0-6) in Linux system for different purpose. The descriptions are given below.

0: Halt System (To shutdown the system)
1: Single user mode
2: Basic multi user mode without NFS
3: Full multi user mode (text based)
4: unused

5: Multi user mode with Graphical User Interface
6: Reboot System

To change the run level, edit the file "/etc/inittab" and change initdefault entry (id:5:initdefault:). If we want to change the run level on the fly, it can be done using 'init' command.

For example, when we type 'init 3' in the commandline , this will move the system from current runlevel to runlevl 3. Current level can be listed by typing the command 'who -r'

66. What is the functionality of a Puppet Server ?

Puppet is an open-source and enterprise application for configuration management toll in UNIX like operating system. Puppet is an IT automation software used to push configuration to its clients (puppet agents) using code. Puppet code can do a variety of tasks from installing new software, to check file permissions, or updating user accounts and lots of other tasks.

67. What is SeLinux?

SELinux is an acronym for Security-enhanced Linux. It is an access control implementation and security feature for the Linux kernel. It is designed to protect the server against misconfigurations and/or compromised daemons. It put limits and instructs server daemons or programs what files they can access and what actions they can take by defining a security policy.

68. What is crontab and explain the fields in a crontab ?

The cron is a deamon that executes commands at specific dates and times in linux. You can use this to schedule activities, either as one-time events or as recurring tasks. Crontab is the program used to install, deinstall or list the tables used to drive the cron daemon in a server. Each user can have their own crontab, and though these are files in /var/spool/cron/crontabs, they are not intended to be edited directly. Here are few of the command line options for crontab.

crontab -e Edit your crontab file.
crontab -l Show your crontab file.
crontab -r Remove your crontab file.

Traditional cron format consists of six fields separated by white spaces:

<Minute> <Hour> <Day_of_the_Month> <Month_of_the_Year> <Day_of_the_Week> <command/program to execute>

The format is explained in detail below.

```
* * * * * *
| | | | | |
| | | | | +-- Year (range: 1900-3000)
| | | | +---- Day of the Week (range: 1-7, 1 standing for Monday)
| | | +------ Month of the Year (range: 1-12)
| | +-------- Day of the Month (range: 1-31)
| +---------- Hour (range: 0-23)
+------------ Minute (range: 0-59)
```

69. What are inodes in Linux ? How to find the inode associated with a file ?

An inode is a data structure on a filesystem on Linux and other Unix-like operating systems that stores all the information about a file except its name and its actual data. When a file is created, it is assigned both a name and an inode number, which is an integer that is unique within the filesystem. Both the file names and their corresponding inode numbers are stored as entries in the directory that appears to the user to contain the files. The concept of inodes is particularly important to the recovery of damaged filesystems. When parts of the inode are lost, they appear in the lost+found directory within the partition in which they once existed.

The following will show the name of each object in the current directory together with its inode number:

ls -i

The avialble number inodes in a filesystem can be found using the below command :

df -i

The other way we can get the inode details of a file by using the stat commmand.

Usage : # stat <file name>

Example :

```
-sh-4.1$ stat note.txt
File: `note.txt'
Size: 4 Blocks: 8 IO Block: 4096 regular file
Device: fd05h/64773d Inode: 8655235 Links: 1
Access: (0644/-rw-r--r--) Uid: (69548/nixuser) Gid: (25000/ UNKNOWN)
Access: 2014-06-29 15:27:56.299214865 +0000
Modify: 2014-06-29 15:28:28.027093254 +0000
Change: 2014-06-29 15:28:28.027093254 +0000
```

70. What is the difference between name based virtual hosting and IP based virtual hosting ? Explain the scenario where name based virtual hosting seems useful ?

Virtual hosts are used to host multiple domains on a single apache instance. You can have one virtual host for each IP your server has, or the same IP but different ports, or the same IP, the same port but different host names. The latter are called "name based vhosts".

n IP-based virtual hosting, we can run more than one web site on the same server machine, but each web site has its own IP address while In Name-based virtual hosting, we host multiple websites on the same IP address. But for this to succeed, you have to put more than one DNS record for your IP address in the DNS database.

In the production shared webhosting environment, getting a dedicated IP address for every domains hosted in the server is not feasible in terms of cost. Most of the customers wont be able to afford the cost of having a dedicated IP address. Here is the place where the concepts of Name based virtual hosting find its place.

71. What is network bonding in Linux and where the important configuration files involved? What is the advantage of Network Bonding ?

Network Bonding is a Linux kernel feature that allows to aggregate multiple network interfaces into a single virtual link. This is a great way to achieve redundant links, fault tolerance or load balancing networks in production system. If one of the physical NIC is down or unplugged, it will automatically move traffic to the other NIC card. Similar way the bonding will increase the interface throughput to handle the traffic it it is configured in active-active mode.

There are 7 modes starting from 0 to 6 which decides how the bonding configuration behaves.

mode=0 (balance-rr) - Round-robin policy

It the default mode. It transmits packets in sequential order from the first available slave through the last.
This mode provides load balancing and fault tolerance.

mode=1 (active-backup)

Active-backup policy: In this mode, only one slave in the bond is active. The other one will become active, only when the active slave fails. The bond’s MAC address is externally visible on only one port (network adapter) to avoid confusing the switch. This mode provides fault tolerance.

mode=2 (balance-xor)

Transmit the traffic based on [(source MAC address XOR'd with destination MAC address) modulo slave count]. This selects the same slave for each destination MAC address. This mode provides load balancing and fault tolerance.

mode=3 (broadcast)

Broadcast policy: transmits everything on all slave interfaces. This mode provides fault tolerance.

mode=4 (802.3ad)

Creates aggregation groups that share the same speed and duplex settings. Utilizes all slaves in the active aggregator according to the 802.3ad specification.

mode=5 (balance-tlb) - Adaptive transmit load balancing

channel bonding that does not require any special switch support. The outgoing traffic is distributed according to the current load (computed relative to the speed) on each slave. Incoming traffic is received by the current slave. If the receiving slave fails, another slave takes over the MAC address of the failed receiving slave.

mode=6 (balance-alb) - Adaptive load balancing

It includes balance-tlb plus receive load balancing (rlb) for IPV4 traffic, and does not require any special switch support. The receive load balancing is achieved by ARP negotiation.

Important Configuration Files involved :

/etc/sysconfig/network-scripts/ifcfg-bond0
/etc/modprobe.d/bonding.conf
/etc/sysconfig/network-scripts/ifcfg-eth[0-4]
/proc/net/bonding/bond0

72. Explain briefly the procedure for re-installing Grub in Linux ?

1) Download Ubuntu Installation / Live cd

2) Boot from Ubuntu Installation / Live cd - usb, burned cd etc.
3) During boot select "Try Ubuntu" , Don't select install !
4) Mount your Linux root partition
sudo mount /dev/sda6 /mnt (Assuming /dev/sda6 is the Linux root partition)
5) Install / reinstall grub
$ sudo grub-install --root-directory=/mnt/ /dev/sda (where /dev/sda is your primary disk)
Installation finished. No error reported.

6) Reboot your system, remove bootable CD and we should have the boot menu ready when the system starts.

73. Explain the fields in /etc/passwd and /etc/shadow ?

The /etc/shadow file stores actual password in encrypted format with some additional properties related to user password.It mainly holds athe account aging parameters. All fields are separated by a colon (:) symbol. It contains one entry per line for each user listed in /etc/passwd file Generally, shadow file entry looks as below.

steve:1XOdE07rn$WA6qFm4W5UIqNfaqE5Uub.:13775:0:99999:7:::

Here is the explanation of each field.

User name : Your login name

Password: Your encrypted password.

Last password change : Days since Jan 1, 1970 that password was last changed

Minimum: The minimum number of days required between password changes.

Maximum: The maximum number of days the password is valid.

Warn : The number of days before password is to expire that user is warned that his/her password must be changed

Inactive : The number of days after password expires that account is disabled

Expire : days since Jan 1, 1970 that account is disabled. It indicates an absolute date specifying when the login may no longer be used

The /etc/passwd file stores essential information, which is required during login /etc/passwd is a text file, that contains a list of user account related parameters like user ID, group ID, home directory, shell, etc.

Here is the sample entry from /etc/passwd file

steve:x:6902:6902::/home/steve:/bin/bash

Username: User's login name.

Password: An x character indicates that encrypted password is stored in /etc/shadow file.

User ID (UID): Each user must be assigned a user ID (UID). UID 0 (zero) is reserved for root.

Group ID (GID): The primary group ID

User Info: The comment field. It allow you to add extra information about the user.

Home directory: The absolute path to the directory the user will be in when they log in.

Command/shell: The absolute path of a command or shell (/bin/bash).

74. How do you boot your system into the following modes, when you are in some trouble ?

a) Rescue mode
b) Single user mode
c) Emergency mode

Rescue mode provides the ability to boot a small Linux environment from an external bootable device like a CD-ROM, or USB drive instead of the system's hard drive.Rescue mode is provided to help you with your system from repairing the file system or fixing certain issues which prevent your normal operations.

In order to get into the rescue mode, change the BIOS settings of the machine to boot from the external media. Once the system started booting using bootable disk, add the keyword rescue as a kernel parameter or else you can give the parameter "linux rescue" in the graphical boot interface.

In single-user mode, the system boots to runlevel 1, but it will have many more additional functionalities compared to switching to runlevel 1 from other levels.

The local file systems can be mounted in this mode, but the network is not activated.

Use the following steps to boot into single-user mode:

1) At the GRUB splash screen during the booting process, press any key to enter the GRUB interactive menu.
2) Select the proper version of kernel that you wish to boot and type "a" to append the line.
3) Go to the end of the line and type "single" as a separate word.
4) Press Enter to exit edit mode and type "b" to boot into single usermode now.

In emergency mode, you are booting into the most minimal environment possible. The root file system is mounted read-only and almost nothing is set up. The main advantage of emergency mode over single-user mode is that the init files are not loaded. If the init is corrupted , you can still mount file systems to recover data that could be lost during a re-installation. To boot into emergency mode, use the same method as described for single-user mode, with one exception, replace the keyword single with the keyword "emergency".

75. In the ps results few of the processes are having process state as "D" . What does it mean ? Briefly explain different process states ?

To have a dynamic view of a process in Linux, always use the top command. This command provides a real-time view of the Linux system in terms of processes. The eighth column in the output of this command represents the current state of processes. A process state gives a broader indication of whether the process is currently running, stopped, sleeping etc.

A process in Linux can have any of the following four states…

Running – A process is said to be in a running state when either it is actually running/ executing or waiting in the scheduler's queue to get executed (which means that it is ready to run). That is the reason that this state is sometimes also known as 'runnable' and represented by (R).

Waiting or Sleeping – A process is said to be in this state if it is waiting for an event to occur or waiting for some resource-specific operation to complete. So, depending upon these scenarios, a waiting state can be subcategorised into an interruptible (S) or uninterruptible (D) state respectively.

Stopped – A process is said to be in the stopped state when it receives a signal to stop. This usually happens when the process is being debugged. This state is represented by (T).

Zombie – A process is said to be in the zombie state when it has finished execution but is waiting for its parent to retrieve its exit status. This state is represented by (Z).

Apart from these four states, the process is said to be dead after it crosses over the zombie state; ie when the parent retrieves its exit status. 'Dead' is not exactly a state, since a dead process ceases to exist.

76. What is drop cache in Linux and how do you clear it ?

Cache in Linux memory is where the Kernel stores the information it may need later, as memory is incredible faster than disk.

It is great that the Linux Kernel takes care about that.Linux Operating system is very efficient in managing your computer memory, and will automatically free the RAM and drop the cache if some application needs memory.

Kernels 2.6.16 and newer provide a mechanism to have the kernel drop the page cache and/or inode and dentry caches on command, which can help free up a lot of memory. Now we can throw away that script that allocated a ton of memory just to get rid of the cache.

To free pagecache:

```
# echo 1 > /proc/sys/vm/drop_caches
```

To free dentries and inodes:

echo 2 > /proc/sys/vm/drop_caches

To free pagecache, dentries and inodes:

echo 3 > /proc/sys/vm/drop_caches

This is a non-destructive operation in normal scenarios and will only free things that are completely unused. Dirty objects will continue to be in use until written out to disk and are not freeable. However it is always preferred to run "sync" first to flush useful things out to disk.

77. Password based authentication is disabled in your infrastructure. So how do you login to the servers ?

To improve the system security even further, most of the organizations turned to use keybased authentications instead of Password based authentication. We can enforce the key-based authentication by disabling the standard password authentication, which involves a public key private key pair. The public key is added in the server configuration file while private key is kept kept confidential on the client side.

Below listed is the procedure, to set up keybased authentication.

1) Generating Key Pairs

a) Generate an RSA key pair by typing the following at a shell prompt:

$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/steve/.ssh/id_rsa):

b) Press Enter to confirm the default location (that is, ~/.ssh/id_rsa) for the newly created key.

c) Enter a passphrase, and confirm it by entering it again when prompted to do so.

d) Copy the content of ~/.ssh/id_rsa.pub into the ~/.ssh/authorized_keys on the machine to which you want to connect,
appending it to its end if the file already exists.

e) Change the permissions of the ~/.ssh/authorized_keys file using the following command:

$ chmod 600 ~/.ssh/authorized_keys

2) Now on your client side, open the remote connection agent like putty and browse your public key and try SSH to the server, you should be able to login without a password now.

```
# ssh server1.myserver.com
The authenticity of host 'server1.myserver.com (192.168.44.2)' can't be established.
RSA key fingerprint is e3:c3:89:37:4b:94:37:d7:0c:d5:6f:9a:38:62:ce:1b.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added 'server1.myserver.com' (RSA) to the list of known hosts.
Last login: Tue July 13 12:40:34 2014 from server2.myserver.com
```

3) Public key authentication can prevent brute force SSH attacks, but only if all password-based authentication methods are disabled. Once public key authentication has been confirmed to be working, disable regular password authentication by editing /etc/ssh/sshd_config and set the following option to "no".

```
PasswordAuthentication no
```

78. Explain the different Scenarios involved in TCP 3 way handshake ?

The TCP three way handshake is the process for establishing a TCP connection.We can explain 3 way handshake with a simple scenario where we assume a client computer is contacting a server to send it some information.

a) The client sends a packet with the SYN bit set and a sequence number of N.
b) The server sends a packet with an ACK number of N+1, the SYN bit set and a sequence number of X.
c) The client sends a packet with an ACK number of X+1 and the connection is established.
d) The client sends the data.

The first three steps in the above process is called the three way handshake.

79. As the disk space utilization was so high in the server, the Administrator has removed few files from the server but still the disk utilization is showing as high. What would be the reason ?

In Linux even if we remove a file from the mounted file system, that will still be in use by some application and for this application it remains available. Its because file descriptor in /proc/ filesystem is held open..So if there are such open descriptors to files already removed, space occupied by them considered as used. You find this difference by checking them using the "df" and "du" commands. While df is to show the file system usage, du is to report the file space usage. du works from files while df works at filesystem level, reporting what the kernel says it has available.

You can find all unlinked but held open files with:

lsof | grep '(deleted)'

This will list the filename which is open witht he pid in which it is running. We can kill those Pids and which will stop these process and will recover the disk space responsible for this file.

80. What is rDNS and explain its benefits in the Linux Domain Name Systems ?

A typical DNS lookup is used to determine which IP address is associated with a hostname, and this is called Forward DNS lookup. A reverse DNS lookup is used for the opposite, to determine which hostname is associated with an IP address. Sometimes reverse DNS lookups are required for diagnostic purposes. Today, reverse DNS lookups are used mainly for security purposes to trace a hacker or spammer. Many modern mailing systems use reverse mapping to provide simple authentication using dual lookup: hostname-to-address and address-to-hostname. The rDNS (reverse DNS) is implemented using a specialized zone record for reverse lookups called PTR record. PTR records always resolve to names, never IP addresses.

81. What is sosreport, how do you generate it while working with your Redhat Support Team in production ?

Sosreport is a command-line utility in Redhat based linux destros (RHEL / CentOS) which collects system configuration and diagnostic information of your linux box like running kernel version, loaded modules, and system and service configuration files. This command also runs external programs to collect further information, and stores this output in the resulting archive. Sosreport is required when you have open a case with redhat for technical support. Redhat support Engineers will require sosreport of your server for troubleshooting purpose. To run sosreport, sos package should be installed. Sos package is part of default installation in most of linux. If for any reason this package is no installed , then use below yum command to install it manually :

yum install sos

Generate the report

Open the terminal type sosreport command :

sosreport

This command will normally complete within a few minutes. Depending on local configuration and the options specified in some cases the command may take longer to finish. Once completed,

sosreport will generate a compressed a file under /tmp folder. The file should be provided to Redhat support representative as an attachment to open a support case.

82. What is swappiness in Linux Memory Management and how do we configure that ?

The swappiness parameter controls the tendency of the kernel to move processes out of physical memory and onto the swap disk. Because disks are much slower than RAM, this can lead to slower response times for system and applications if processes are too aggressively moved out of memory.

swappiness can have a value of between 0 and 100

swappiness=0 tells the kernel to avoid swapping processes out of physical memory for as long as possible

swappiness=100 tells the kernel to aggressively swap processes out of physical memory and move them to swap cache

The default setting in Redhat/Ubuntu based Linux distros is swappiness=60. Reducing the default value of swappiness will probably improve overall performance for a typical Ubuntu desktop installation.

```
~$ cat /proc/sys/vm/swappiness
60
```

If we have enough RAM, we can turn that down to 10 or 15. The swap file will then only be used when the RAM usage is around 80 or 90 percent.

To change the system swappiness value, open /etc/sysctl.conf as root. Then, change or add this line to the file:

```
vm.swappiness = 10
```

Reboot for the change to take effect

You can also change the value while your system is still running

```
sysctl vm.swappiness=10
```

We can also clear swap by running swapoff -a and then swapon -a as root instead of rebooting to achieve the same effect.

83. What is git ?

Git is a very popular and efficient open source Version Control System. It tracks content such as files and directories. It stores the file content in BLOBs - binary large objects. The folders are represented as trees. Each tree contains other trees (subfolders) and BLOBs along with a simple text file which consists of the mode, type, name and Secure Hash Algorithm of each blob and subtree entry. During repository transfers, even if there are several files with the same content and different names, the GIT software will transfer the BLOB once and then expand it to the different files.

84. What is inode ? Briefly explain the structure ?

Inode is a data structure that keeps track of all the information about a file. When we keep our information in a file and the OS stores the information about a file in an inode. Information about files is sometimes called metadata. We can say that an inode is metadata of the data. In a file system, inodes consist roughly of 1% of the total disk space, whether it is a whole storage unit or a partition on a storage unit. The inode space is used to ?track? the files stored on the hard disk. The inode entries store metadata about each file, directory or object, but only points to these structures rather than storing the data. Each entry is 128 bytes in size. The metadata contained about each structure can include the following:

Inode number
Access Control List (ACL)
Extended attribute
Direct/indirect disk blocks
Number of blocks
File access, change and modification time
File deletion time
File generation number
File size
File type
Group
Number of links
Owner
Permissions
Status flags

Inode structure of a directory consists of a name to inode mapping of files and directories in that directory.In a directory, You can find the inode number corresponding to the files using the command "ls -i"

786727 -rw------- 1 root root 4226530 May 29 13:17 sudo.log
786437 -rw-------. 1 root root 32640 Jun 23 20:11 tallylog

786440 -rw-rw-r--. 1 root utmp 276096 Jul 20 06:45 wtmp
786741 -rw------- 1 root root 9653 Jul 17 09:38 yum.log

Similar way, the number of inodes allocated, used and free in a Filesystem can be listed using "df -i" command

df -i /root
Filesystem Inodes IUsed IFree IUse% Mounted on
/dev/mapper/RootVol-lvmroot
524288 80200 444088 16% /

85. User's Primary Group id is listed in which file, at the time of creation of the user (On a standard Unix system)

a) /etc/passwd
b) /etc/groups
c) /etc/login
d) /etc/profile

Ans:- a) /etc/passwd

86. The encrypted password of a user is stored in

a) /etc/shadow
b) /etc/enpasswwd
c) /etc/.passwd
d) /etc/passwd

Ans:- a) /etc/shadow

87. A user can change the default log-in shell using

a) chmod
b) chsh

c) rmsh
d) tchsh

Ans:- b) chsh

88. Which of the following identifiers associated with a process decide its privilege level

a) uid
b) suid
c) euid
d) gid

Ans:- c) euid

89. The /etc/passwd file doesn't contain

a) userid
b) home directory for a user
c) login shell name
d) None of the above

Ans:- d) None of the above

90. User id 0 is

a) An innvalid user id
b) The id of the root user
c) The id of a user when the user's account is deleted
d) None of the above

Ans:- b) The id of the root user

91. The login shell is

a) The shell program that runs when the user logs in
b) The shell program that authenticates the user while logging in
c) Common shell for all the users that belong to the same group
d) None of the above

Ans:- a) The shell program that runs when the user logs in

92. Which of the following command can be used to change the user password?

a) User can't change the password
b) passwd
c) passd
d) pwd

Ans:- b) passwd

93. What does the following command do ?
who | wc –l

a) List the number of users logged in
b) List the users
c) List the number of users in the system
d) Display the content of who command

Ans:- a) List the number of users logged in

94. By default, a Linux user falls under which group?

a) staff
b) others
c) same as userid (UPG)
d) system

Ans:- c) same as userid (UPG)

95. Which of the following files need to be referred for user's secondary group?

a) /etc/passwd
b) /etc/shadow
c) /etc/group
d) /etc/profile

Ans:- c) /etc/group

Q:96 What is the default port & configuration file of SSH Server ?

Ans: 22 is default port for ssh and '/etc/ssh/sshd_config' is the configuration file.

Q:97 How to change the default ssh port in linux ?

Ans: To change the port , first edit the file ' /etc/ssh/sshd_config ' , change the vaule of parameter ' port 22 ', now restart the ssh service.

Q:98 What is the configuration file of ssh client ?

Ans: ' /etc/ssh/ssh_config ' is configuration file for ssh client.

Q:99 What is SSH port forwarding ?

Ans: SSH Port Forwarding, sometimes called SSH Tunneling, which allows you to establish a secure SSH session and then tunnel arbitrary TCP connections through it. Tunnels can be created at any time, with almost no effort and no programming.

Syntax : ssh -L localport:host:hostport user@ssh_server -N

where:
-L – port forwarding parameters
localport – local port (chose a port that is not in use by other service)
host – server that has the port (hostport) that you want to forward
hostport – remote port

-N – do not execute a remote command, (you will not have the shell, see below)
user – user that have ssh access to the ssh server (computer)
ssh_server – the ssh server that will be used for forwarding/tunneling

Without the -N option you will have not only the forwarding port but also the remote shell.

Q:100 How to disable the root login in linux server ?

Ans: Open the file ‘ /etc/ssh/sshd_config ‘ and chnage the paramenter ‘PermitRootLogin yes’ to ‘PermitRootLogin no’ & restart the ssh service.

Q:101 How to allow only specific users to ssh your linux server ?

Ans: Open the file ‘/etc/ssh/sshd_config’ and add the parameter ‘AllowUsers user1 user2’ **& then restart the ssh serve**r.

Q:102 How to enable debugging in ssh command ?

Ans: To enable debugging in ssh command use ‘-v’ option like ‘ssh root@www.linuxtechi.com -v’. To increase the debugging level just increase the number of v’s.

Q:103 What is the difference between ssh & Telnet ?

Ans: In ssh communication between client & server is encrypted but in telnet communication between the client & server is in plain text . We can also say SSH uses a public key for authentication while Telnet does not use any authentication.SSH adds a bit more overhead to the bandwidth compared to Telnet.Default port of ssh is 22 and for telnet 23.

Q:104 What is use of sshpass command in linux ?

Ans: sshpass is a command which allows us to automatically supply password to the command prompt so that automated scripts can be run as desired by users. sshpass supplies password to ssh prompt using a dedicated tty , fooling ssh to believe that a interactive user is supplying password.

Q:105 What is the use of scp command ?

Ans: SCP stands for Secure Copy ,it copies files between hosts over a network. It uses ssh for data transfer & uses the same authentication and provides the same security as ssh. Unlike rcp, scp will ask for passwords or passphrases if they are needed for authentication.

Q:106 What is the use of blowfish options in scp command ?

Ans: Using blowfish options in scp command , we can increase the speed, by default scp uses the Triple-DES cipher to encrypt the data being copied.
Example : scp -c blowfish /home/itstuff.txt
root@mail.linuxtechi.com:/opt/

Q:107 How to limit the bandwidth used by scp command ?

Ans: We can limit the bandwidth used by the scp command using the -l option as shown in the syntax.'#scp -l bandwidth_limit filename username@remote-host:/folder-name' , where bandwidth_limit is numeric to be specified in kilobits per second.

Q:108 How to enable passwordless ssh authentication in Linux ?

Ans: To Implement passwordless or Keys based authentication we have to generate Public and Private keys , Copy the Pubic keys to

remote Linux servers either manually or by ssh-copy-id command.ssh-copy-id command will automatically copy the contents of id_rsa.pub file to '~/.ssh/authorized_keys' file of remote linux server.

Q:109 How to check SSH server's Version ?

Ans: Using the command ' ssh -V ' we can find the ssh server's version.

Q:110 How to Copy the file "server.txt" from the local machine to a remote host using port 2751 in /opt folder.

Ans: scp -P 2751 /home/server.txt root@mail.linuxtechi.com:/opt

111. Is it possible to increase the logical volume on fly ?

Answer: Yes.We can increase the logical volume without umount it.

1. How to reduce the logical volume ? is it possible to reduce on fly ?
   Answer: No.You can't reduce the logical volume on fly. Here is the steps to reduce the logical volume on redhat Linux.

- Un-mount the filesystem
- Run e2fsck on the volume device
- Reduce the Filesystem.(resize2fs)
- Reduce the logical Volume(lvreduce)
- Mount the filesystem back for production.

112 . How to do you scan the new LUN or disk for LVM physical volume ?

Answer:Use "pvscan" to scan existing physical volume from newly connected SAN or DISKS.

113 . How to scan disks for existing volume group ?

Answer:Use “vgscan” to scan existing volume group from newly connected SAN or DISKS.
But you should use “pvscan” prior to executing this command.

114 . How to scan a logical volume from exising volume group?

Answer: lvscan

115 . How to stop the logical volume ? or deactivate the logical volume ?

Answer: “lvchange -an /dev/vg_name/lv_name”

116 . How to activated the logical volume which in deactivated state ?

Answer: “lvchange -ay /dev/vg_name/lv_name” .

117 . How to disable the volume group ? or Deactivate the volume group ?

Answer:”vgchange -an volume_group_name” .

118 . How to enable the volume group ? or Activate the volume group ?

Answer:”vgchange -ay volume_group_name” .

119 . How do you find that what are the disks are used for logical volume mirroring ?

Answer: use “lvs -a -o +devices”

120 . What are steps to perform in order to increase the logical volume on fly ?

Answer:

- Extend the logical volume
- Increase the Filesystem size
- Verify the status using df command or lvs command.

121 . How to list the imported volume groups ?

Answer: Use “vgs” command to display the imported volume group.

122 . How to list the available logical volumes on the system?

Answer: Use “lvs” command to list the available logical volumes on the system.

123 . How to list the available physical volumes in LVM?

Answer: Use “pvs” command to list the available physical volumes.

124 .How to see the detailed volume group information ?

Answer: Use “vgdisplay vg_name”

125 . How to see the detailed logical volume information ?

Answer: Use “lvdisplay /dev/vg_name/lv_name”

126 .How to see the detailed physical volume information ?

Answer: Use “pvdisplay /dev/disk_name” Ex: pvdisplay /dev/sde

127 . How to rename volume Group ? can we rename the VG on fly ?

Answer:Yes.Its possible to rename the volume group on fly.But
the mounted volumes will not reflect the same unless you re-mount the volume with new VG name.
Need to update the /etc/fstab with new VG name to mount the volumes across the system reboot.

128 . How to take a LVM configuration backup ?

Answer:Use “vgcfgbackup vg_name” to take the latest configuration backup of volume group.The default volume group backup location is “/etc/lvm/backup”

129 . How to re-create the device files for LVM volumes ?

Answer:Run “vgmknodes” to recreate the LVM devices files.

130 . What is lvmdump ?

Answer: ”lvmdump” is tool for LVM2 to collect the various information for diagnostic purposes.By default, it creates a tarball suitable for submission along with a problem report

131 . How are snapshots in LVM2 different from LVM1 in Redhat Linux?

Answer:LVM1 snapshots are readonly by default where LVM2 snapshots were read/write.

132 . What are the steps involved to create the logical volume from scratch ?

Answer:

i. Create a physical volume using pvcreate command.
   #pvcreate /dev/sdc
ii. Create a volume group using “vgcreate” command
   #vgcreate vg02 /dev/sdc
iii. Create a logical volume using “lvcreate” command
   #lvcreate -L 100M -n vol1 vg02
iv. Create a filesystem on logical volume using mkfs command.
   #mkfs -t ext4 /dev/vg02/vol1

v.Mount the filesystem using mount command for use.
#mount -t ext4 /dev/vg02/vol1 /vol1

133 . How to extent the volume group ?

Answer:Using “vgextend” we can increase the volume group.

134 . Assume Volume group “vg02” is already exists.How do you extend the volume group with 50GB ? Provide all the steps with commands.

Answer:

1.Get the 50GB lun from SAN team.(/dev/sdd)
2.Create physcical volume (# pvcreate /dev/sdd)
2.Extend the volume group (# vgextend vg02 /dev/sdd)

135 . If the vg02 has two physical volumes called /dev/sdc/ & /dev/sdd. How do you remove /dev/sdd from vg02.

Answer: “vgreduce vg02 /dev/sdd/”

136 . How to decommission/remove LVM completely from the host ?

Answer:

1. Un-mount all the logical filesystems
2. Remove the logical volumes using “lvremove” command.
3. Destroy the volume group using “vgremove” command.
4. Use “pvremove” command remove the physical volumes from the system.

137 . Why is LVM is required ?

Ans: LVM stands for Logical Volume Manager , to resize filesystem’s size online we required LVM partition in Linux. Size of LVM partition can be extended and reduced using the lvextend & lvreduce commands respectively.

138 . How To check Memory stats and CPU stats ?

Ans: Using ‘free’ & ‘vmstat’ command we can display the physical and virtual memory statistics respectively.With the help of ‘sar’ command we see the CPU utilization & other stats.

139 . What does Sar provides and at which location Sar logs are stored ?

Ans: Sar Collect, report, or save system activity information. The default version of the sar command (CPU utilization report) might be one of the first facilities the user runs to begin system activity investigation, because it monitors major system resources. If CPU utilization is near 100 percent (user + nice + system), the workload sampled is CPU-bound.

By default log files of Sar command is located at /var/log/sa/sadd file, where the dd parameter indicates the current day.

140 . How to increase the size of LVM partition ?

Ans: Below are the Logical Steps :

- Use the lvextend command (lvextend -L +100M /dev/<Name of the LVM Partition> , in this example we are extending the size by 100MB.
- resize2fs /dev/<Name of the LVM Partition>
- check the size of partition using ‘df -h’ command

141 . How to reduce or shrink the size of LVM partition ?

Ans: Below are the logical Steps to reduce size of LVM partition :
-Umount the filesystem using umount command,
-use resize2fs command , e.g resiz2fs /dev/mapper/myvg-mylv 10G
-Now use the lvreduce command , e.g lvreduce -L 10G /dev/mapper/myvg-mylv
Above Command will shrink the size & will make the filesystem size 10GB.

142 . How to create partition from the raw disk ?

Ans: Using fdisk utility we can create partitions from the raw disk.Below are the steps to create partition from the raw dsik :
- fdisk /dev/hd* (IDE) or /dev/sd* (SCSI)
- Type n to create a new partition
- After creating partition , type w command to write the changes to the partition table.

142. Name the Linux specific partition types?

Ans. 0x82 > Linux swap
0x83 > Linux
0x8e > Linux LVM
0xfd > Linux RAID auto
0x5 > Extended
0xf > Windows partition

143. How many partitions are supported by Linux?

Ans. The maximum number of partitions supported by Linux kernel is:
63 or IDE drives , 15 for SCSI drives

144. Why partitions required?

Ans. Separate partitions improve performance by keeping data together which reduces the disk head seek.

145. What are the tools used to create or manage partitions?

Ans. Fdisk, sfdisk parted (GNU) - An advanced partition manipulation tool (create, copy, resize etc.)

145. What is the function of partprobe?

Ans. Reinitializes the kernel's in memory copy of the partition table.

146. How to create a filesystem?

Ans. The mkfs command is used to create the filesystem.

mkfs.ext2 / mkfs.ext3	>	To create ext2/ext3 filesystem
mkfs.minix	>	minix filesystem
mkfs.msdos	>	MS-DOS filesystem

147. What is the difference between ext2 and ext3 filesystem?

Ans. ext3 filesystem supports journaling, where as ext2 does not.

148. What are the journaling modes supported by ext3 filesystem?

Ans. Ordered > Journals only metadata (This is the default)
Journaled > Journals data as well as metadata
Writeback > Journal updates are not atomic, but this gives better performance.

149. How to convert an ext2 filesystem to ext3?

Ans Change the /etc/fstab to specify ext3 for desired filesystem(s)
Create the ext3 journal on the ext2 filesystem(s) as:
tune2fs -j

If the kernel needs to have access to the ext3 module at boot time, create a new initial ramdisk as:
mkinitrd /boot/initrd-.img

150. How to connect a file system to a Linux box?

Ans. Mount command is used.

151. How to find currently mounted and available filesystems?

Ans. Use mount command without arguments.

152. Which filesystem is referred by mount when displaying mounted & available filesystems?

Ans. When mount command is invoked without any arguments it referes to the /etc/mtab filesystem.

153. How to disconnect a filesystem from a Linux box?

Ans. The umount command is used.

154. Which command is used to display information about the processes using a filesystem?

Ans. The fuser command is used.

155. How to display who/what is acting on a filesystem?

Ans. # fuser -v mnt_point

156. How to kill all actions on a filesystem?

Ans. # fuser -km mnt_point

157. How to view all the mounted and unmounted partitions?

Ans. Use the fdisk -l command to view all the mounted and unmounted filesystems available on your system.

158. How to view only the mounted filesystems?

Ans. Use the df -k command, which shows only mounted filesystems but has the big advantage of giving you the mount points too.

159. What is the function of e2label command?

Ans. With the help of e2label command a filesystem label can be written into the superblock of ext2/ext3 filesystem.
Eg:- #e2label /dev/hda3 datadisk3
Will create a label of datadisk3 on the filesystem on partition /dev/hda3.

160. How to display the label for a device (/dev/hdb2)?

Ans. # e2lable /dev/hdb2

161. What is the function of /etc/fstab?

Ans. This file is referenced each time the system boots to create the desired filesystem hierarchy.

162. What are the fields /etc/fstab containd?

Ans. The /etc/fstab fields are:

Device	>	Special device/filesystem label dev to mount
Mount_Point	>	The path used to access the filesystem
fs_type	>	The filesystem type
Options	>	List of Options (each is separated by a comma)
Dump frequency	>	Level 0 dump freq (0 = never dump, 1 = daily & 2 = every other day)
fsck Order	>	0 = ignore, 1 = first (root should have this value), 2 to 9 = 2nd, 3rd and so on. NFS & CD-ROM s should be ignored (i.e., value 0).

163. How to list the ext2/ext3 filesystem attributes?

Ans. lsattr command displays file attributes

164. How to change the ext2/ext3 filesystem attributes?

Ans. chattr command changes the file attributes.

165. What is swap space?

Ans. Swap space is supplement to system RAM.

166. How to set up swap partition?

Ans. Steps involved in setting up swap partition.
Create a partition using a partitioning program (fdisk/sfdisk/parted)
Set partition id type to 0x82.
Create the signature on the partition using the mkswap command
mkswap -v1 /dev/hdb3
Add an entry for the swap in /etc/fstab file as:
/dev/hdb3 swap swap defaults 0 0
Activate the swap partition using
swapon -a
Check the swap partition status using
swapon -s

167. How to create a swap file?

Ans. Create a file as
dd if=/dev/zero of=swapfile bs=512 count=N
(Where N is the file size in KB)
Run the mkswap to create signature
Activate the swap file with swapon command (OR) initialize it in the startup script /etc/rc.d/rc.local

168. Why fsck used?

Ans. fsck (filesystem check) is used to maintain filesystem consistency.

169. When the system runs the fsck and which script invokes it?

Ans. When the system boots, the rc.sysinit script runs the fsck on any filesystems marked for checking in /etc/fstab file. If any of these filesystems are markes as dirty or have data in the journal, fsck will attempt to repair them. If it succeeds, the filesystems will be mounted and boot process continues, else rc.sysinit will run sulogin and will report that fsck needs to be run manually.

170. What is e2fsck command?

Ans. fsck is a front end tool for e2fsck.

32. What is journaling?
33. Types of journals and their function?

Ans. Ordered : This is the default and journal only meta-data
Journaled : Journals data and meta-data
Writeback : Journal updates are not atomic.

171 . Differentiate ext2 and ext3 file systems?

Ans. ext3 filesystem supports journaling concept where ext2 does not.

172 . What is the function of tune2fs command?

Ans. This is used to modify the filesystem attributes (Like converting ext2 to ext3).

173 Q) What is Linux and why is it so popular?
Answer - Linux is an operating system that uses UNIX like Operating system.....

174 Q) What is LILO?
Answer - LILO is Linux Loader is a boot loader for Linux. It is used to load Linux into the memory and start the Operating system.....

175 Q) What is the difference between home directory and working directory?

Answer - Home directory is the default working directory when a user logs in. On the other hand, working directory is the user's current directory.....

176 Q) What is the difference between internal and external commands?

Answer - Internal commands are commands that are already loaded in the system. They can be executed any time and are independent.....

177 Q) Explain the difference between a static library and a dynamic library.

Answer - Static libraries are loaded when the program is compiled and dynamically-linked libraries are loaded in while.....

178 Q) What is LD_LIBRARY_PATH?

Answer - LD_LIBRARY_PATH is an environment variable. It is used for debugging a new library or a non standard library.....

179 Q) What is the file server in Linux server?

Answer - File server is used for file sharing. It enables the processes required fro sharing.....

180 Q) What is NFS? Q) What is its purpose?

Answer - NFS is Network File system. It is a file system used for sharing of files over a network.....

How do I send email with linux?

Answer - Email can be sent in Linux using the mail command. .....

181 Q) Explain RPM (Red Hat Package Manager) features.

Answer - RPM is a package managing system (collection of tools to manage software packages).....

182 Q) What is Kernel? Q) Explain the task it performs.

Answer - Kernel is used in UNIX like systems and is considered to be the heart of the operating system.....

183 Q) What is Linux Shell? Q) What is Shell Script?

Answer - Linux shell is a user interface used for executing the commands. Shell is a program the user.....

184 Q) What are Pipes? Q) Explain use of pipes.

Answer - A pipe is a chain of processes so that output of one process (stdout) is fed an input (stdin) to another.....

185 Q) Explain trap command; shift Command, getopts command of linux.

Answer - Trap command: controls the action to be taken by the shell when a signal is received. .....

186 Q) What Stateless Linux server? Q) What feature it offers?

Answer - A stateless Linux server is a centralized server in which no state exists on the single workstations. .....

187 Q) What does nslookup do? Q) Explain its two modes.

Answer - Nslookup is used to find details related to a Domain name server. Details like IP addresses of a machine, MX records,.....

188 Q) What is Bash Shell?

Answer - Bash is a free shell for UNIX. It is the default shell for most UNIX systems. It has a combination of the C and Korn shell features. .....

189 Q) Explain some Network-Monitoring Tools in Linux: ping, traceroute, tcpdump, ntop

Answer - Network monitoring tools are used to monitor the network, systems present on the network, traffic etc.....

How does the linux file system work?

Answer - Linux file structure is a tree like structure. It starts from the root directory, represented by '/', and then expands into sub-directories.....

190 Q) What are the process states in Linux?

Answer - Process states in Linux.....

191 Q) What is a zombie?

Answer - Zombie is a process state when the child dies before the parent process. In this case the structural information of the process is still in the process table.....

192 Q) Explain each system calls used for process management in linux.

Answer - System calls used for Process management.....

193 Q) Which command is used to check the number of files and disk space used and the each user's defined quota?

repquota command is used to check the status of the user’s quota along with the disk space and number of files used. This command gives a summary of the user’s quota that how much space and files are left for the user. Every user has a defined quota in Linux. This is done mainly for the security, as some users have only limited access to files. This provides a security to the files from unwanted access. The quota can be given to a single user or to a group of users.

194 Q) What is the name and path of the main system log?

By default the main system log is /var/log/messages. This file contains all the messages and the script written by the user. By default all scripts are saved in this file. This is the standard system log file, which contains messages from all system software, non-kernel boot issues, and messages that go to 'dmesg'. dmesg is a system file that is written upon system boot.

195 Q) How secured is Linux? Q) Explain.

Security is the most important aspect of an operating system. Due to its unique authentication module, Linux is considered as more secured than other operating systems. Linux consists of PAM. PAM is Pluggable Authentication Modules. It provides a layer between applications and actual authentication mechanism. It is a library of loadable modules which are called by the application for authentication. It also allows the administrator to control when a user can log in. All PAM applications are configured in the directory "/etc/pam.d" or in a file "/etc/pam.conf". PAM is controlled using the configuration file or the configuration directory.

196 Q) Can Linux computer be made a router so that several machines may share a single Internet connection? How?

Yes a Linux machine can be made a router. This is called "IP Masquerade." IP Masquerade is a networking function in Linux similar to the one-to-many (1: Many) NAT (Network Address Translation) servers found in many commercial firewalls and network routers. The IP Masquerade feature allows other "internal" computers connected to this Linux box (via PPP, Ethernet, etc.) to also reach the Internet as well. Linux IP Masquerading allows this functionality even if the internal computers do not have IP addresses.
The IP masquerading can be done by the following steps:

1. The Linux PC must have an internet connection and a connection to LAN. Typically, the Linux PC has two network interfaces-an Ethernet card for the LAN and a dial-up PPP connection to the Internet (through an ISP).

2. All other systems on your LAN use the Linux PC as the default gateway for TCP/IP networking. Use the same ISP-provided DNS addresses on all systems.

3. Enable IP forwarding in the kernel. By default the IP forwarding is not enabled. To ensure that IP forwarding is enabled when you reboot your system, place this command in the /etc/rc.d/rc.local file.

4. Run /sbin/iptables-the IP packet filter administration program-to set up the rules that enable the Linux PC to masquerade for your LAN.

197 Q) What is the minimum number of partitions you need to install Linux?

Minimum 2 partitions are needed for installing Linux. The one is / or root which contains all the files and the other is swap. Linux file system is function specific which means that files and folders are organized according to their functionality. For example, all executables are in one folder, all devices in another, all libraries in another and so on. / or 'root' is the base of this file system. All the other folders are under this one. / can be consider as C: .Swap is a partition that will be used as virtual memory. If there is no more available RAM a Linux computer will use an area of the hard disk, called swap, to temporarily store data. In other words it is a way of expanding your computers RAM.

Which command is used to review boot messages?

dmesg command is used to review boot messages. This command will display system messages contained in the kernel ring buffer. We can use this command immediately after booting to see boot messages. A ring buffer is a buffer of fixed size for which any new data added to it overwrites the oldest data in it. Its basic syntax is

dmesg [options]

Invoking dmesg without any of its options causes it to write all the kernel messages to standard output. This usually produces far too many lines to fit into the display screen all at once, and thus only the final messages are visible. However, the output can be redirected to the less command through the use of a pipe, thereby allowing the startup messages to be viewed on one screen at a time
dmesg | less

Which utility is used to make automate rotation of a log?

logrotate command is used to make automate rotation of log.
Syntax of the command is:
logrotate [-dv] [-f|] [-s|] config_file+
It allows automatic rotation, compression, removal, and mailing of log files. This command is mainly used for rotating and compressing log files. This job is done every day when a log file becomes too large. This command can also be run by giving on command line. We can done force rotation by giving –f option with this command in command line. This command is also

used for mailing. We can give –m option for mailing with this command. This option takes two arguments one is subject and other is recipient name.

198 Q) What are the partitions created on the mail server hard drive?

The main partitions are done firstly which are root, swap and boot partition. But for the mail server three different partitions are also done which are as follows:
1. /var/spool- This is done so that if something goes wrong with the mail server or spool than the output cannot overrun the file system.
2. /tmp- putting this on its own partition prevents any user item or software from overrunning the system files.
3. /home- putting this on its own is useful for system upgrades or reinstalls. It allow not to wipe off the /home hierarchy along with other areas.

199 Q) How documentation of an application is stored?

When a new application is installed its documentation is also installed. This documentation is stored under the directory named for application. For example if my application name is App1 then the path of the documentation will be /user/doc/App1. It contains all the information about the application. It contains date of creating application, name of application and other important module of the application. We can get the basic information of application from the documentation.

200 Q) How shadow passwords are given?

pwconv command is used for giving shadow passwords. Shadow passwords are given for better system security. The pwconv command creates the file /etc/shadow and changes all passwords to ‘x’ in the /etc/passwd file. First, entries in the shadowed file which don't exist in the main file are removed. Then, shadowed entries which don't have `x' as the password in the main file are updated. Any missing shadowed entries are added. Finally, passwords in the main file are replaced with `x'. These programs can be used for initial conversion as well to update the shadowed file if the main file is edited by hand.

201 Q) How do you create a new user account?

useradd command is used for creating a new user account. When invoked without the -D option, the useradd command creates a new user account using the values specified on the command line and the default values from the system. The new user account will be entered into the system files as needed, and initial files copied, depending on the command line options. This

command uses the system default as home directory. If –m option is given then the home directory is made.

202 Q) Which password package is installed for the security of central password?

Shadow password packages are used for security of central passwords. Security is the most important aspect of every operating system. When this package is not installed the user information including passwords is stored in the /etc/passwd file. The password is stored in an encoded format. These encoded forms can be easily identified by the System crackers by randomly encoding the passwords from dictionaries. The Shadow Package solves the problem by relocating the passwords to another file (usually /etc/shadow). The /etc/shadow file is set so that it cannot be read by just anyone. Only root will be able to read and write to the /etc/shadow file.

203 Q) Which shell do you assign to a POP3 mail-only account?

POP3 mail only account is assigned to the /bin/false shell. However, assigning bash shell to a POP3 mail only gives user login access, which is avoided. /bin/nologin can also be used. This shell is provided to the user when we don't want to give shell access to the user. The user cannot access the shell and it reject shell login on the server like on telnet. It is mainly for the security of the shells. POP3 is basically used for downloading mail to mail program. So for illegal downloading of emails on the shell this account is assigned to the /bin/false shell or /bin/nologin. These both shells are same they both do the same work of rejecting the user login to the shell. The main difference between these two shells is that false shell shows the incorrect code and any unusual coding when user login with it. But the nologin shell simply tells that no such account is available. So nologin shell is used mostly in Linux.

204 Q) Which daemon is responsible for tracking events on Linux system?

syslogd is responsible for tracking system information and save it to the desired log files. It provides two system utilities which provide system logging and kernel message trapping. Internet and UNIX domain sockets support enable this utility package to support both local and remote logging. Every logged message contains at least a time and a hostname field, normally a program name field, too. So to track these information this daemon is used. syslogd mainly reacts to the set of signals given by the user. These are the signals given to syslogd: SIGHUP: This lets syslogd perform a re-initialization. All open files are closed, the configuration file (default is /etc/syslog.conf) will be reread and the syslog facility is started again. SIGTERM: The syslogd will die. SIGINT, SIGQUIT: If debugging is enabled these are ignored, otherwise syslogd will die. SIGUSR1: Switch debugging on/off. This option can only be used if syslogd is started with the - d debug option. SIGCHLD: Wait for Childs if some were born, because of waiting messages.

205 Q) Which daemon is used for scheduling of the commands?

The crontab command is used for scheduling of the commands to run at a later time. SYNTAX
crontab [-u user] file
crontab [-u user] { -l | -r | -e }

Options
-l List - display the current crontab entries.

-r Remove the current crontab.

-e Edit the current crontab using the editor specified by the VISUAL or EDITOR environment variables.
When user exits from the editor, the modified crontab will be installed automatically. Each user can have their own crontab, and though these are files in /var, they are not intended to be edited directly. If the –u option is given than the crontab gives the name of the user whose crontab is to be tweaked. If it is given without this then it will display the crontab of the user who is executing the command.

206 Q: What Is a Port Number?

A **port number** is part of the addressing information used to identify the senders and receivers of messages. Port numbers are most commonly used with TCP/IP connections. Home network routers and computer software work with ports and sometimes allow you to configure port number settings. These port numbers allow different applications on the same computer to share network resources simultaneously.

207 Q. What is the range of ports or how many ports are there?

Ans:- Port numbers can vary from 0 to 65535, so total we can get 65536 ports

208 Q. Why port numbers are just 65536?

Ans:- This is because limitation in TCP/IP stack where the port field is just 16bit size. So we get only 2^16 port which is equal to 65536 ports

Port numbers have a range of 0..65535 (although often 0 has special meaning). In the original BSD TCP implementation, only `root` can bind to ports 1..1023, and dynamically assigned ports were assigned from the range

1024..5000; the others were available for unprivileged static assignment. These days 1024..5000 is often not enough dynamic ports, and IANA has now officially designated the range 49152..65535 for dynamic port assignment. However even that is not enough dynamic ports for some busy servers, so the range is usually configurable (by an administrator). On modern Linux and Solaris systems (often used as servers), the default dynamic range now starts at 32768. Mac OS X and Windows Vista default to 49152..65535.

```
[root@desktop12 ~]# cat /proc/sys/net/ipv4/ip_local_port_range
32768   61000

solaris$ /usr/sbin/ndd /dev/tcp tcp_smallest_anon_port tcp_largest_anon_port
32768

65535

macosx$ sysctl net.inet.ip.portrange.first net.inet.ip.portrange.last
net.inet.ip.portrange.first: 49152
net.inet.ip.portrange.last: 65535

vista> netsh int ipv4 show dynamicport tcp
Protocol tcp Dynamic Port Range

Start Port : 49152
Number of Ports : 16384
```

209 Q.What are the well-known ports?

Ans:- Well known ports are from 0 to 1023(total 2^10=1024 ports)

210 Q.What are the Registered Ports, and (Range: 1024 to 49151)

Ans:- Used by specific service upon applications such as Oracle database listener (1521), MySql (3306), Microsoft Terminal server (3389) etc.

211 Q.What are the Dynamic and/or Private Ports. (Range: 49152 to 65535)

Ans:- These ports can’t be registered by IANA. This is used for custom or temporary purposes and for automatic allocation of short-lived (or

ephemeral) ports which is used internally by application/processes. You can see these ports by running ‘netstat’ command under “Local address” column.

212 Q.What do you mean by default port?

Default port is a designated port for particular well-known server.

213 Q.Can we change default port for a service(example Apache, squid)?

Ans: - Yes, we can change

214 Q.What are the protocol numbers for TCP and UDP?

Ans:-
TCP protocol number:6
UDP protocol number:17

215 Q.How to find which ports are open?

Ans:-
You can find the ports in your linux server with the nmap command
#netstat -ntlp
here n -> display the host by numbers
t ->shows tcp protocols
u ->to check udp protocols
l ->listening ports
p -> displays which process controls the port

216 Q.How to find printer open ports ?

Ans: -

```
[root@desktop13 Desktop]# netstat -ntlp |grep cups
tcp        0      0  127.0.0.1:631            0.0.0.0:*              LISTEN      1525/cupsd
tcp        0      0  ::1:631                  :::*                   LISTEN      1525/cupsd
```

Q. How to detect remote server openports?

[root@desktop13 Desktop]# nmap -sT 192.168.0.250
it will show like this
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
25/tcp open smtp
80/tcp open http

217 Q. How to change SSH port number?

Ans: -
[root@desktop13 Desktop]# netstat -ntlp | grep ssh
tcp 0 0 0.0.0.0:22 0.0.0.0:* LISTEN 1646/sshd
tcp 0 0 :::22 :::* LISTEN 1646/sshd
go to the configuration file
[root@desktop13 Desktop]# vim /etc/ssh/sshd_config
port=2200
save and quit

[root@desktop13 Desktop]# service sshd restart

Stopping sshd: [OK]
Starting sshd: [OK]

[root@desktop13 Desktop]# netstat -ntlp | grep ssh
tcp 0 0 0.0.0.0:2200 0.0.0.0:* LISTEN 11904/sshd
tcp 0 0 :::2200 :::* LISTEN 11904/sshd

now port num changed successfully

218 Q. rempte loggin with port number?

ssh -p 22 root@192.168.0.111

219 Q.Some important port numbers:

20-FTP Data (For transferring FTP data)

21-FTP Control (For starting FTP connection)

22-SSH(For secure remote administration which uses SSL to encrypt the transmission)

23-Telnet (For insecure remote administration

25-SMTP(Mail Transfer Agent for e-mail server such as SEND mail)

53- DNS(Special service which uses both TCP and UDP)

68-DHCP

69-TFTP(Trivial file transfer protocol uses udp protocol for connection less transmission of data)

80 -HTTP/WWW(apache)

88-Kerberos

110-POP3(Mail delivery Agent)

123-NTP(Network time protocol used for time syncing uses UDP protocol)

137-NetBIOS(nmbd)

139,138,445-SMB-Samba(smbd)

143-IMAP

161-SNMP(For network monitoring)

389-LDAP(For centralized administration)

443-HTTPS(HTTP+SSL for secure web access)

514-Syslogd(udp port)

995-POP3s

3260-ISCSI

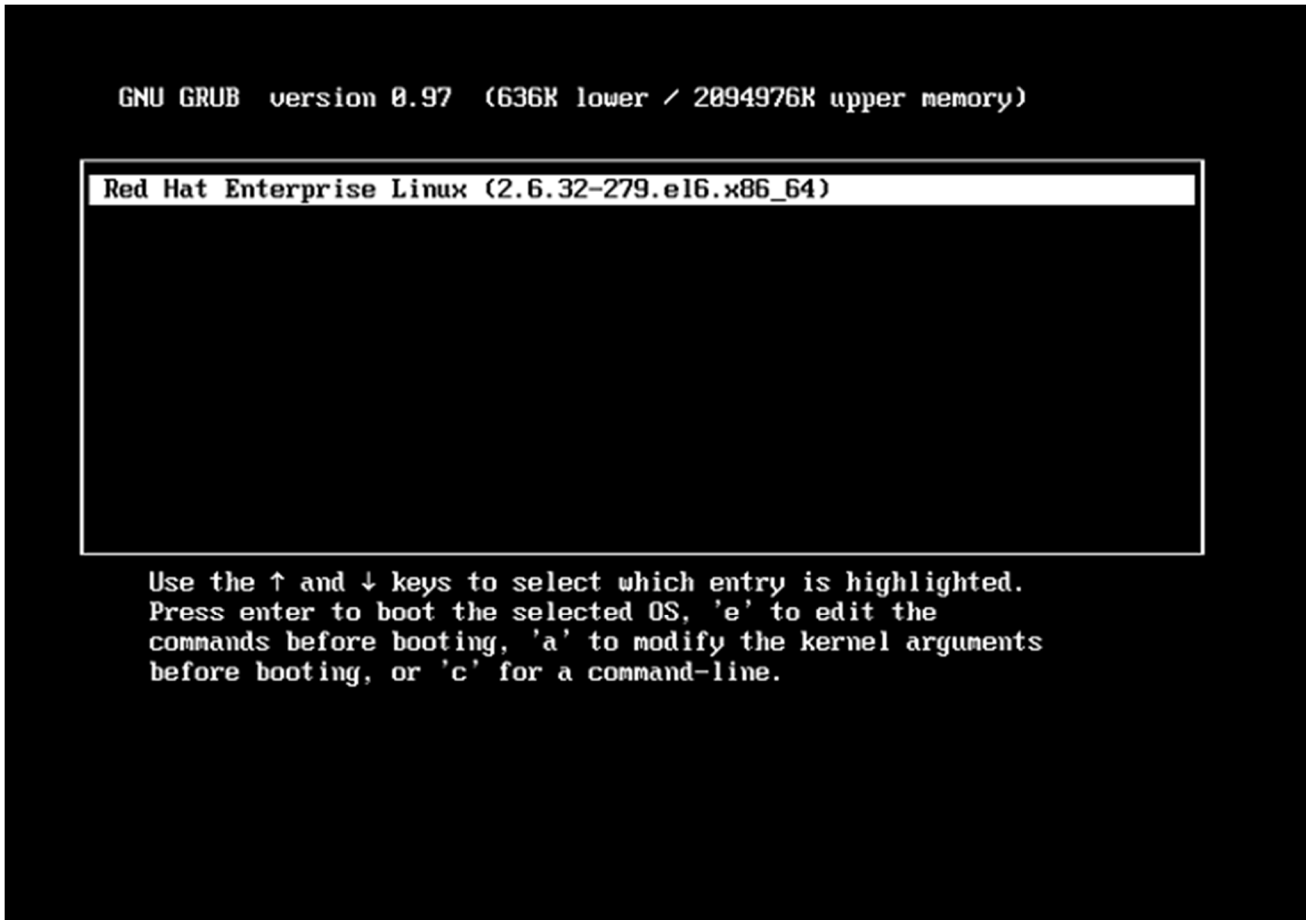
3128-squid proxy

631-Printers(cups)

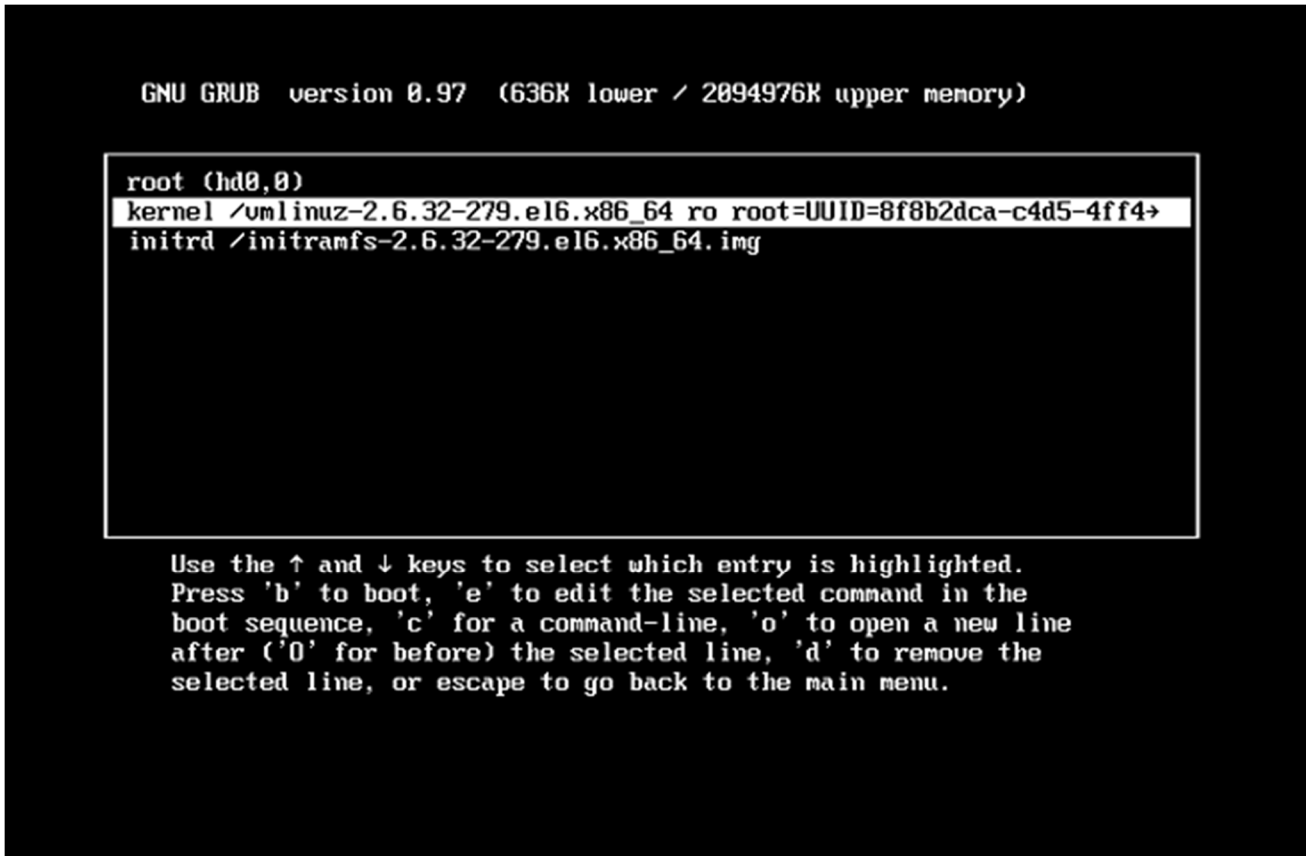
2049-NFS(nfsd, rpc.nfsd, rpc, portmap)

220 Q. How to Break Root Password in RHEL 6

Step 1: Hard reboot your Server and configure in GRUB, Type “e” to edit



Step 2: Choose kernel line and type “e” to edit that line



Step 3: In kernel configuration , we have 2 methods to change to single mode. 1 we can use number “1”, or 2 we can use character “S”. Enter after done.

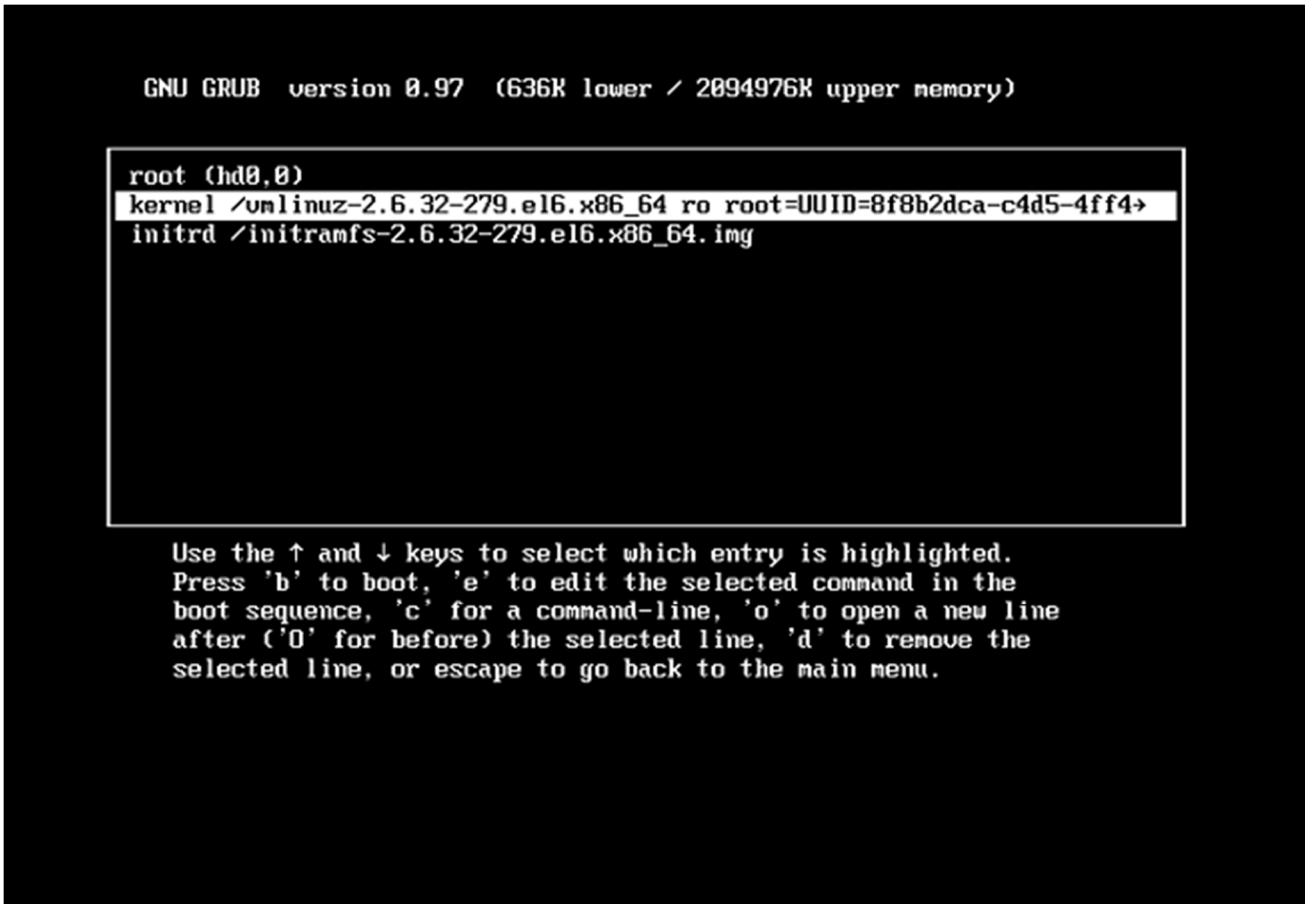
```
 [ Minimal BASH-like line editing is supported.  For the first word, TAB
   lists possible command completions.  Anywhere else TAB lists the possible
   completions of a device/filename.  ESC at any time cancels.  ENTER
   at any time accepts your changes.]

<=pc KEYTABLE=us rd_NO_DM rhgb quiet 1
```

```
[ Minimal BASH-like line editing is supported.  For the first word, TAB
  lists possible command completions.  Anywhere else TAB lists the possible
  completions of a device/filename.  ESC at any time cancels.  ENTER
  at any time accepts your changes.]

<=pc KEYTABLE=us rd_NO_DM rhgb quiet S
```

Step 4: Choose kernel which we just made change and type “b” to boot



Step 5: Now we are in single mode, we can reset password and reboot Server with new root password.

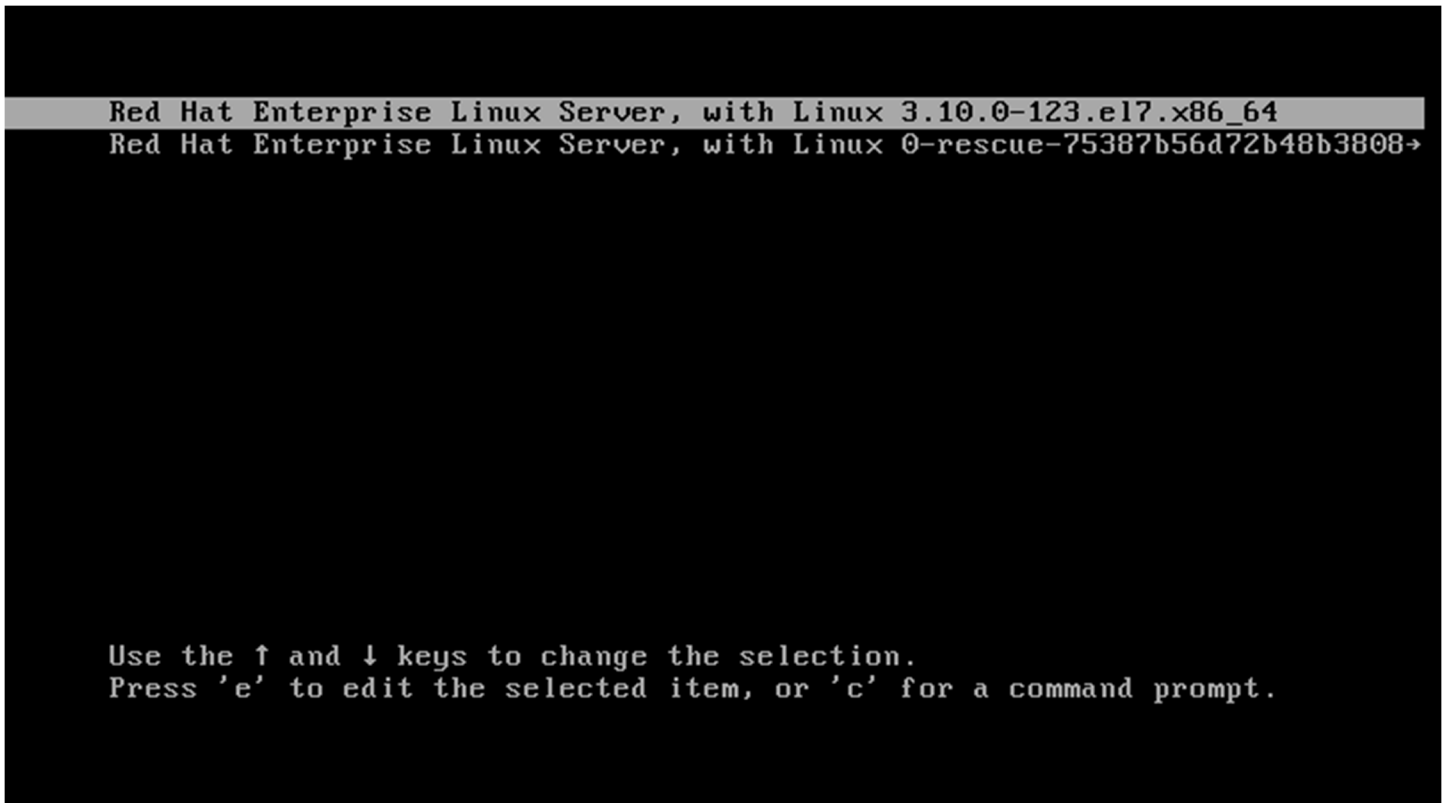
```
                                                          [  OK  ]
Remounting root filesystem in read-write mode:            [  OK  ]
Mounting local filesystems:                               [  OK  ]
Enabling local filesystem quotas:                         [  OK  ]
Enabling /etc/fstab swaps:                                [  OK  ]
                Welcome to Red Hat Enterprise Linux Server
Starting udev:                                            [  OK  ]
Setting hostname FileServer.ithelpblog.com:               [  OK  ]
Setting up Logical Volume Management:   No volume groups found
                                                          [  OK  ]
Checking filesystems
/dev/sda3: clean, 98092/512064 files, 621260/2044672 blocks
/dev/sda1: recovering journal
/dev/sda1: clean, 39/76912 files, 46505/307200 blocks
                                                          [  OK  ]
Remounting root filesystem in read-write mode:            [  OK  ]
Mounting local filesystems:                               [  OK  ]
Enabling local filesystem quotas:                         [  OK  ]
Enabling /etc/fstab swaps:                                [  OK  ]
[root@FileServer /]# passwd
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@FileServer /]# _
```

Step 6: Login Server with new root password

221 Q. How to Break Root Password in RHEL 7

1. Edit GRUB2 boot menu

Start your system and once you see your GRUB2 boot menu use e key to edit your default boot item.
Usually it is the first line:



Once you hit e key you will see a screen similar to the one below:

```
setparams 'Red Hat Enterprise Linux Server, with Linux 3.10.0-123.el7.x86_64'

        load_video
        set gfxpayload=keep
        insmod gzio
        insmod part_msdos
        insmod xfs
        set root='hd0,msdos1'
        if [ x$feature_platform_search_hint = xy ]; then
          search --no-floppy --fs-uuid --set=root --hint-bios=hd0,msdos1 --hin\
t-efi=hd0,msdos1 --hint-baremetal=ahci0,msdos1 --hint='hd0,msdos1'  ae1320ff-2\
96b-4747-bc28-1b1af6161b1e
        else
          search --no-floppy --fs-uuid --set=root ae1320ff-296b-4747-bc28-1b1a\
f6161b1e                                                                      ↓

      Press Ctrl-x to start, Ctrl-c for a command prompt or Escape to
      discard edits and return to the menu. Pressing Tab lists
      possible completions.
```

Depending on you terminal screen size you may see more or less information. In case you have a small terminal screen size note the little down pointing arrow on the right edge of your screen. The arrow means that more text is available when scrolling down. Scroll down and locate a line with `rhgb quiet` keywords:

```
        insmod xfs                                                          ↑
        set root='hd0,msdos1'
        if [ x$feature_platform_search_hint = xy ]; then
          search --no-floppy --fs-uuid --set=root --hint-bios=hd0,msdos1 --hin↘
t-efi=hd0,msdos1 --hint-baremetal=ahci0,msdos1 --hint='hd0,msdos1'  ae1320ff-2↘
96b-4747-bc28-1b1af6161b1e
        else
          search --no-floppy --fs-uuid --set=root ae1320ff-296b-4747-bc28-1b1a↘
f6161b1e
        fi
        linux16 /vmlinuz-3.10.0-123.el7.x86_64 root=UUID=d28446ab-eb57-493f-b7↘
0e-0ef0efe513f9 ro rd.lvm.lv=rhel/root crashkernel=auto  rd.lvm.lv=rhel/swap v↘
console.font=latarcyrheb-sun16 vconsole.keymap=us rhgb quiet LANG=en_AU.UTF-8
        initrd16 /initramfs-3.10.0-123.el7.x86_64.img


      Press Ctrl-x to start, Ctrl-c for a command prompt or Escape to
      discard edits and return to the menu. Pressing Tab lists
      possible completions.
```

Move your cursor (HINT: move to end of the line with CTRL+E) on `rhgb quiet` keywords and replace them with `init=/bin/bash` as show below:

```
        insmod xfs                                                          ↑
        set root='hd0,msdos1'
        if [ x$feature_platform_search_hint = xy ]; then
          search --no-floppy --fs-uuid --set=root --hint-bios=hd0,msdos1 --hin↘
t-efi=hd0,msdos1 --hint-baremetal=ahci0,msdos1 --hint='hd0,msdos1'  ae1320ff-2↘
96b-4747-bc28-1b1af6161b1e
        else
          search --no-floppy --fs-uuid --set=root ae1320ff-296b-4747-bc28-1b1a↘
f6161b1e
        fi
        linux16 /vmlinuz-3.10.0-123.el7.x86_64 root=UUID=d28446ab-eb57-493f-b7↘
0e-0ef0efe513f9 ro rd.lvm.lv=rhel/root crashkernel=auto  rd.lvm.lv=rhel/swap v↘
console.font=latarcyrheb-sun16 vconsole.keymap=us init=/bin/bash_LANG=en_AU.UT↘
F-8
        initrd16 /initramfs-3.10.0-123.el7.x86_64.img                       ↓

      Press Ctrl-x to start, Ctrl-c for a command prompt or Escape to
      discard edits and return to the menu. Pressing Tab lists
      possible completions.
```

Once you edit the boot line as show above press `CTRL + x` to start booting your RHEL 7 system into a single mode. At the end of the system boot you will enter a single mode:

```
GiB)
[    1.142272] sd 2:0:0:0: [sda] Write Protect is off
[    1.143832] sd 2:0:0:0: [sda] Write cache: enabled, read cache: enabled, does
n't support DPO or FUA
[    1.147157]  sda: sda1 sda2
[    1.149176] sd 2:0:0:0: [sda] Attached SCSI disk
[    1.288058] tsc: Refined TSC clocksource calibration: 2762.841 MHz
[    1.288567] Switching to clocksource tsc
[    1.337150] bio: create slab <bio-1> at 1
[  OK  ] Started dracut initqueue hook.
         Mounting /sysroot...
[    1.494617] SGI XFS with ACLs, security attributes, large block/inode numbers
, no debug enabled
[    1.498590] XFS (dm-0): Mounting Filesystem
[    1.524223] XFS (dm-0): Ending clean mount
[  OK  ] Mounted /sysroot.
[  OK  ] Reached target Initrd Root File System.
         Starting Reload Configuration from the Real Root...
[  OK  ] Started Reload Configuration from the Real Root.
[  OK  ] Reached target Initrd File Systems.
[  OK  ] Reached target Initrd Default Target.
[    1.665365] systemd-journald[81]: Received SIGTERM
[    1.670298] systemd-cgroups-agent[410]: Failed to get D-Bus connection: Faile
d to connect to socket /run/systemd/private: No such file or directory
bash-4.2# _
```

2. Read&Write root partition remount

Once you enter a single your root partition is mounted as Read Only `ro`. You ca confirm it with the following command:

```
# mount | grep root
```

In order to mount our partition with Read/Write flag we use `mount` with a `remount` option as follows:

```
# mount -o remount,rw /
```

Next, confirm that the root file system is mounted Read/Write `rw`:

```
# mount | grep root
```

All the above steps are show below:

```
[    1.149176] sd 2:0:0:0: [sda] Attached SCSI disk
[    1.288058] tsc: Refined TSC clocksource calibration: 2762.841 MHz
[    1.288567] Switching to clocksource tsc
[    1.337150] bio: create slab <bio-1> at 1
[  OK  ] Started dracut initqueue hook.
         Mounting /sysroot...
[    1.494617] SGI XFS with ACLs, security attributes, large block/inode numbers
, no debug enabled
[    1.498590] XFS (dm-0): Mounting Filesystem
[    1.524223] XFS (dm-0): Ending clean mount
[  OK  ] Mounted /sysroot.
[  OK  ] Reached target Initrd Root File System.
         Starting Reload Configuration from the Real Root...
[  OK  ] Started Reload Configuration from the Real Root.
[  OK  ] Reached target Initrd File Systems.
[  OK  ] Reached target Initrd Default Target.
[    1.665365] systemd-journald[81]: Received SIGTERM
[    1.670298] systemd-cgroups-agent[410]: Failed to get D-Bus connection: Faile
d to connect to socket /run/systemd/private: No such file or directory
bash-4.2# mount | grep root
/dev/mapper/rhel-root on / type xfs (ro,relatime,attr2,inode64,noquota)
bash-4.2# mount -o remount,rw /
bash-4.2# mount | grep root
/dev/mapper/rhel-root on / type xfs (rw,relatime,attr2,inode64,noquota)
bash-4.2# _
```

3. Change root's password

Still in the single mode we can proceed with the actual root password recovery. To do this we use `passwd` command:

```
# passwd
```

You will need to enter your password twice as shown below:

```
         Mounting /sysroot...
[    1.494617] SGI XFS with ACLs, security attributes, large block/inode numbers
, no debug enabled
[    1.498590] XFS (dm-0): Mounting Filesystem
[    1.524223] XFS (dm-0): Ending clean mount
[  OK  ] Mounted /sysroot.
[  OK  ] Reached target Initrd Root File System.
         Starting Reload Configuration from the Real Root...
[  OK  ] Started Reload Configuration from the Real Root.
[  OK  ] Reached target Initrd File Systems.
[  OK  ] Reached target Initrd Default Target.
[    1.665365] systemd-journald[81]: Received SIGTERM
[    1.670298] systemd-cgroups-agent[410]: Failed to get D-Bus connection: Faile
d to connect to socket /run/systemd/private: No such file or directory
bash-4.2# mount | grep root
/dev/mapper/rhel-root on / type xfs (ro,relatime,attr2,inode64,noquota)
bash-4.2# mount -o remount,rw /
bash-4.2# mount | grep root
/dev/mapper/rhel-root on / type xfs (rw,relatime,attr2,inode64,noquota)
bash-4.2# passwd
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
bash-4.2# _
```

4. SELinux relabeling

The additional step which needs to be taken on SELinux enables Linux system is to relabel SELinux context. If this step is ommited you will not be able to login with your new root password. The following command will ensure that the SELinux context for entire system is relabeled after reboot:

```
# touch /.autorelabel
```

```
[    1.494617] SGI XFS with ACLs, security attributes, large block/inode numbers
, no debug enabled
[    1.498590] XFS (dm-0): Mounting Filesystem
[    1.524223] XFS (dm-0): Ending clean mount
[  OK  ] Mounted /sysroot.
[  OK  ] Reached target Initrd Root File System.
         Starting Reload Configuration from the Real Root...
[  OK  ] Started Reload Configuration from the Real Root.
[  OK  ] Reached target Initrd File Systems.
[  OK  ] Reached target Initrd Default Target.
[    1.665365] systemd-journald[81]: Received SIGTERM
[    1.670298] systemd-cgroups-agent[410]: Failed to get D-Bus connection: Faile
d to connect to socket /run/systemd/private: No such file or directory
bash-4.2# mount | grep root
/dev/mapper/rhel-root on / type xfs (ro,relatime,attr2,inode64,noquota)
bash-4.2# mount -o remount,rw /
bash-4.2# mount | grep root
/dev/mapper/rhel-root on / type xfs (rw,relatime,attr2,inode64,noquota)
bash-4.2# passwd
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
bash-4.2# touch /.autorelabel
bash-4.2# _
```

5. Reboot System

The final step when resetting your lost root password on RHEL 7 linux system is to reboot. This can be done with a following command:

```
# exec /sbin/init
```

After reboot you will be able to use your new root password.

223 : Why to use NFS ?

Ans: A Network File System (NFS) allows remote machine to mount file systems over a network and interact with those file systems as though they are mounted locally. This enables system administrators to consolidate resources onto centralized servers over the network.

224 Q: What is the default port of NFS server ?

Ans: By default NFS uses 2049 TCP port.

225 Q: What are different versions of NFS Server ?

Ans: Currently, there are three versions of NFS. NFS version 2 (**NFSv2**) is older and widely supported. NFS version 3 (**NFSv3**) supports safe asynchronous writes and is more robust at error handling than NFSv2; it also supports **64-bit file sizes** and

offsets, allowing clients to access more than **2Gb** of file data.

NFS version 4 (**NFSv4**) works through firewalls and on the Internet, no longer requires an **rpcbind** service, supports ACLs, and utilizes stateful operations. Red Hat Enterprise Linux 6.X & Centos 6.X supports NFSv2,NFSv3, and NFSv4 clients. When mounting a file system via NFS, Red Hat Enterprise Linux uses NFSv4 by default, if the server supports it.

226 Q: What are configuration files of NFS server ?

Ans: **'/etc/exports'** is the main configuration file that controls which file systems are exported to remote hosts and specifies options. **'/etc/sysconfig/nfs**' is the file through which we can fix ports for **RQUOTAD_PORT, MOUNTD_PORT, LOCKD_TCPPORT, LOCKD_UDPPORT and STATD_PORT**

227 Q: What are different options used in /etc/exports file ?

Ans: Below are list of options used in /etc/exports file :

- **ro:** The directory is shared read only; the client machine will not be able to write to it. This is the default.
- **rw:** The client machine will have read and write access to the directory.
- **root_squash**: By default, any file request made by user root on the client machine is treated as if it is made by user nobody on the server. (Exactly which UID the request is mapped to depends on the UID of user "nobody" on the server, not the client.)
- **no_root_squash** : if this option is used , then root on the client machine will have the same level of access to the files on the system as root on the server. This can have serious security implications, although it may be necessary if you want to perform any administrative work on the client machine that involves the exported directories. You should not specify this option without a good reason.
- **no_subtree_check** : If only part of a volume is exported, a routine called subtree checking verifies that a file that is requested from the client is in the appropriate part of the volume. If the entire volume is exported, disabling this check will speed up transfers.
- **sync** : Replies to the NFS request only after all data has been written to disk. This is much safer than async, and is the default in all nfs-utils versions after 1.0.0.
- **async :** Replies to requests before the data is written to disk. This improves performance, but results in lost data if the server goes down.
- **no_wdelay :** NFS has an optimization algorithm that delays disk writes if NFS deduces a likelihood of a related write request soon arriving. This saves disk writes and can speed performance
- **wdelay :** Negation of no_wdelay , this is default
- **nohide :** Normally, if a server exports two filesystems one of which is mounted on the other, then the client will have to mount both filesystems explicitly to get

access to them. If it just mounts the parent, it will see an empty directory at the place where the other filesystem is mounted. That filesystem is "hidden". Setting the nohide option on a filesystem causes it not to be hidden, and an appropriately authorised client will be able to move from the parent to that filesystem without noticing the change.

- **hide :** Negation of nohide This is the default

228 Q: How to list available nfs share on local machine & remote machine ?

Ans: **'showmount -e localhost'** : Shows the available shares on your local machine **'showmount -e <Remote-server-ip or hostname>**': Lists the available shares at the remote server.

229 Q: What is pNFS ?

Ans: Parallel NFS (pNFS) as part of the NFS v4.1 standard is available as of Red Hat Enterprise Linux 6.4. The **pNFS** architecture improves the scalability of NFS, with possible improvements to performance. That is, when a server implements pNFS as well, a client is able to access data through multiple servers concurrently. It supports three storage protocols or layouts: **files, objects, and blocks.**

230 Q: What is the difference between Hard mount & Soft mount in nfs ?

Ans: Difference between soft mount and hard mount is listed below :

- **Soft Mount :** Consider we have mounted a NFS share using 'soft mount' . When a program or application requests a file from the NFS filesystem, NFS client daemons will try to retrieve the data from the NFS server. But, if it doesn't get any response from the NFS server (due to any crash or failure of NFS server), the NFS client will report an error to the process on the client machine requesting the file access. The advantage of this mechanism is "fast responsiveness" as it doesn't wait for the NFS server to respond. But, the main disadvantage of this method is data corruption or loss of data. So, this is not a recommended option to use.
- **Hard Mount :** Suppose we have mounted the NFS share using hard mount, it will repeatedly retry to contact the server. Once the server is back online the program will continue to execute undisturbed from the state where it was during server crash. We can use the mount option "intr" which allows NFS requests to be interrupted if the server goes down or cannot be reached. Hence the recommended settings are hard and intr options.

231 **Q: How to check iostat of nfs mount points ?**

Ans: Using command '**nfsiostat**' we can list iostat of nfs mount points. Use the below command :
nfsiostat <interval> <count> <mount_point>
<interval> : specifies the amount of time in seconds between each report. The first report contains statistics for the time since each file system was mounted. Each subsequent report contains statistics collected during the interval since the previ-ous report.
<count> : If the <count> parameter is specified, the value of <count> determines the number of reports generated at seconds apart. if the interval parameter is specified without the <count> parameter, the command generates reports continuously.
<mount_point> : If one or more <mount point> names are specified, statistics for only these mount points will be displayed. Otherwise, all NFS mount points on the client are listed.

232 Q: How to check nfs server version ?

Ans: **'nfsstat -o all'** command shows all information about active versions of NFS.

233 Q: What is portmap?

Ans: The **portmapper** keeps a list of what services are running on what ports. This list is used by a connecting machine to see what ports it wants to talk to access certain services.

234 Q: How to reexport all the directories of '/etc/exports' file ?

Ans: Using the command ' **exportfs -r** ' , we can reexport or refresh entries of '/etc/exports' file without restarting nfs service.

235 Q: **How to retrieve a list of clients connected to the NFS server ?**

To retrieve a list of clients connected to the NFS server, use the showmount command from a shell prompt. To also show the directories the clients are connected to, use the showmount -a command.

236 Q: **What is meaning of "no_root_squash" option ?**

Treat remote root user as local root. Do not map requests from root to the anony-mous user and group ID.

237 Q: **Can we grant access by Username and password for nfs share?**

No, access is granted only for IP address.

238 Q: **What is the role of "all_squash" option?**

Treat all client users as anonymous users. Map all user and group IDs to the anonymous user and

group ID.

239 Q: - What is the role of "root_squash" option?

All requests from the user root are translated or mapped as if they came from the user anonymous (default).

240 Q: - Explain option "all_squash"?

The UID and GID of exported files are mapped to the user anonymous. It is good for public directories.

241 Q: - Explain "exportfs" command?

The exportfs command is used to maintain the current table of exported file systems for NFS.

242 Q: - Explain command "/usr/sbin/exportfs -f"?

It will flush everything out of the kernels export table. Any clients that are active will get new entries added by mountd when they make their next request.

243 Q: What is the role of "all_squash" option?

ANS:Treat all client users as anonymous users. Map all user and group IDs to the anonymous user
and group ID.

244 Q: - What is the role of "root_squash" option?

ANS:All requests from the user root are translated or mapped as if they came from the user anonymous (default).

245 Q: - Explain option "all_squash"?

ANS:The UID and GID of exported files are mapped to the user anonymous. It is good for public directories.

246 Q: - Explain "exportfs" command?

ANS:The exportfs command is used to maintain the current table of exported file systems for NFS.

247 Q: - Explain command "/usr/sbin/exportfs -f"?

ANS:It will flush everything out of the kernels export table. Any clients that are active will get

new entries added by mountd when they make their next request.

248 Q: - Which option is used with exportfs command to display the current export list, also displays the list of export options?

ANS:exportfs -v

249 Q: - Which option is used with exportfs command to re-export all directories?

ANS:exportfs -r

250 Q: - How you will export directory (/data) to host 192.168.1.51, allowing asynchronous writes without adding the entry in /etc/exports file?

ANS:# exportfs -o async 192.168.1.51:/data

251 Q: - Is rpc.mountd daemon supports TCP_WRAPPERS?

ANS:Yes, The rpc.mountd daemon is protected by the tcp_wrappers. You have to give the clients access to rpc.mountd if they should be allowed to use NFS Server.

252 Q: - Explain "nfsstat" command?

ANS:The nfsstat command displays the statistics about NFS client and NFS server activity.

253 Q: - What do you understand by "nfsstat -o all -234" command?

ANS:It will Show all information about all versions of NFS.

254 Q: - What do you understand by "nfsstat --nfs --server -3" command?

ANS:It will show statistics for NFS version 3 server.

255 Q: - Can NFS share mounted on Window XP and Justify your answer?

ANS:No, Window XP operating system doesn't support nfs protocol.

256 Q: - 192.168.1.51:/data is exported by NFS Server and i want to add this NFS share to client /etc/fstab file. How you will add this entry in /etc/fstab file?

ANS:# device mount-point fs-type options dump fsckorder
192.168.1.51:/data /mnt nfs defaults 0 0

257 Q: - Explain "Soft Mounting" option at NFS Client?

ANS:if a file request fails, the NFS client will report an error to the process on the client machine requesting the file access. if it cannot be satisfied (for example, the server is down), then it quits. This is called soft mounting.

258 Q: - Explain "Hard Mounting" option at NFS Client?

ANS:If a file request fails, the NFS client will report an error to the process on the client machine requesting the file access. if it cannot be satisfied, then it will not quit until the request is satisfied. This is called Hard mounting.

259 Q: -How you will check "portmap" service is running or not?

ANS:rpcinfo -p

260 Q. what is samba?
Ans:-
"Samba is an Open Source/Free Software suite that provides seamless file and print services to SMB/CIFS clients." Samba is freely available, unlike other SMB/CIFS implementations, and allows for interoperability between Linux/Unix servers and Windows-based clients.

261 Q. What is CIFS?
The Common Internet File System (CIFS), also known as Server Message Block (SMB), is a network protocol whose most common use is sharing files on a Local Area Network (LAN). The protocol allows a client to manipulate files just as if they were on the local computer. Operations such as read, write, create, delete, and rename are all supported – the only difference being that the files are not on the local computer and are actually on a remote server.

262 Q. configuration file for samba?

Ans:-

/etc/samba/smb.conf

263 Q: - Which SELinux security context used for SAMBA ?

Ans:-
samba_share_t

264 Q: - On which ports SAMBA server works ?

Ans:-

- UDP port 137 for netbios-ns, the NETBIOS Name Service
- UDP port 138 for netbios-dgm, the NETBIOS Datagram Service
- TCP port 139 for netbios-ssn, the NETBIOS session service
- TCP port 445 for microsoft-ds, the Microsoft Domain Service

265 Q: - What are the Secrity or Authentication Mode for SAMBA server?
Ans:-
ADS
DOMAIN
SERVER
USER
SHARE

266 Q: - How to Manually Create Machine Trust Accounts ?

Ans:-
/usr/sbin/useradd -g machines -d /var/lib/nobody -c "machine nickname" -s /bin/false machine_name$
passwd -l machine_name$

267 Q: - What are the SAMBA server Types ?

Ans:-

- Primary Domain Controller (PDC)
- Backup Domain Controller (BDC)
- ADS Domain Controller

268 Q: - Which protocol SAMBA server uses ?
Ans:-

SMB, which stands for Server Message Block, is a protocol for sharing files, printers, serial ports, and communications abstractions such as named pipes and mail slots between computers.

269 Q: - How Does a Workstation find its Domain Controller?

Ans:-

There are two different mechanisms to locate a domain controller: one method is used when NetBIOS over TCP/IP is enabled and the other when it has been disabled in the TCP/IP network configuration. Where NetBIOS over TCP/IP is disabled, all name resolution involves the use of DNS, broadcast messaging over UDP, as well as Active Directory communication technologies.

270 Q: - Can Samba Be a Backup Domain Controller to an NT4 PDC?

Ans:-

No. The native NT4 SAM replication protocols have not yet been fully implemented.

271 Q: - How Do I Replicate the smbpasswd File?

Ans:-

Replication of the smbpasswd file is sensitive. It has to be done whenever changes to the SAM are made. Every user's password change is done in the smbpasswd file and has to be replicated to the BDC. So replicating the smbpasswd file very often is necessary.As the smbpasswd file contains plaintext password equivalents, it must not be sent unencrypted over the wire. The best way to set up smbpasswd replication from the PDC to the BDC is to use the utility rsync. rsync can use ssh as a transport. ssh itself can be set up to accept only rsync transfer without requiring the user to type a password.As said a few times before, use of this method is broken and awed. Machine trust accounts will go out of sync, resulting in a broken domain. This method is not recommended. Try using LDAP instead.

272 Q: - Can Samba fully replace my Windows NT server that is not a Primary Domain Controller (PDC)?

Ans:-
Samba can completely serve files and printers to Windows, just as a Windows NT server would.

273 Q Can Samba replaces my Windows NT PDC?

Ans:-
Not completely. Samba domain control capabilities for a Windows 9x client are
solid and complete, and so these clients would probably never know the difference. The domain control support for Windows NT/2000 clients is still being developed. Currently, enough has been implemented to allow a Windows NT client to join a Samba-controlled domain, but there is more to

domain control than that. The most conspicuous absence is the lack of support for Windows NT trust relationships and the SAM replication protocol used between NT PDCs and Backup Domain Controllers (BDCs).

274 Q What TCP and UDP ports required for NetBIOS over TCP/IP use?
Ans:-
The NBT name service uses port 137/udp, the NBT session service uses port 139/tcp, and the NBT datagram service uses port 138/udp.

275 Q: - How SMB protocol works?
Ans:-
There will be three stages in creating an SMB connection between a client and a specific share on a server.
The first stage in connecting to an SMB share is to negotiate the SMB protocol dialect to use. In the request packet, the client sends a text listing of all the SMB dialects that it understands. The server selects the most advanced protocol that it knows and responds to the client, specifying the protocol number from the list. At this point, the client and server have agreed that SMB commands can be used for the remainder of the conversation.
The second stage is to create a session connection between the client and server. To do this, the client issues a session setup request, which includes a sername and some proof of validity, such as a password. The server attempts to validate requesting user. If successful, the server then returns a session UID to client. This UID is unique for each session and has no relation to the server internal representation of users.
The third stage before access to files on a remote share is allowed is for the client to make a successful tree connection to the shared resource. The client sends to the server a tree connect request, which includes the UID previously issued by the server. At this stage the server verifies that the authenticated user is authorized to access the requested resource. If the user has sufficient privileges to access the share, the client is issued a tree connection ID (TID). The TID is used in all requests to access files

contained in the resource to which the TID refers. In this way SMB protocol works.

276 Q: - How man sections samba configuration file (smb.conf) contains?
Ans:-
smb.conf file contains three sections.
1. [global] Contains settings that determine Samba overall behavior.
2. [homes] A default share for providing a home directory for all users.
3. [printers] A default share for exporting all printers on the host via CIFS.

277 Q: - If a netbios name is not defined in smb.conf, than what will be netbios name?
Ans:-
If a netbios name is not defined, Samba will use the IP hostname of the server by default.

278 Q: - I want to use User level security for my samba server than what i have to add in smb.conf file?
Ans:-
security = user

279 Q: - How you will verify that your smb.conf file doesn't have any mistakes and misspellings?
Ans:-
"testparm " tool that verifies the syntax of a configuration file(smb.conf).
testparm -s smb.conf

280 Q: - What is the use of "smbclient" command?
Ans:-
"smbclient" is used to display the list of shares on your server. This verifies that smbd is running and functioning correctly. The -L option instructs smbclient to enumerate the shares on the server rather than actually connecting to one. The -N switch instructs smbclient to use an anonymous

login rather than the login name of the current user.

smbclient -L localhost -N
Antother use of "smbclient" command to connect the samba share.
smbclient //<server>/<share> -U <username>

281 Q: - Explain "smbstatus" command?
Ans:-
The smbstatus utility displays information about connected users and currently locked files.

282 Q: - Is it possible for Samba to share file systems that have been mounted using NFS?
Ans:-
Yes. However, this can be problematic if the NFS server that provides the file system fails, causing the Samba server to hang. It is always safer to use Samba to share a local file system.

283 Q: - How many simultaneous connections can a Samba server support?
Ans:-
In theory, there is no limit. In practice, the limit is determined by the server's hardware, specifically the total amount of available RAM and the CPU power. It might also depend on the amount of activity from the smbd processes.

284 Q: - Can Samba be a member of more than one workgroup at the same time?
Ans:-
No, Samba can be a member of only one workgroup.

285 Q: - What is SWAT?
Ans:-
SWAT is GUI Based administration tool for samba server.

286 Q: - I am trying to use SWAT, but I keep getting the message There was no response. The server could be down or not responding. What is the problem?
Ans:-
The most likely cause is that SWAT is not listening to connections, or you have used the wrong URL in trying to connect to SWAT. SWAT usually lives behind port 901, so the URL you should use is
http://ID_ADDRESS_OF_SERVER:901/
: - Can i set empty password for samba user?
yes, If you want to set the value to an empty password, you must change
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX:XXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXX
to
NOPASSWORDXXXXXXXXXXXXXXXXXXXXXXX:XXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXX

in your smbpasswd file.
Note: - if you have edited the smbpasswd file by hand, make sure that the LAN Manager and NT password fields contain exactly 32 characters, no more and no fewer. If these fields do not have exactly 32 characters, Samba will not be able to correctly read the entry.
or You can modify by "smbpasswd" command.
smbpasswd -n USER_NAME
Also you have to set the null passwords parameter to yes in the [global] section of smb.conf:
null passwords = yes

287 Q: - Does Samba support PAM?

Ans:-
Yes

288 Q: - What is role of "NTLM"?
Ans:-

The challenge/response authentication protocol available to Windows clients and servers for validating connection requests.

289 Q: - Explain "force group" parameter used in smb.conf?
Ans:-
It will define the group id to be used for all file access in the place of the user's primary group.

290 Q: - Explain "force user" parameter used in smb.conf?
Ans:-
It will define the user id to be used for all file access.

291 Q: - Explain "write list" parameter used in smb.conf?
Ans:-
A list of users and/or groups that should be given write access even if the read only parameter has been enabled.

292 Q: - My clients are getting the error message that the Disk is Full when trying to print to my Samba server, but there is plenty of space. What is the problem?
Ans:-
If smbd is unable to write the spooled file to the directory defined by the path parameter for a printer if the write permission were denied, for example it would respond to the client with the message, Disk is Full. Samba will also return this error message if the amount of free disk space in the spool directory has fallen below the value specified by the min print space parameter.

293 Q: - When I click on my Samba server in the network neighborhood, I am continually prompted for a password to the IPC$ share no matter what I enter.
Ans:
The Windows client is attempting to use encrypted passwords. However, the Samba server is configured to support only clear-text passwords. You should either enable encrypted passwords on the server or enable clear-text passwords on the Windows client.

294 Q: - Why is security = domain better than security = server?
Ans:-
There are three reasons why security = domain is better. The first is because this method enables the Samba server to participate in domain trust relationships. This is impossible with server-level security. The second reason is that, under server-level security, each smbd process must keep an open connection with the authentication server. This can drain a Windows NT PDC quickly. Under domain-level security, this connection is maintained only long enough to perform the validation, thus conserving valuable resources. The final reason is that, as a domain member, the Samba server has access to much more information about user accounts, which can be used to automate the creation and deletion of user accounts upon demand.

295 Q: - what is nmbd daemon?
Ans:-
This daemon handles all name registration and resolution requests. It is the primary vehicle involved in network browsing. It handles all UDP-based protocols. The nmbd daemon should be the first command started as part of the Samba startup process.

296 Q: - What is smdb daemon?
Ans:-
This daemon handles all TCP/IP-based connection services for file- and print-based operations. It also manages local authentication. It should be started immediately following the startup of nmbd.

297 Q: - What is winbindd daemon?
Ans:-
This daemon should be started when Samba is a member of a Windows NT4 or ADS domain. It is also needed when Samba has trust relationships with another domain. The winbindd daemon will check the smb.conf file for the presence of the idmap uid and idmap gid parameters. If they are found, winbindd will use the values specified for UID and GID allocation. If these parameters are not specified, winbindd will start but it will not be able to allocate UIDs or GIDs.

298 Q: - Explain the parameter "wins support = Yes" used in smb.conf?
Ans:-

If the Samba server was configured to provide WINS support ("wins support = Yes"), then the WINS server is able to provide name resolution for all of the hosts that are not listed in the /etc/hosts file or within the DNS. Making this adjustment in the Name Service Switch configuration file (/etc/nsswitch.conf) allows the Linux system to query the WINS server for local name resolution. This saves manual adjustments to host files.

299 Q: - How to automate SMB share mounting during system startup?

Ans:-

Add smb share entry in /etc/fstab file.

```
//IP_ADDRESS_OF_SERVER/Shared   /shared   smbfs   noauto,defaults
 0  0
```

300 Q: - how to start and stop samba server?

Ans:-

```
/etc/init.d/smb  restart
          Or
Service smb restart
```

301 Q: - How to start sendmail server ?

Ans:-

service sendmail restart

302 Q: - On which ports sendmail and senmail with SSL works ?

Ans:-

By default, Sendmail uses TCP and UDP port 25 for non-encrypted transfers. If the
Sendmail server is configured to use SSL for encrypting email sent and received, it uses
port 465.

303 Q: - Explain use of "trusted-users" file ?

Ans:-

List of users that can send email as other users without a warning including system users such as apache for the Apache HTTP Server.

304 Q: - Explain the use of "local-host-names" file ?

Ans:-

If the email server should be known by different hostnames, list the host-names in this file, one line per hostname. Any email sent to addresses at these
hostnames is treated as local mail. The FEATURE option must
be enabled in the sendmail.mc file for this file to be referenced.

305 Q: - explain the use of /etc/aliases file ?

Ans:-

/etc/aliases, can be used to redirect email from one user to another. By default, it includes redirects for system accounts to the root user. It can then be used to redirect all email for the root user to the user account for the system administrator.

306 Q: - Can we use SSL Encryption with Sendmail ?
Ans:-

Yes, Sendmail can be configured to encrypt email sent and received using SSL (secure sockets layer).

307 Q: - What is Sendmail ?
Ans:-

Sendmail is an MTA, meaning it accepts email messages sent to it using the SMTP proto-col and transports them to another MTA email server until the messages reach their destinations. It also accepts email for the local network and delivers them to local mail spools, one for each user.

308 Q: - What is the role of MUA ?

Ans:-

An MUA (Mail User Agent) with access to the mailbox file, directly or through a network file system, can read messages from the disk and display them for the user. This is generally a console or webmail application running on the server.

309 Q: - Which are the important configuration files for Sendmail server ?

Ans:-

The /etc/mail/ directory contains all the Sendmail configuration files, with **sendmail.cf** and**submit.cf** being the main configuration files. The sendmail.cf file includes options for the mail transmission agent and accepts SMTP connections for sending email. The submit.cf file configures the mail submission program.

310 Q: - How to configure sendmail to accept mail for local delivery that is addressed to other hosts?

Ans:-

Create a /etc/mail/local-host-names file. Put into that file the hostnames and domain names for which sendmail should accept mail for local delivery. Enter the names with one hostname or domain name per line. And also make sure that Sendmail configuration file should contain "use_cw_file" option.

dnl Load class $=w with other names for the local host

FEATURE(`use_cw_file')

311 Q: - When an organization stores aliases on an LDAP server, how you will configure sendmail to read aliases from the LDAP server?

Ans:-

Use "sendmail -bt -d0" command to check the sendmail compiler options. If sendmail was not compiled with LDAP support, recompile and reinstall sendmail.
Add an ALIAS_FILE define, containing the string ldap to the sendmail configuration.

```
# Set the LDAP cluster value
define(`confLDAP_CLUSTER', `wrotethebook.com')
# Tell sendmail that aliases are available via LDAP
define(`ALIAS_FILE', `ldap:')
```

312 Q: - How to forward emails of a local user to external address?
Ans:-
Add an alias to the aliases file for each user whose mail must be forwarded to another system. The recipient field of the alias entry must be a full email address that includes the host part. After adding the desired aliases, rebuild the aliases database file with the newaliases command.

313 Q: - You have been asked to create a sendmail configuration that sends all local mail to a mail hub, while directly delivering mail addressed to external systems.
Ans:-
Create a sendmail configuration containing the MAIL_HUB define to identify the mail relay host for local mail. Use the LOCAL_USER command to exempt the root user's mail from relaying.

```
dnl Define a relay server for local mail
define(`MAIL_HUB', `smtp.test.com')
dnl Users whose mail is not passed to the mail hub
LOCAL_USER(root)
```

Rebuild and reinstall sendmail.cf, and then restart sendmail.

314 Q: - How to configure multiple mail queues?
Ans:-

```
mkdir /var/spool/mqueue/queue.1
mkdir /var/spool/mqueue/queue.2
mkdir /var/spool/mqueue/queue.3
chmod 700 /var/spool/mqueue/queue.1
chmod 700 /var/spool/mqueue/queue.2
chmod 700 /var/spool/mqueue/queue.3
```

Add the QUEUE_DIR define to the sendmail configuration to use the new queue directories. dnl Declare the queue directory path
define(`QUEUE_DIR', `/var/spool/mqueue/queue.*')

315 Q: - How to disable certain SMTP commands?
Ans:-
Add the confPRIVACY_FLAGS define to the sendmail configuration to set Privacy Options that disable unwanted, optional SMTP commands. Here we will disables the EXPN, VRFY, VERB, and ETRN commands.
dnl Disable EXPN, VRFY, VERB and ETRN
define(`confPRIVACY_FLAGS', `noexpn,novrfy,noverb,noetrn')
Rebuild and reinstall sendmail.cf, and then restart sendmail.
316 Q: - In which Sendmail configuration file we have to make changes?
Ans:-
we will make the changes only in the sendmail.mc file, and the changes will be moved into the sendmail.cf file for us.

317 Q: - What does /etc/mail/access file contains?
Ans:-
The access database ("/etc/mail/access") is a list of IP addresses and domainnames of allowable connections.
FEATURE(`access_db',`hash -T<TMPF> -o /etc/mail/access.db')dnl
and cat /etc/mail/access
localhost.localdomain RELAY
localhost RELAY
127.0.0.1 RELAY
192.168.0 RELAY
test.com RELAY

318 Q: - How to restrict sendmail to sending a big file?
Ans:-
define(`confMAX_MESSAGE_SIZE',`52428800')dnl
or If you are using a PHP based webmail application like SquirrelMail, you can adjust the max file size in php.ini file.
vi php.ini
post_max_size = 50M
upload_max_filesize = 50M
memory_limit = 64M

319 Q: - How to set 25 recipients for each email?
Ans:-
define(`confMAX_RCPTS_PER_MESSAGE',`50')dnl

320 Q: - Which antivirus you have integrated with sendmail ?
Ans:-
ClaimAV

321 Q: - What is Clamav-Milter?
Ans:-
Clamav-Milter is a tool to integrate sendmail and clamAV antivirus.

322 Q: - Which configuration files are required to integrate sendmail and ClaimAV antivirus?
Ans:-
milter.conf and clamav-milter

323 Q: - How to test sendmail integration with ClaimAV?
Ans:-
grep Milter /var/log/maillog
You have to get following type of messages.
sendmail: Milter add: header: X-Virus-Scanned: ClamAV version 0.88.2, clamav-milter version 0.88.2 on mail.test.com
sendmail: Milter add: header: X-Virus-Status: Clean

324 Q: - Which tool you have used to block spamming?
Ans:-
SpamAssassin

325 Q: - What does "/etc/mail/" directory contains?
Ans:-
The /etc/mail/ directory contain all the Sendmail configuration files, with sendmail.cf and submit.cf being the main configuration files.

326 Q: - Explain the use of /etc/mail/relay-domains file?
Ans:-

The /etc/mail/relay-domains file is used to determine domains from which it will relay mail. The contents of the relay-domains file should be limited to those domains that can be trusted not to originate spam.

327 Q: - What is the name of spamassassin configuration file?
Ans:-
/etc/mail/spamassassin/local.cf

328 Q: - How to check mail Queue of sendmail?
/usr/lib/sendmail -bp
or
mailq

329 Q: - How to use m4 macro processor to generate a new sendmail.cf?
Ans:-
m4 /etc/mail/sendmail.mc > /etc/mail/sendmail.cf

330 : - What is location of log files for Apache server ?
Answer: /var/log/httpd

331 Q:-What is virtual hosting?
Answer: Virtual hosting is a method for hosting multiple domain names on a server using a single IP address. Virtual hosting allows one server to share its resources, such as memory and processor cycles, in order to use its resources more efficiently.

332 Q: - What are the types of virtual hosts ?
Answer:
name-based and IP-based.
Name-based virtual host means that multiple names are running on each IP address.
IP-based virtual host means that a different IP address exists for each website served. Most configurations are named-based because it only requires one IP address.

333 Q: - How to restart Apache web server ?
Answer: service httpd restart
/etc/init.d/httpd restart

334 Q: - How to check the version of Apache server ?
Answer: rpm -qa |grep httpd

335 Q: - What is meaning of "Listen" in httpd.conf file ?
Answer: Port number on which to listen for nonsecure (http) transfers. Default port number of httpd is 80

336 Q: - What is DocumentRoot ?
Answer:it is a location of files which are accessible by clients. By default, the Apache HTTP server in RedHat Enterprise Linux is configured to serve files from the /var/www/html/ directory.

337 Q: - On which port Apache server works ?
Answer: http - port 80
https - port 443

338 Q: - Tell me name of main configuration file of Apache server ?
Answer:/etc/httpd/conf/httpd.conf

339 Q: - What do you mean by a valid ServerName directive?
Answer: The DNS system is used to associate IP addresses with domain names. The value of ServerName is returned when the server generates a URL. If you are using a certain domain name, you must make sure that it is included in your DNS system and will be available to clients visiting your site.

340 Q: - What is the main difference between <Location> and <Directory> sections?
Answer: Directory sections refer to file system objects; Location sections refer to elements in the address bar of the Web page

241 Q:-What is the difference between a restart and a graceful restart of a web server?
Answer: During a normal restart, the server is stopped and then started, causing some requests to be lost. A graceful restart allows Apache children to continue to serve their current requests until they can be replaced with children running the new configuration.

342 Q: - If you have added “loglevel Debug” in httpd.conf file, than what will happen?
Answer: It will give you more information in the error log in order to debug a problem.

343 Q: - Can you record the MAC (hardware) address of clients that access your server?
Answer: No

: - Can you record all the cookies sent to your server by clients in Web Server logs?
Answer: Yes, add following lines in httpd.conf file.
CustomLog logs/cookies_in.log "%{UNIQUE_ID}e %{Cookie}i"
CustomLog logs/cookies2_in.log "%{UNIQUE_ID}e %{Cookie2}i"

344 Q: - Can we do automatically roll over the Apache logs at specific times without having to shut down and restart the server?
Answer: Yes
Use CustomLog and the rotatelogs programs
Add following line in httpd.conf file. CustomLog "| /path/to/rotatelogs/path/to/logs/access_log.%Y-%m-%d 86400" combined

345 Q: - What we can do to find out how people are reaching your site?
Answer: Add the following effector to your activity log format. %{Referer}

346 Q: - If you have only one IP address, but you want to host two web sites on your server. What will you do?
Answer: In this case I will use Name Based Virtual hosting.

```
ServerName 10.111.203.25
NameVirtualHost *:80
<VirtualHost *:80>
ServerName web1.test.com
DocumentRoot /var/www/html/web1
</VirtualHost>

<VirtualHost *:80>
ServerName web2.test2.com
DocumentRoot /var/www/html/web2
</VirtualHost>
```

347 Q: - Can I serve content out of a directory other than the DocumentRootdirectory?
Answer: Yes, by using “Alias” we can do this.

348 Q: - If you have to more than one URL map to the same directory but you don't have multiple Alias directives. What you will do?
Answer: In this case I will use “AliasMatch” directives.
The AliasMatch directive allows you to use regular expressions to match arbitrary patterns in URLs and map anything matching the pattern to the desired URL.

349 Q: - How you will put a limit on uploads on your web server?
Answer: This can be achieved by LimitRequestBody directive.

```
<Directory "/var/www/html/data_uploads">
LimitRequestBody 100000
</Directory>
```

Here I have put limit of 100000 Bytes

350 Q: - I want to stop people using my site by Proxy server. Is it possible?
Answer: <Directory proxy:http://www.test.com/myfiles>
Order Allow,Deny
Deny from all
Satisfy All
</Directory>

351 Q: - What is mod_evasive module?
Answer: mod_evasive is a third-party module that performs one simple task, and performs it very well. It detects when your site is receiving a Denial of Service (DoS) attack, and it prevents that attack from doing as much damage. mod_evasive detects when a single client is making multiple requests in a short period of time, and denies further requests from that client. The period for which the ban is in place can be very short, because it just gets renewed the next time a request is detected from that same host.

352 Q: - How t to enable PHP scripts on your server?
Answer: If you have mod_php installed, use AddHandler to map .php and .phtml files to the PHP handler. AddHandler application/x-httpd-php .phtml .php

353 Q: - Which tool you have used for Apache benchmarking?
Answer: ab (Apache bench)
ab -n 1000 -c 10 http://www.test.com/test.html

354 Q: - Can we cache files which are viewed frequently?
Answer: Yes we can do it by using mod_file_cache module.
CacheFile /www/htdocs/index.html

355 Q:- **How do you check for the httpd.conf consistency and any errors in it?**
Answer: httpd –t

351 Q:- **Apache runs as which user? and location of main config file?.**
Answer: Apache runs with the user “nobody” and httpd daemon. Apache main configuration file: /etc/httpd/conf/httpd.conf (CentOS/RHEL/Fedora) and /etc/apache2.conf (Ubuntu/Debian).

352 Q:- **On which port Apache listens http and https both?**
Answer : By default Apache runs on http port 80 and https port 443 (for SSL certificate). You can also use netstat command to check ports.

353 Q: - which are the important configuration files for DNS server ?

Ans:-

BIND uses /etc/named.conf as its main configuration file, the /etc/rndc.conf file as the configuration file for name server control utility rndc, and the /var/named/ directory for zone files and the like.

354 Q: - What is BIND ?

Ans:-
BIND stands for Berkeley Internet Name Domain which is the most commonly used Domain Name System (DNS) server on the Internet.

355 Q: - On which version of bind u have worked ?

Ans:-
BIND 9

356 Q: - What is the role of DNS ?

Ans:-
A DNS server, or name server, is used to resolve an IP address to a hostname or vice versa.

357 Q: - What is Name Server?

Ans:-

A name server keeps information for the translation of domain names to IP addresses and IP addresses to domain names. The name server is a program that performs the translation at the request of a resolver or another name server.

358 Q: - What is Primary name server or primary master server?

Ans:-

Primary name server/primary master is the main data source for the zone. It is the authoritative server for the zone. This server acquires data about its zone from databases saved on a local disk. The primary server must be published as an authoritative name server for the domain in the SOA resource record, while the primary master server does not need to be published.

359 Q: - What is Secondary name server/slave name server?

Ans:-

Secondary name server/slave name server acquires data about the zone by copying the data from the primary name server (respectively from the master server) at regular time intervals. It makes no sense to edit these databases on the secondary name servers, although they are saved on the local server disk because they will be rewritten during further copying.

360 Q: - what is Root name server?

Ans:-

Root name server is an authoritative name server for the root domain (for the dot). Each root name server is a primary server, which differentiates it from other name servers.

361 Q: - what is Stealth name server?

Ans:-

Stealth name server is a secret server. This type of name server is not published anywhere. It is only known to the servers that have its IP address statically listed in their configuration. It is an authoritative server. It acquires the data for the zone with the help of a zone transfer. It can be the main server for the zone. Stealth servers can be used as a local backup if the local servers are unavailable.

362 Q: - What do you mean by "Resource Records"?

Ans:-

Information on domain names and their IP addresses, as well as all the other information distributed via DNS is stored in the memory of name servers as Resource Records (RR).

363 Q: - Explain "TTL"?

Ans:-

Time to live. A 32-bit number indicating the time the particular RR can be kept valid in a server cache. When this time expires, the record has to be considered invalid. The value 0 keeps nonauthoritative servers from saving the RR to their cache memory.

364 Q: - Tell me 5 Types of DNS records?

Ans:-

A, NS, CNAME, SOA, PTR, MX.

365 Q:- explain "SOA Record"?

Ans:-

The Start of Authority (SOA) record determines the name server that is an authoritative source of information for the particular domain. There is always only one SOA record in the file, and it is placed at the beginning of the file of authoritative resource records.

366 Q: - what is "A Record"

Ans:-

A (Address) records assign IP addresses to domain names of computers. The IP address cannot have a dot at the end.

367 Q: - Explain "CNAME Record"?

Ans:-

Synonyms to domain names can be created using CNAME records. This is often referred to as 'creating aliases for computer names'.

368 Q: - What are "HINFO and TXT Records"?

Ans:-

HINFO and TXT records are for information only. An HINFO record has two items in its data part. The first item is information about hardware, and the second one is information about software. A TXT record contains a general data string in its data part.Example :test.com IN SOA .....mail IN A 192.1.1.2IN HINFO My_Server UNIXIN TXT my server

369 Q: - what are "MX Records"?

Ans:-

MX records specify the mailing server of the domain. An MX record shows to which computer a mail of a particular domain should be sent. The MX record also includes a priority number, which can be used to determine several computers where the mail for the domain can be sent. The first attempt is to deliver the mail to the computer with the highest priority (lowest value). If this attempt fails, the mail goes to the next computer (with a higher priority value), and so on.test.com IN SOA .....mail

```
test.com IN SOA .....mail
IN      A       192.1.1.2
IN      HINFO   AlphaServer UNIX
IN      TXT     my  server
IN      MX  30   mail2.nextstep4it.com
IN      MX  20   mail3.nextstep4it.com
IN      MX  10   mail2.nextstep4it.com
```

370 Q: - Explain "PTR Records"?

Ans:-

A Pointer Record (PTR) is used to translate an IP address into a domain name.

371 Q: - What is Dynamic DNS?

Ans:-

Dynamic DNS a method of keeping a domain name linked to a changing IP address as not all computers use static IP addresses. Typically, when a user connects to the Internet, the user's ISP assigns an unused IP address from a pool of IP addresses, and this address is used only for the duration of that specific connection. This method of dynamically assigning addresses extends the usable pool of available IP addresses. A dynamic DNS service provider uses a special program that runs on the user's computer, contacting the DNS service each time the IP address provided by the ISP changes and subsequently updating the DNS database to reflect the change in IP address.

372 Q: - What is the role of "named-checkconf Utility"?

Ans:-

The named-checkconf utility checks the syntax of the named.conf configuration file.
Syntax: named-checkconf [-t directory] [filename]

373 Q: - what is the role of "named-checkzone Utility"?

Ans:-

The named-checkzone utility checks the syntax and consistency of the zone file.Syntax: named-checkzone [-dgv] [-c class] zone [filename]

374 . What is MySQL?

MySQL is an open source DBMS which is built, supported and distributed by MySQL AB (now acquired by Oracle)

375 . Why MySQL is used?

MySQL database server is reliable, fast and very easy to use. This software can be downloaded as freeware and can be downloaded from the internet.

376 . What are Heap tables?

HEAP tables are present in memory and they are used for high speed storage on temporary

basis.

• BLOB or TEXT fields are not allowed

• Only comparison operators can be used =, <,>, = >,=<

• AUTO_INCREMENT is not supported by HEAP tables

• Indexes should be NOT NULL

377 . What is the default port for MySQL Server?

The default port for MySQL server is 3306.

378 . What are the advantages of MySQL when compared with Oracle?

- MySQL is open source software which is available at any time and has no cost involved.
- MySQL is portable
- GUI with command prompt.
- Administration is supported using MySQL Query Browser

379 . Differentiate between FLOAT and DOUBLE?

Following are differences for FLOAT and DOUBLE:

• Floating point numbers are stored in FLOAT with eight place accuracy and it has four bytes.

• Floating point numbers are stored in DOUBLE with accuracy of 18 places and it has eight bytes.

380 . Differentiate CHAR_LENGTH and LENGTH?

CHAR_LENGTH is character count whereas the LENGTH is byte count. The numbers are same for Latin characters but they are different for Unicode and other encodings.

381 . How to represent ENUMs and SETs internally?

ENUMs and SETs are used to represent powers of two because of storage optimizations.

382 . What is the usage of ENUMs in MySQL?

ENUM is a string object used to specify set of predefined values and that can be used during table creation.

MySQL

```
Create table size(name ENUM('S
```

1 Create table size(name ENUM('Small', 'Medium','Large');

383 . Define REGEXP?

REGEXP is a pattern match in which matches pattern anywhere in the search value.

384 . Difference between CHAR and VARCHAR?

Following are the differences between CHAR and VARCHAR:

- CHAR and VARCHAR types differ in storage and retrieval
- CHAR column length is fixed to the length that is declared while creating table. The length value ranges from 1 and 255
- When CHAR values are stored then they are right padded using spaces to specific length. Trailing spaces are removed when CHAR values are retrieved.

385 . Give string types available for column?

The string types are:

- SET
- BLOB
- ENUM
- CHAR
- TEXT
- VARCHAR

386 . How to get current MySQL version?

```
SELECT VERSION ();
```

1 SELECT VERSION ();

is used to get the current version of MySQL.

387 . What storage engines are used in MySQL?

Storage engines are called table types and data is stored in files using various techniques.

Technique involves:

- Storage mechanism
- Locking levels
- Indexing
- Capabilities and functions.

388 . What are the drivers in MySQL?

Following are the drivers available in MySQL:

- PHP Driver
- JDBC Driver
- ODBC Driver
- C WRAPPER
- PYTHON Driver
- PERL Driver
- RUBY Driver
- CAP11PHP Driver
- Ado.net5.mxj

389 . What does a TIMESTAMP do on UPDATE CURRENT_TIMESTAMP data type?

TIMESTAMP column is updated with Zero when the table is created. UPDATE CURRENT_TIMESTAMP modifier updates the timestamp field to current time whenever there is a change in other fields of the table.

390 . What is the difference between primary key and candidate key?

Every row of a table is identified uniquely by primary key. There is only one primary key for a table.

Primary Key is also a candidate key. By common convention, candidate key can be designated as primary and which can be used for any foreign key references.

391 . How do you login to MySql using Unix shell?

We can login through this command:

[mysql dir]/bin/mysql -h hostname -u <UserName> -p <password>

392 . What does myisamchk do?

It compress the MyISAM tables, which reduces their disk or memory usage.

393 . How do you control the max size of a HEAP table?

Maximum size of Heal table can be controlled by MySQL config variable called max_heap_table_size.

394 . What is the difference between MyISAM Static and MyISAM Dynamic?

In MyISAM static all the fields will have fixed width. The Dynamic MyISAM table will have fields like TEXT, BLOB, etc. to accommodate the data types with various lengths.

MyISAM Static would be easier to restore in case of corruption.

395 . What are federated tables?

Federated tables which allow access to the tables located on other databases on other servers.

396 . What, if a table has one column defined as TIMESTAMP?

Timestamp field gets the current timestamp whenever the row gets altered.

397 . What happens when the column is set to AUTO INCREMENT and if you reach maximum value in the table?

It stops incrementing. Any further inserts are going to produce an error, since the key has been used already.

398 . How can we find out which auto increment was assigned on Last insert?

LAST_INSERT_ID will return the last value assigned by Auto_increment and it is not required to specify the table name.

399 . How can you see all indexes defined for a table?

Indexes are defined for the table by:

```
SHOW INDEX FROM <tablename>;
```

400 . What do you mean by % and _ in the LIKE statement?

% corresponds to 0 or more characters, _ is exactly one character in the LIKE statement.

401 . How can we convert between Unix & MySQL timestamps?

UNIX_TIMESTAMP is the command which converts from MySQL timestamp to Unix timestamp

FROM_UNIXTIME is the command which converts from Unix timestamp to MySQL timestamp.

402 . What are the column comparisons operators?

The = , <>, <=, <, >=, >,<<,>>, <=>, AND, OR, or LIKE operators are used in column comparisons in SELECT statements.

403 . How can we get the number of rows affected by query?

Number of rows can be obtained by

```
SELECT COUNT (user_id) FROM
```

1 SELECT COUNT (user_id) FROM users;

404 . Is Mysql query is case sensitive?

No.

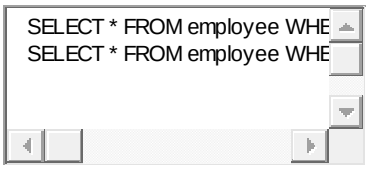
```
SELECT VERSION(), CURRENT_
SeLect version(), current_date;
seleCt vErSiOn(), current_DATE
```

1 SELECT VERSION(), CURRENT_DATE;
2 SeLect version(), current_date;
3 seleCt vErSiOn(), current_DATE;

All these examples are same. It is not case sensitive.

405 . What is the difference between the LIKE and REGEXP operators?

LIKE and REGEXP operators are used to express with ^ and %.



1 SELECT * FROM employee WHERE emp_name REGEXP "^b";
2 SELECT * FROM employee WHERE emp_name LIKE "%b";

406 . What is the difference between BLOB AND TEXT?

A BLOB is a binary large object that can hold a variable amount of data. There are four types of BLOB –

- TINYBLOB
- BLOB
- MEDIUMBLOB and
- LONGBLOB

They all differ only in the maximum length of the values they can hold.

A TEXT is a case-insensitive BLOB. The four TEXT types

- TINYTEXT
- TEXT
- MEDIUMTEXT and
- LONGTEXT

They all correspond to the four BLOB types and have the same maximum lengths and storage requirements.

The only difference between BLOB and TEXT types is that sorting and comparison is performed in case-**sensitive** for BLOB values and case-**insensitive** for TEXT values.

407 . What is the difference between mysql_fetch_array and mysql_fetch_object?

Following are the differences between mysql_fetch_array and mysql_fetch_object:

mysql_fetch_array() -Returns a result row as an associated array or a regular array from database.

mysql_fetch_object – Returns a result row as object from database.

408 . How can we run batch mode in mysql?

Following commands are used to run in batch mode:

```
mysql ;
mysql mysql.out
```

1 mysql ;
2 mysql mysql.out

409 . Where MyISAM table will be stored and also give their formats of storage?

Each MyISAM table is stored on disk in three formats:

- The '.frm' file stores the table definition
- The data file has a '.MYD' (MYData) extension
- The index file has a '.MYI' (MYIndex) extension

410 . What are the different tables present in MySQL?

Total 5 types of tables are present:

- MyISAM
- Heap
- Merge
- INNO DB
- ISAM

MyISAM is the default storage engine as of MySQL .

411 . What is ISAM?

ISAM is abbreviated as Indexed Sequential Access Method.It was developed by IBM to store and retrieve data on secondary storage systems like tapes.

412 . What is InnoDB?

lnnoDB is a transaction safe storage engine developed by Innobase Oy which is a Oracle Corporation now.

413 . How MySQL Optimizes DISTINCT?

DISTINCT is converted to a GROUP BY on all columns and it will be combined with ORDER BY clause.

```
SELECT DISTINCT t1.a FROM t1
```

1 SELECT DISTINCT t1.a FROM t1,t2 where t1.a=t2.a;

414 . How to enter Characters as HEX Numbers?

If you want to enter characters as HEX numbers, you can enter HEX numbers with single quotes and a prefix of (X), or just prefix HEX numbers with (Ox).

A HEX number string will be automatically converted into a character string, if the expression context is a string.

415 . How to display top 50 rows?

In MySql, top 50 rows are displayed by using this following query:

```
SELECT * FROM
LIMIT 0,50;
```

1 SELECT * FROM
2 LIMIT 0,50;

416 . How many columns can be used for creating Index?

Maximum of 16 indexed columns can be created for any standard table.

417 . What is the different between NOW() and CURRENT_DATE()?

NOW () command is used to show current year,month,date with hours,minutes and seconds.

CURRENT_DATE() shows current year,month and date only.

418 . What are the objects can be created using CREATE statement?

Following objects are created using CREATE statement:

- DATABASE
- EVENT
- FUNCTION

- INDEX
- PROCEDURE
- TABLE
- TRIGGER
- USER
- VIEW

419 . How many TRIGGERS are allowed in MySql table?

SIX triggers are allowed in MySql table. They are as follows:

- BEFORE INSERT
- AFTER INSERT
- BEFORE UPDATE
- AFTER UPDATE
- BEFORE DELETE and
- AFTER DELETE

420 . What are the nonstandard string types?

Following are Non-Standard string types:

- TINYTEXT
- TEXT
- MEDIUMTEXT
- LONGTEXT

421 . What are all the Common SQL Function?

CONCAT(A, B) – Concatenates two string values to create a single string output. Often used to combine two or more fields into one single field.

FORMAT(X, D) – Formats the number X to D significant digits.

CURRDATE(), CURRTIME() – Returns the current date or time.

NOW() – Returns the current date and time as one value.

MONTH(), DAY(), YEAR(), WEEK(), WEEKDAY() – Extracts the given data from a date value.

HOUR(), MINUTE(), SECOND() – Extracts the given data from a time value.

DATEDIFF(A, B) – Determines the difference between two dates and it is commonly used to calculate age

SUBTIMES(A, B) – Determines the difference between two times.

FROMDAYS(INT) – Converts an integer number of days into a date value.

422 . Explain Access Control Lists.

An ACL (Access Control List) is a list of permissions that is associated with an object. This list is the basis for MySQL server's security model and it helps in troubleshooting problems like users not being able to connect.

MySQL keeps the ACLs (also called grant tables) cached in memory. When a user tries to authenticate or run a command, MySQL checks the authentication information and permissions against the ACLs, in a predetermined order.

423. What is NFS?

Ans: NFS stands for Network File System. NFS was originally developed by Sun Microsystems in the 1980's. NFS allows remote hosts to mount file systems over a network and interact with those file systems as though they are mounted locally. This enables system administrators to consolidate resources onto centralized servers on the network.

424 Q. Which NFS versions are available?

NFS Version 2
NFS Version 3
NFS Version 4

425 Q. What is different between NFS Version 2 & 3 ?

Ans: NFS 2 default 8kb transfer rate, it did not check the authentication at the time connection. Client wants to access unauthorized file it shows error messages like "write error","read error" nfs 3 32kb transfer rate. It check at the time connection- ACL Support

426 Q. What is the role of "all_squash" option?

Ans: Treat all client users as anonymous users. Map all user and group IDs to the anonymous user and group ID.

427 Q. What is the role of "root_squash" option?

Ans: All requests from the user root are translated or mapped as if they came from the user anonymous (default).

428 Q. Explain "exportfs" command?

Ans: The exportfs command is used to maintain the current table of exported file systems for

429 Q. Difference between nfs soft and hard mouniting points?

Ans: Hard mount option:- If the client fails to access the server, then connection hang it, after the system up then it access the server. Soft:- If the client failed to connect the server it Immediately gives the error report, and close the connection.

To retrieve a list of clients connected to the NFS server use show mount command from Shell prompt.

Showmount -a

430 Q. Name of the configuration file for NFS server?

Ans: /etc/exports/

431 Q. Can we grant access based on user name and password?

Ans: No, access is granted only for IP address

432 Q. Explain command "/usr/sbin/exportfs -f" ?

Ans: It will flush everything out of the kernel export tab

433 Q. Explain Hard and Soft Mounting in NFS?

Ans : If a file request fail, the NFS client will report an error to the process on the client machine requesting the file access. If it cannot be satisfied the it quite. This is called Soft mounting.

If a file request fails, NFS client will report an error to the process on the client machine requesting the file access. If cannot be satisfied, then it will not quit until the request if satisfied. This is called hard mounting.

434 Q. What is portmap?

Ans: The portmapper keeps a list of what services are running on what [ports .The list is used by a connecting machine to see what ports it wants to talk to access certain services.

435 Q. Which SELinux ports Samaba server works?

Ans: Samba_share_t

436 Q. On which Ports SAMBA server works?

Ans:

UDP port 137 for NetBIOS-ns, NETBIOS name Service
UDP port 138 for netbios-dgm, NetBIOS Datagram service
TCP port 139 for netbios-ssn , NETBIOS session service
TCP port 445 for microsoft-d, Microsoft domain service

437 Q.What are the Security or Authentication mode for SAMBA server?

Ans:
ADS
DOMAIN
SERVER
USER
SHARE

438 Q. What are the SAMBA server types ?

Ans: Primary Doman controller
Backup Domain controller
ADS domain controller

439 Q.Which protocol SAMBA Server uses?

Ans: SMB, which stands for server Message block, is a protocol for sharing printers, serial ports and communication abstraction such as named pipes and mail slots between computers.

440 Q. How SMP protocol works?

Ans: There will be three stages in creating SM connection between a client and specific shares on the server

The first stage in connecting to an SMB share is to negotiate the SMB protocol dialect to use. IN the request packet the client send a text listing all the SMB dialects that it understands. The server selects the most advanced protocol that it knows and responds to the client specifying the protocol number from the list .At this point the client and server have agreed SMB commands can be used for the remainder of the conversation.

The second stage is to create a session connection between the client and server. To do this the client issues a session setup request .which included a surname and some proof of validity such as password .The server attempts to validate requesting user. IF successful the server then returns a session UID to client .This UID is unique for each session and has no relation to the server internal representation of user.
Third stage before access to file on a remote share is allowed si for the client to make successful tree connection to the shared resource .The client sends to the server a tre connect request, which includes the UID previously issued by the server .At this stage the server verifies that the authenticated user is authorized to access the requested resource. If the user has sufficient privileges to access the share, the client is issued a tree connection ID. The TID is used in all request files contained in the resource to which the TID referees.

441 Q. How many Section Samba Configuration File (Smb.conf) contains?

1. [Global] - Contains Settings that determines samba overall behavior
2. [Homes}- A default share for providing a home directory for all users
3. [Printers] - A default share for Exporting All printer on the host via CIFS.

442 Q.What is SWAT?

Ans: Swat is GUI based administration toll for Samba

443 Q.What is the roles for NTLM?

The chanllange/repsonse authentication protocol available to windows clients and server for validating connection request.

444 Q. What is nbd daemon?

445 Ans: This daemon handles all name registration and resolution request .It is the primary vehicle involved in network browsing .It handles all UDP based protocol. The nmbd daemon should be first command to start as a part of the samba start up process.

446 Q. What is smdb daemon?

Ans: This daemon handles all TCP/IP based connection services for file and print based operation .It also manages local authentication. It should be started immediately following the start up of nmbd.

447 Q. what are the daemons running In NIS? how to do u create NIS users and update Maps?

Ans: Below daemons will run the NIS

1) ypservd
2) ypxfrd
3) yppasswd
4) ypbind

448 Q. What is the difference between TFTP and FTP servers?

Ans: Both are file transfer servers but slight difference is TFTP server uses UDP protocol and FTP uses TCP protocol. TFTP is good for slow connection paths.

449 Q. What is the port no for FTP?

Ans: 20 for Data and 21 for Control(normally when an interviewer ask's ftp port number just say 21.

450 Q. What is the port no for TFTP?

Ans: 69.

451 Q. How to restrict users to their home directories?

Ans: By setting up "chroot_local_user=YES".

452 Q. How to restrict total number of users accessing FTP server?

Ans: If Vsftpd is running under xinted service, then you can use xineted to get per-service per IP connection limits or

If you run vsftpd in "standalone" mode with the "setting listen=YES", then

you can investigate the setting (e.g.) : "max_clients=10"

453 Q.I want to copy multiple files with out prompting for any info, how can I do that one?

Ans: Simply do "ftp -i ftpserver" this command will suppress any info displayed on ftp server or you can just type prompt ftp prompt to suppress info messages have a look here.

454 Q. Some times Local users cannot log in. How to resolve this issue?

Ans: Check "local_enable=YES" in your /etc/vsftpd/vsftpd. conf to allow local users to log in.

455 Q.In which port telnet is listening?

Ans: 23

456 Q. How do we integrate with LDAP(Lightweight Directory Access Protocol)users and login?

Ans: Use vsftpd's(Very Secure FTPD) PAM(Pluggable Authentication Modules) integration to do this, and have PAM authenticate against an LDAP repository.

457 Q. What is sand witch configuration in qmail?

Ans: Qmail + Clam + Spamassassin- This is normally called Sandwitch configuration in qmail.

458 Q. Advantages of Qmail?

Ans: More secure, better designed, modular, faster, more reliable, easier to configure, don't have to upgrade it every few months or worry about being vulnerable to something due to some obscure feature being enabled qmail supports host and user masquerading, full host hiding, virtual domains, null clients, list-owner rewriting, relay control, double-bounce recording, arbitrary RFC 822 address lists, cross-host mailing list loop detection, per-

recipient checkpointing, downed host backoffs, independent message retry schedules, etc. qmail also includes a drop-in “sendmail” wrapper so that it will be used transparently by your current UAs.

459 Q.What is the difference between POP3 and IMAP ?

Ans: The Difference

POP3 works by reviewing the inbox on the mail server, and downloading the new messages to your computer. IMAP downloads the headers of the new messages on the server, then retrieves the message you want to read when you click on it.

When using POP3, your mail is stored on your PC. When using IMAP, the mail is stored on the mail server. Unless you copy a message to a “Local Folder” the messages are never copied to your PC.

Scenarios of Use

POP3

· You only check e-mail from one computer.

· You want to remove your e-mail from the mail server.

IMAP

· You check e-mail from multiple locations.

· You use Webmail.

460 Q. How to block and open relay?

Ans: Open relays are e-mail servers that are configured to accept and transfer e-mail on behalf of any user anywhere, including unrelated third parties.

The qmail-smtpd daemon will consult the rcpthosts control file to determine valid destination addresses, and reject anything else.

461 Q. Which Groups and Users are required to start QMAIL Server?

Groups :- nofiles

qmail

Users :- qmaild

alias

qmaill

qmailp

qmailq

qmailr

qmails

462 Q.What is the role of "qmail-send" process ?

Ans: qmail-send - deliver mail messages from the queue

463 Q. How to check, qmail server is up or down?

Ans: We can use qmailctl command to check the status of qmail server.

```
# qmailctl stat

/service/qmail-send: up (pid 30303) 187 seconds
/service/qmail-send/log: up (pid 30304) 187 seconds
/service/qmail-smtpd: up (pid 30305) 187 seconds
/service/qmail-smtpd/log: up (pid 30308) 187 seconds
messages in queue: 0
messages in queue but not yet preprocessed: 0
```

464 Q. On which port Send mail and SSL works?

Ans: By default Sendmail uses TCP and UDP port 25 for non encrypted transfers. If the sendmail server is configured to use SLL for encrypting email Send and received it uses port 465.

465 Q.Explain use of trusted-user file?

Ans: List of users that can send email as other users with out a warning including system users such as apache for the Apache http server

466 Q. Explain the use for “local-host -name" ?

Ans: If the mail server should be known by different hostnames, list the host names in this file, one line per hostname.

467 Q. Explain the use of access file?

Ans: Can be used to redirect email from one user to another. By default it included redirects for system accounts to the root user.

468 Q. What is the role of MUA?

Ans: An MUA (Mail user agent) with access to the mail box file, directly or through a network file system, can read message form the dist and display them for the user.

469 Q. What is the location of postfix mail server queue?

Ans: By Default the post-fix mail ques are /var/spool/postfix directory. Message queue is created as a separate sub directory within this directory .Each message is stored as a separate file in the sub directory, using unique identifier for the file name.

470 Q.What is LMTP?

Ans: The local Mail Transport Protocol is different mail transport protocol described in RFC 2033.LMTP utilizes as a set protocol similar to SMTP for delivering messages to the local host .Postfix can be configured to deliver messages to local users.

471 Q.What is canonical Table?

Ans: The clean up program uses the canonical table to rewrite message address contained in the message header .The mail administrator can use one canonical lookup table for both received messages and sent messages or separate tables for each.

472 Q. What is the difference between Postfix and Sendmail?

Ans: The mail difference between Postfix and Sendmail is Postfix Modularity .Just as the Unix system broke up email functionality between modules, postfix extends that practice to the MTA program. This allows each modular program to be smaller and quicker than on e large.

Postfix is more secure than sendmail.Postfix uses separate userID to be adds to the mail server. Each module runs under this user id.

473 Q: Define Process?

Ans: In computing, a process is an instance of a computer program that is being executed. It contains the program code and its current activity. Depending on the operating system (OS), a process may be made up of multiple threads of execution that execute instructions concurrently

474 Q.Explain THREADS vs Processes ?.

Ans: THREADS, LIKE PROCESSES, ARE A MECHANISM TO ALLOW A PROGRAM to do more than one thing at a time.As with processes, threads appear to run concurrently; the Linux kernel schedules them asynchronously, interrupting each thread from time to time to give others a chance to execute.

Conceptually, a thread exists within a process.Threads are a finer-grained unit of execution than processes.When you invoke a program, Linux creates a new process and in that process creates a single thread, which runs the program sequentially.That thread can create additional threads; all these threads run the same program in the same process, but each thread may be executing a different part of the program at any given time.

Threads

- Will by default share memory
- Will share file descriptors
- Will share filesystem context
- Will share signal handling

Processes

- Will by default not share memory
- Most file descriptors not shared
- Don't share filesystem context
- Don't share signal handling.

475 Q.What is PID?

Ans: Process (process ID, PID): All software runs within an operating system concept known as a process. Each program running on a system is assigned its own process ID (PID). Users can easily obtain a process list (using Task Manager on Windows or ps on UNIX) in order to see what is running. Key point: Trojans, root kits, and other evil software will attempt to hide themselves from the process list, either by providing replacements to the programs that list processes (like ps), or by hooking the system calls that enumerate processes.

476 Q. What are the different states of process?

Ans: In a multitasking computer system, processes may occupy a variety of states. These distinct states may not actually be recognized as such by the operating system kernel, however they are a useful abstraction for the understanding of processes

Process states in Linux:

Running: Process is either running or ready to run

Incorruptible: a Blocked state of a process and waiting for an event or signal from another process

Uninterruptible: a blocked state. Process waits for a hardware condition and cannot handle any signal

Stopped: Process is stopped or halted and can be restarted by some other process

Zombie: process terminated, but information is still there in the process table.

477 Q. What is a zombie?

Ans: Zombie is a process state when the child dies before the parent process. In this case the structural information of the process is still in the process table. Since this process is not alive, it cannot react to signals. Zombie state can finish when the parent dies. All resources of the zombie state process are cleared by the kernel

478 Q, Explain Daemons?

Ans: Daemons is a computer program that runs as a background process, rather than being under the direct control of an interactive user. Typically daemon names end with the letter d: for example, syslogd is the daemon that implements the system logging facility, or sshd, which services incoming SSH connections.

479 Q. **Explain each system calls used for process management in Linux.**

Ans: System calls used for Process management:

Fork () -Used to create a new process

Exec() -Execute a new program
Wait() **-** wait until the process finishes execution
Exit() - Exit from the process
Getpid() - get the unique process id of the process
Getppid() **-** get the parent process unique id
Nice() -to bias the existing property of process

480 Q. Some Tools to see process and memory status?

Ans: top – Process Activity Command

The top program provides a dynamic real-time view of a running system i.e. actual process activity. By default, it displays the most CPU-intensive tasks running on the server and updates the list every five seconds.

vmstat – System Activity, Hardware and System Information
The command vmstat reports information about processes, memory, paging, block IO, traps, and cpu activity.

481 Q. How to put a job in background & bring it to foreground?

Ans: we can put a job in background by putin & symbol at the end of the line on the command prompt.
Example: #cp -r /var/* /tmp &

482 Q.Explain Memory management and related commands?

Ans: Linux comes with different set of commands to check memory usage. The free command displays the total amount of free and used physical and swap memory in the system, as well as the buffers used by the kernel. The vmstat command reports information about processes, memory, paging, block IO, traps, and cpu activity. Finally, you can use the top command which provides a dynamic real-time view of a running system. It can display system summary information as well as a list of tasks currently being managed by the Linux kernel.

483 Q. What is the major advantage of a hash table?

Ans: The major advantage of a hash table is its speed. Because the hash function is to take a range of key values and transform them into index values in such a way that the key values are distributed randomly across all the indices of a hash table.

484 Q. What are the techniques that you use to handle the collisions in hash tables?

Ans: We can use two major techniques to handle the collisions. They are open addressing and separate chaining. In open addressing, data items that hash to a full array cell are placed in another cell in the array. In separate chaining, each array element consist of a linked list. All data items hashing to a given array index are inserted in that list.

485 Q. Explain file system of linux. The root "/" filesystem, /usr filesystem, /var filesystem, /home filesystem, /proc filesystem.

Ans: Root "/" file system: The kernel needs a root file system to mount at start up. The root file system is generally small and should not be changed often as it may interrupt in booting. The root directory usually does not have the critical files. Instead sub directories are created. E.g. /bin (commands needed during bootup), /etc (config files) , /lib(shared libraries).

/usr filesystem : this file system is generally large as it contains the executable files to be shared amongst different machines. Files are usually the ones installed while installing Linux. This makes it possible to update the system from a new version of the distribution, or even a completely new distribution, without having to install all programs again. Sub directories include /bin, /include, /lib, /local (for local executables)

/var filesystem : this file system is specific to local systems. It is called as var because the data keeps changing. The sub directories include /cache/man (A cache for man pages), /games (any variable data belong to games), /lib (files that change), /log (log from different programs), /tmp (for temporary files)

/home filesystem: - this file system differs from host to host. User specific configuration files for applications are stored in the user's home directory in a file. UNIX creates directories for all users directory. E.g /home/my_name. Once the user is logged in ; he is placed in his home directory.

/proc filesystem : this file system does not exist on the hard disk. It is created by the kernel in its memory to provide information about the system. This information is usually about the processes. Contains a hierarchy of special files which represent the current state of the kernel .Few of the Directories include /1 (directory with information about process num 1, where 1 is the identification number), /cpuinfo (information about cpu), /devices (information about devices installed), /filesystem (file systems

configured), /net (information about network protocols), /mem (memory usage)

486 Q. How do I use chmod and chown command under Linux?

Ans. Use chown command to change file owner and group information. Use chmod command to change file access permissions such as read, write etc.

487 Q. what is the command for finding the highest memory occupied file in Linux?
#du -sh * | sort -nr

488 Q. How to create swap if you don't have free partition?
this is some what tricky question if you are new to Linux Administration here is the command to create swap if you don't have free partition before doing this you have switch off the swap

swapoff -a
dd if=/dev/zero of=/root/swapfile bs=1k count=1024
swapon filename

489 Q. How to see swap details?

Ans:
cat /proc/swap
free

490 Q. How to see/get info about RAM in your system

Ans:
free
cat /proc/meminfo

491 Q. What is the difference between ext2 and ext3

Ans:

ext3=ext2+journaling
"Journaling" is an add-on to a file system that records changes as they are made.
ext2 files system is fast ,less disk writes

ext2 file system in less stable/secure .ext2 file system required to run fsck command if the system get crashed

492 Q. How to convert ext2 file system to ext3?

Ans: tune2fs -j /dev/hda1

493 Q. How convert ext3 file system to ext2

Ans: tune2fs -O^has-journal /dev/hda1

494 Q. What is default block/chunk size for PV

Ans: 4MB

495 Q. What is the command to update the disk quota on file system

Ans: mount -o remount,rw /home

496 Q.What is the partition type number for swap,RAID,LVM?

Ans: 82(swap),fd(RAID),8e(LVM)

497 Q. What is RAID?

Ans: RAID, stands for Redundant Array of Inexpensive Disks. RAID is a method by which same data or information is spread across several disks, using techniques such as disk striping (RAID Level 0), disk mirroring (RAID Level 1), and disk striping with parity (RAID Level 5) to achieve redundancy, lower latency, increased bandwidth, and maximized ability to recover from hard disk crashes.

498 Q. Explain RAID 0?

Ans: RAID level 0 works on “striping” technique. In RAID 0 the array is broken down into strips and data is written into strips. RAID 0 allows high I/O performance but provides no redundancy. RAID 0 Array Size is equal to sum of disks in array. If one drive fails then all data in the array is lost.

499 Q: - Explain RAID 1?

Ans: RAID Level 1 is based on Mirroring technique. Level 1 provides redundancy by writing identical data to each member disk of the array. The storage capacity of the level 1 array is equal to the capacity of one of the mirrored hard disks in a Hardware RAID or one of the mirrored partitions in a Software RAID. RAID 1 provides redundancy means good protection against disk failure. In RAID 1 write speed is slow but read speed is good.

500 Q . Explain RAID 5?

Ans: RAID Level 5 is based on rotating parity with striping technique. RAID-5 stores parity information but not redundant data (but parity information can be used to reconstruct data). The storage capacity of Software RAID level 5 is equal to the capacity of the member partitions, minus the size of one of the partitions if they are of equal size. The performance of RAID 5 is based on parity calculation process but with modern CPUs that usually is not a very big problem. In RAID 5 read and write speeds are good.

501 Q.What is LVM?

Ans: LVM stands for Logical Volume Manager. LVM, is a storage management solution that allows administrators to divide hard drive space into physical volumes (PV), which can then be combined into logical volume groups (VG), which are then divided into logical volumes (LV) on which the file system and mount point are created.

502 Q. What are the steps to create LVM?

Ans:

A. Create physical volumes by “pvcreate” command

#pvcreate /dev/sda2

B. Add physical volume to volume group by “vgcreate” command

#vgcreate VLG0 /dev/sda2

C. Create logical volume from volume group by “lvcreate” command.

#lvcreate -L 1G -n LVM1 VLG0

D. Now create file system on /dev/sda2 partition by “mke2fs” command.

#mke2fs -j /dev/VLG0/LVM1

503 Q. What is Volume group (VG)?

Ans: The Volume Group is the highest level abstraction used within the LVM. It gathers together a collection of Logical Volumes and Physical Volumes into one administrative unit.

504 Q.Tell me all steps to remove a LVM?

Ans: To remove a logical volume from a volume group, first unmount it with the umount command:

umount /dev/<vgname>/<lvname>
and then use the lvremove command:
lvremove /dev/<vgname>/<lvname>

505 Q. What is disk Quota?

Ans: Disk quota is nothing but restricting the disk-space usage to the users.

Disk quota can be implemented in two ways,

- One on INODE
- two on BLOCK

Soft limit - this is the disk limit where the user gets just a warning msg saying that your disk quota is going to expire. This is just a warning, no restriction on data creation

Hard limit - this is the disk limit where user gets error message, I repeat user gets error message stating that unable to create data

506 Q. How SSL works?

Ans: The Secure Sockets Layer (SSL) is a commonly-used protocol for managing the security for transmission of messages on the Internet. It not only encrypts the data but determines whether both client and server has the expected authentication.

To request a secure page, the client uses HTTPS.
The server sends the client its public key and certificate.
The client checks that the certificate was issued by a trusted party (usually a trusted Certificate Authority) that the certificate is still valid, and that the certificate is related to the contacted site.
The client uses the public key to encrypt a random symmetric encryption key and sends it to the server, along with the encrypted URL required and other encrypted HTTP data.
The server decrypts the symmetric encryption key using its private key and uses the symmetric key to decrypt the URL and HTTP data.

The server sends back the requested HTML document and HTTP data that are encrypted with the symmetric key.
The client decrypts the HTTP data and HTML document using the symmetric key and displays the information.

Working of SSL:

* Web Browser checks for the certificate to make sure that the site you are connecting to is the real site and not someone intercepting.

*Determine encryption types that the browser and website server can both use to understand each other.

*Unique codes are send to the browser & server which are used for encryption (server side) & decryption (browser side)

*The browser and Server start communicating; the web browser shows the encrypting icon and web pages are processed secured.

SSL has recently been succeeded by Transport Layer Security (TLS), which is based on SSL. The TLS protocol allows client/server applications to communicate across a network in a way designed to prevent eavesdropping and tampering.

Lets take an E.g ,You are accessing your company secured site.

When you visit your company website, the company server will automatically redirect you to its secure site using the HTTPS protocol before you can log in. This results in your browser and the company site negotiating a secure channel using SSL.This negotiation goes a little like this (note that I've simplified it greatly).
The browser sends a message stating what the latest version of SSL it can support and a list of symmetric algorithms it can use.The web server sends back a message with the version of SSL and the algorithm that will be used.It sends its certificate as well. The client verifies the certificate using the known certificates that came with the browser; in other words, it checks that it has been signed by a trusted CA and that it hasn't expired. If the certificate is valid, the browser generates a one-time key for the session, encrypts it with the server's public key (it's part of the certificate), and sends it to the server. The server decrypts the key, then uses that key together with the agreed symmetric algorithm for the rest of the session.Let's take stock. Your browser is certain that the server is who it says it is (your browser is trying to access Your company.com, and the certificate says it's valid for Your Company.com - and the CA agrees). The browser has generated a cryptographic key that will be used for one time only: this particular session. It'll be thrown away after you log out.The key was sent encrypted with Your Company.com's public key, which only Your company.com can decrypt with its private key. There are a couple of other messages sent that check your browser and the web server have agreed on the same key (if anything went wrong, the session is dropped).Once Your company.com has presented you with a login screen (which will be sent over HTTPS, if the company knows what it's doing) and you've filled it in, it'll know who you are. Your id and password will have been sent encrypted over the secure channel that you've both established. Eve and Mallory

are completely out of the loop.SSL uses both symmetric and asymmetric encryption algorithms. Symmetric algorithms use the same key to encrypt and decrypt data. They are faster than asymmetric algorithms but can be insecure. Asymmetric algorithms use a pair of keys. Data encrypted using one key can only be decrypted using the other. Typically, one of the keys is kept private while the other is made public. Because one key is always kept private, asymmetric algorithms are generally secure; however, they are much slower than symmetric algorithms. To reap the benefits of both algorithms, SSL encapsulates a symmetric key that is randomly selected each time inside a message that is encrypted with an asymmetric algorithm. After both the client and server possess the symmetric key, the symmetric key is used instead of the asymmetric ones.

Working: of TLS:

A TLS client and server negotiate on various parameters used to establish the connection securily by using a handshaking procedure.

The handshake begins when a client connects to a TLS-enabled server requesting a secure connection and presents a list of supported ciphers and hash functions.

From this list of supported ciphers & has functions, the server chooses the strongest cipher and hash function that it can support and notifies the client about decision.

The server sends back its identification in the form of a digital certificate. The certificate usually contains the server name, the trusted certificate authority (CA) and the server's public encryption key.

Now the client confirms the validity of the certificate by verifing it with CA

The client encrypts a random number with the server's public key and sends the result to the server. Only the server should be able to decrypt it, with its private key. This random number, is used for encryption (server side) & decryption (browser side)

If any one of the above steps fails, the TLS handshake fails and the connection is not created.

TLS and SSL are an integral part of most Web browsers (clients) and Web servers. If a Web site is on a server that supports SSL, SSL can be enabled and specific Web pages can be identified as requiring SSL access. Any Web server can be enabled by using Netscape's SSLRef program library which can be downloaded for noncommercial use or licensed for commercial use. TLS and SSL are not interoperable. However, a message sent with TLS can be handled by a client that handles SSL but not TLS.

507 Q. What is a SAN?

Ans: SAN is short for Storage Area Network. It is a high-speed network of storage elements, similar in form and function to a LAN that establishes direct and indirect connections between multiple servers and multiple storage elements. The SAN is an extension of the server's storage bus

508 Q. What does a SAN do?

Ans: SANs create connectivity. SANs offer a method of attaching storage that improves data reliability, availability and performance
SAN overcomes traditional network bottlenecks by connecting in three ways:
· Server-to-storage (direct attached storage)
· Server-to-server (network attached storage)
· Storage-to-storage (SAN Attached Storage)

509 Q. Name some of the SAN topologies and Explain each of them ?

Ans: Point-to-point, arbitrated loop, and switched fabric topologies

a) Point-to-Point

A point-to-point connection is the simplest topology. It is used when there are
exactly two nodes and future expansion is not predicted. There is no sharing of the media, which allows the devices to use the total bandwidth of the link. A simple link initialization is needed before communications can begin.

b) Arbitrary Loop

Our second topology is Fiber Channel Arbitrated Loop (FC-AL). FC-AL is more
useful for storage applications. It is a loop of up to 126 nodes (NL_Ports) that is managed as a shared bus. Traffic flows in one direction, carrying data frames and primitives around the loop with a total bandwidth of 400 MBps (or 200 MBps for a loop based on 2 Gbps technology).

c) Switched Fabric Loop

It applies to switches and directors that support the FC-SW standard, that is, it is not limited to switches as its name suggests. A Fibre Channel fabric is one or more fabric switches in a single, sometimes extended, configuration. Switched fabrics provide full bandwidth per port compared to the shared bandwidth per port in arbitrated loop
Implementations.

510 Q.What's the need for separate network for storage why LAN cannot be used?

Ans: LAN hardware and operating systems are geared to user traffic, and LANs are tuned for a fast user response to messaging requests. With a SAN, the storage units can be secured separately from the servers and totally apart from the user network enhancing storage access in data blocks (bulk data transfers), advantageous for server-less backups.

511 Q .What is FCP?

Ans: The Fibre Channel Protocol (FCP) is the interface protocol of SCSI on Fibre
Channel. It is a gigabit speed network technology primarily used for Storage Networking. Fibre Channel is standardized in the T11 Technical Committee of the
InterNational Committee for Information Technology Standards (INCITS), an American National Standard Institute (ANSI) accredited standards committee. It
started for use primarily in the supercomputer field, but has become the standard
connection type for storage area networks in enterprise storage. Despite its name, Fibre Channel signaling can run on both twisted-pair copper wire and fiber
optic cables.

512 Q.What is iSCSI ?

Ans: Internet SCSI (iSCSI) is a transport protocol that carries SCSI commands from
an initiator to a target. It is a data storage networking protocol that transports
standard Small Computer System Interface (SCSI) requests over the standard
Transmission Control Protocol/Internet Protocol (TCP/IP) networking technology.
iSCSI enables the implementation of IP-based storage area networks (SANs),
enabling customers to use the same networking technologies — for both storage
and data networks. As it uses TCP/IP, iSCSI is also well suited to run over almost
any physical network. By eliminating the need for a second network technology

just for storage, iSCSI has the potential to lower the costs of deploying networked
storage.

513 Q.What is FCIP ?

Ans: Fibre Channel over IP (FCIP) is also known as Fibre Channel tunneling or
storage tunneling. It is a method to allow the transmission of Fibre Channel information to be tunnelled through the IP network. FCIP encapsulates Fibre Channel block data and subsequently transports it over a TCP socket. TCP/IP services are utilized to establish connectivity between remote SANs. Any congestion control and management, as well as data error and data loss recovery, is handled by TCP/IP services, and does not affect FC
fabric services. The major point with FCIP is that is does not replace FC with IP, it simply allows
deployments of FC fabrics using IP tunneling

514 Q. What is iFCP

Ans: Internet Fibre Channel Protocol (iFCP) is a mechanism for transmitting data to and from Fibre Channel storage devices in a SAN, or on the Internet using TCP/IP. iFCP gives the ability to incorporate already existing SCSI and Fibre Channel networks into the Internet. iFCP is able to be used in tandem with existing Fibre Channel protocols, such as FCIP, or it can replace them. Whereas FCIP is a tunneled solution, iFCP is an FCP routed solution.iFCP is a gateway-to-gateway protocol, and does not simply encapsulate FC block data. Gateway devices are used as the medium between the FC initiators and targets. As these gateways can either replace or be used in tandem with existing FC fabrics, iFCP could be used to help migration from a
Fibre Channel SAN to an IP SAN, or allow a combination of both

515 Q. What is FICON ?

Ans: FICON is a protocol that uses Fibre Channel as its physical medium. FICON
channels are capable of data rates up to 200 MBps full duplex, they extend the

channel distance (up to 100 km), increase the number of control unit images per link, increase the number of device addresses per control unit link, and retain the topology and switch management characteristics of ESCON.

516 Q. What is FICON address ?

Ans :FICON generates the 24-bit FC port address field in yet another way. When communication is required from the FICON channel port to the FICON CU port, the FICON channel (using FC-SB-2 and FC-FS protocol information) will provide both the address of its port, the source port address identifier (S_ID), and the address of the CU port, the destination port address identifier (D_ID) when the communication is from the channel N_Port to the CU N_Port.

517 Q. What is FSPF ?

Ans: FSPF keeps track of the links on all switches in the fabric and associates a cost with each link. The cost is always calculated as being directly proportional to the number of hops. The protocol computes paths from a switch to all other switches in the fabric by adding the cost of all links traversed by the path, and choosing the path that minimizes the cost.

518 Q. How FSPF works

Ans: The collection of link states (including cost) of all switches in a fabric constitutes the topology database (or link state database). The topology database is kept in all switches in the fabric, and they are maintained and synchronized to each other. There is an initial database synchronization, and an update mechanism.
 .The initial database synchronization is used when a switch is initialized, or when an ISL comes up. The update mechanism is used when there is a link state

change. This ensures consistency among all switches in the fabric.

519 Q. What is Network Attached Storage (NAS) ?

Ans: Network Attached Storage (NAS) is basically a LAN-attached file server that
serves files using a network protocol such as Network File System (NFS). NAS is
a term used to refer to storage elements that connect to a network and provide
file access services to computer systems. A NAS storage element consists of an
engine that implements the file services (using access protocols such as NFS or
CIFS), and one or more devices, on which data is stored. NAS elements may be
attached to any type of network. From a SAN perspective, a SAN-attached NAS
engine is treated just like any other server, but a NAS does not provide any of the
activities that a server in a server-centric system typically provides, such as
e-mail, authentication, or file management.

520 Q. How is Fiber Channel Different from iSCSI?

Ans: Fibre Channel and iSCSI each have a distinct place in the IT infrastructure as SAN alternatives to DAS. Fibre Channel generally provides high performance and high availability for business-critical applications, usually in the corporate data center. In contrast, iSCSI is generally used to provide SANs for business applications in smaller regional or departmental data centers.

521 Q. What is Frames?

Ans: Fibre Channel places a restriction on the length of the data field of a frame at 528

transmission words, which is 2112 bytes. (See Table 3-2 on page 52.) Larger

amounts of data must be transmitted in several frames. This larger unit that

consists of multiple frames is called a sequence. An entire transaction between

two ports is made up of sequences administered by an even larger unit called an

exchange.

A frame consists of the following elements:

_ SOF delimiter

_ Frame header

_ Optional headers and payload (data field)

_ CRC field

_ EOF delimiter

522 Q. What is Loop address ?

Ans : An NL_Port, like an N_Port, has a 24-bit port address. If no switch connection

exists, the two upper bytes of this port address are zeroes (x'00 00') and referred

to as a private loop. The devices on the loop have no connection with the outside

world. If the loop is attached to a fabric and an NL_Port supports a fabric login,

the upper two bytes are assigned a positive value by the switch. We call this

mode a public loop.

523 Q. What is LUN?

Ans: LUN unique number that is assigned to each storage device or partition of the storage that the storage can support

524 Q. What is LUN Masking?

Ans: A method used to create an exclusive storage area and access control. And this can be achieved by storage device control program.

525 Q. What is WWN?

Ans: WWN is a 64bit address that is hard coded into a fiber channel HBA and this is used to identify individual port (N_Port or F_Port) in the fabric.

526 Q. What is metaLUN?

Ans: A metaLUN is a type of LUN whose maximum capacity can be the combined capacities of all the LUNs that compose it. The metaLUN feature lets you dynamically expand the capacity of a single LUN (base LUN) into a larger unit called a metaLUN. You do this by adding LUNs to the base LUN. You can also add LUNs to a metaLUN to further increase its capacity. Like a LUN, a metaLUN can belong to Storage Group, and can participate in Snap View, Mirror View and SAN copy sessions. MetaLUNs are supported only on CX-Series storage systems. A metaLUN may include multiple sets of LUNs and each set of LUNs is called a component. The LUNs within a component are striped together and are independent of other LUNs in the metaLUN.

527 Q. What is a HBA?

Ans: Host bus adapters (HBAs) are needed to connect the server (host) to the storage.

528 Q. What is SAN fabric?

Ans: SAN fabric is a hardware device that connects workstations and servers to storage devices in a SAN network. It uses the Fibre Channel switching technology to connect a server to a storage device. The SAN fabric offers a high-speed dedicated network including high availability features, very low latency, and high throughput.

529 Q . What is zoning?

Ans: Fabric management service that can be used to create logical subsets of devices within a SAN. This enables portioning of resources for management and access control purpose.

530 Q. What are the two major classification of zoning?

Ans: Two types of zoning are

a) Software Zoning

b) Hardware Zoning

531 Q.What are different levels of zoning?

Ans: a) Port Level zoning

b) WWN Level zoning

c) Device Level zoning

d) Protocol Level zoning

e) LUN Level zoning

532 Q. What are the 3 prominent characteristics of SAS Protocol?

a) Native Command Queuing (NCQ)

b) Port Multiplier

c) Port Selector

533 Q.What is the purpose of disk array?

Ans: Probability of unavailability of data stored on the disk array due to single point failure is totally eliminated.

534 Q. What is disk array?

Ans: Set of high performance storage disks that can store several terabytes of data. Single disk array can support multiple points of connection to the network.

535 Q.What are the advantages of RAID?

Ans: “Redundant Array of Inexpensive Disks”

Depending on how we configure the array, we can have the

- data mirrored [RAID 1] (duplicate copies on separate drives)

- striped [RAID 0] (interleaved across several drives), or

- parity protected [RAID 5](extra data written to identify errors).

These can be used in combination to deliver the balance of performance and reliability that the user requires.

536 Q. How is a SAN managed?

Ans: There are many management software’s used for managing SAN's to name a few

Santricity

- IBM Tivoli Storage Manager.

- CA Unicenter.

- Veritas Volumemanger.

537 Q. Which one is the Default ID for SCSI HBA?

Ans: Generally the default ID for SCSI HBA is 7.

SCSI- Small Computer System Interface

HBA - Host Bus Adaptor

538 Q. What is the highest and lowest priority of SCSI?

Ans: There are 16 different ID's which can be assigned to SCSI device 7, 6, 5, 4, 3, 2, 1, 0, 15, 14, 13, 12, 11, 10, 9, 8.

Highest priority of SCSI is ID 7 and lowest ID is 8.

539 Q.what is SRDF ?

Ans: RDF (Symmetrix Remote Data Facility) is a family of EMC products that facilitates the data replication from one Symmetrix storage array to another through a Storage Area Network or IP network.SRDF logically pairs a device or a group of devices from each array and replicates data from one to the other synchronously or asynchronously. An established pair of devices can be split, so that separate hosts can access the same data independently (maybe for backup), and then resynchronized.
In synchronous mode (SRDF/S), the primary array waits until the secondary array has acknowledged each write before the next write is accepted, ensuring that the replicated copy of the data is always as current as the primary. However, the latency due to propagation increases significantly with distance.

Asynchronous SRDF (SRDF/A) transfers changes to the secondary array in units called delta sets, which are transferred at defined intervals. Although the remote copy of the data will never be as current as the primary copy, this method can replicate data over considerable distances and with reduced bandwidth requirements and minimal impact on host performance. Other forms of SRDF exist to integrate with clustered environments and to manage

multiple SRDF pairs where replication of multiple devices must be consistent (such as with the data files and log files of a database application).

540 **Q. EMC control center 5.1**

Ans: The EMC Control Center Web Console uses [Port : 10799] data stored in the Repository to monitor your storage-attached network and manageControlCenter alerts remotely through a Web browser

541 Q. **What Is Emc Power path?**

Ans: EMC PowerPath is a server-resident software solution that enhances performance and information availability. It integrates multiple path I/O capabilities, automatic load balancing, and path failover functions into one comprehensive package for use on open server platforms connected to Symmetrix enterprise storage systems. PowerPath enables you to do more work in a shorter time so you can serve more customers, run more applications, and exploit more business opportunities.

542 Q. Define RAID? Which one you feel is good choice?

Ans: RAID (Redundant array of Independent Disks) is a technology to achieve redundancy with faster I/O. There are Many Levels of RAID to meet different needs of the customer which are: R0, R1, R3, R4, R5,

R10, R6. Generally customer chooses R5 to achieve better redundancy and speed and it is cost effective.

R0 – Striped set without parity/[Non-Redundant Array].

Provides improved performance and additional storage but no fault tolerance. Any disk failure destroys the array, which becomes more likely with more disks in the array. A single disk failure destroys the entire array because when data is written to a RAID 0 drive, the data is broken into fragments. The number of fragments is dictated by the number of disks in the drive. The fragments are written to their respective disks simultaneously on the same sector. This allows smaller sections of the entire chunk of data to be read off the drive in parallel, giving this type of arrangement huge bandwidth.

RAID 0 does not implement error checking so any error is unrecoverable. More disks in the array means higher bandwidth, but greater risk of data loss

R1 - Mirrored set without parity.

Provides fault tolerance from disk errors and failure of all but one of the drives. Increased read performance occurs when using a multi-threaded operating system that supports split seeks, very small performance reduction when writing. Array continues to operate so long as at least one drive is functioning. Using RAID 1 with a separate controller for each disk is sometimes called duplexing.

R3 - Striped set with dedicated parity/Bit interleaved parity.

This mechanism provides an improved performance and fault tolerance similar to RAID 5, but with a dedicated parity disk rather than rotated parity stripes. The single parity disk is a bottle-neck for writing since every write requires updating the parity data. One minor benefit is the dedicated parity disk allows the parity drive to fail and operation will continue without parity or performance penalty.

R4 - Block level parity.

Identical to RAID 3, but does block-level striping instead of byte-level striping. In this setup, files can be distributed between multiple disks. Each

disk operates independently which allows I/O requests to be performed in parallel, though data transfer speeds can suffer due to the type of parity. The error detection is achieved through dedicated parity and is stored in a separate, single disk unit.

R5 - Striped set with distributed parity.

Distributed parity requires all drives but one to be present to operate; drive failure requires replacement, but the array is not destroyed by a single drive failure. Upon drive failure, any subsequent reads can be calculated from the distributed parity such that the drive failure is masked from the end user. The array will have data loss in the event of a second drive failure and is vulnerable until the data that was on the failed drive is rebuilt onto a replacement drive.

R6 - Striped set with dual distributed Parity.

Provides fault tolerance from two drive failures; array continues to operate with up to two failed drives. This makes larger RAID groups more practical, especially for high availability systems. This becomes increasingly important because large-capacity drives lengthen the time needed to recover from the failure of a single drive. Single parity RAID levels are vulnerable to data loss until the failed drive is rebuilt: the larger the drive, the longer the rebuild will take. Dual parity gives time to rebuild the array without the data being at risk if one drive, but no more, fails before the rebuild is complete.

543 Q. **How many minimum drives are required to create R5 (RAID 5) ?**

Ans: You need to have at least 3 disk drives to create R5.

544 Q. What are the advantages of SAN?

Ans: Massively extended scalability

Greatly enhanced device connectivity

Storage consolidation

LAN-free backup

Server-less (active-fabric) backup

Server clustering

Heterogeneous data sharing

Disaster recovery - Remote mirroring

While answering people do NOT portray clearly what they mean & what advantages each of them have, which are cost effective & which are to be used for the client's requirements.

545 Q.What is the difference b/w SAN and NAS?

Ans: The basic difference between SAN and NAS, SAN is Fabric based and NAS is Ethernet based.

SAN - Storage Area Network

It accesses data on block level and produces space to host in form of disk.

NAS - Network attached Storage

It accesses data on file level and produces space to host in form of shared network folder.

546 Q. What is a typical storage area network consists of - if we consider it for implementation in a small business setup?

Ans: If we consider any small business following are essentials components of SAN

- Fabric Switch
- FC Controllers
- JBOD's

547 Q. Can you briefly explain each of these Storage area components?

Ans: Fabric Switch: It's a device which interconnects multiple network devices .There are switches starting from 16 port to 32 ports which connect 16 or 32 machine nodes etc. vendors who manufacture these kind of switches are Brocade, McData.

FC Controllers: These are Data transfer media they will sit on PCI slots of Server; you can configure Arrays and volumes on it.

JBOD: Just Bunch of Disks is Storage Box, it consists of Enclosure where set of hard-drives are hosted in many combinations such SCSI drives, SAS, FC, SATA.

548 Q. What is the most critical component in SAN?

Ans: Each component has its own criticality with respect to business needs of a company.

549 Q.Can you name some of the states of RAID array?

Ans: There are states of RAID arrays that represent the status of the RAID arrays which are given below

a. Online

b. Degraded

c. Rebuilding

d. Failed

550 Q. Name the features of SCSI-3 standard?

\

Ans: QAS: Quick arbitration and selection

Domain Validation

CRC: Cyclic redundancy check

551 Q. Can we assign a hot spare to R0 (RAID 0) array?

Ans: No, since R0 is not redundant array, failure of any disks results in failure of the entire array so we cannot rebuild the hot spare for the R0 array.

552 Q. What is storage virtualization?

Ans: Storage virtualization is amalgamation of multiple n/w storage devices into single storage unit.

553 Q. What is virtualization?

Ans: A technique of hiding the physical characteristics of computer resources from the way in which other system application or end user interact with those resources. Aggregation, spanning or concatenation of the combined multiple resources into larger resource pools.

554 Q.What is Multipath I/O?

Ans: Fault tolerant technique where, there is more than one physical path between the CPU in the computer systems and its main storage devices through the buses, controllers, switches and other bridge devices connecting them.

555 Q. What is RAID?

Ans: Technology that groups several physical drives in a computer into an array that you can define as one or more logical drive. Each logical drive appears to the operating system as single drive. This grouping enhances the performance of the logical drive beyond the physical capability of the drives.

556 Q.What is stripe-unit-size?

Ans:I t is data distribution scheme that complement s the way operating system request data. Granularity at which data is stored on one drive of the array before subsequent data is stored on the next drive of the array. Stripe unit size should be close to the size of the system I/O request.

557 Q. What is the smallest unit of information transfer in FC?

Ans: Frame

558 Q. What is the different between mirroring, Routing and multipathing?

Ans: Redundancy Functions Relationships Role

Mirroring Generates 2 ios to 2 storage targets Creates 2 copies of data

Routing Determined by switches independent of SCSI Recreates n/w route after a

Failure Multipathing Two initiator to one target Selects the LUN initiator pair to

use

559 Q. Briefly list the advantages of SAN?

Ans: SANs fully exploit high-performance, high connectivity network technologies

SANs expand easily to keep pace with fast growing storage needs

SANs allow any server to access any data

SANs help centralize management of storage resources

SANs reduce total cost of ownership (TCO).

iSCSI fundamentals

iSCSI is a protocol defined by the Internet Engineering Task Force (IETF) which enables SCSI commands to be encapsulated in TCP/IP traffic, thus allowing access to remote storage over low cost IP networks.

560 Q.What advantages would using an iSCSI Storage Area Network (SAN) give to your organization over using Direct Attached Storage (DAS) or a Fibre Channel SAN?

Ans: iSCSI is cost effective, allowing use of low cost Ethernet rather than expensive Fibre architecture.

· Traditionally expensive SCSI controllers and SCSI disks no longer need to be used in each server, reducing overall cost.

· Many iSCSI arrays enable the use of cheaper SATA disks without losing hardware RAID functionality.

· The iSCSI storage protocol is endorsed by Microsoft, IBM and Cisco, therefore it is an industry standard.

· Administrative/Maintenance costs are reduced.

· Increased utilisation of storage resources.

· Expansion of storage space without downtime.

· Easy server upgrades without the need for data migration.

· Improved data backup/redundancy.

561 Q.What's MySQL ?

Ans: MySQL (pronounced "my ess cue el") is an open source relational database management system (RDBMS) that uses Structured Query Language (SQL), the most popular language for adding, accessing, and processing data in a database. Because it is open source, anyone can download MySQL and tailor it to their needs in accordance with the general public license. MySQL is noted mainly for its speed, reliability, and flexibility. ...

562 Q. What is DDL, DML and DCL ?

Ans: If you look at the large variety of SQL commands, they can be divided into three large subgroups. Data Definition Language deals with database schemas and descriptions of how the data should reside in the database, therefore language statements like CREATE TABLE or ALTER TABLE belong to DDL. DML deals with data manipulation, and therefore includes most common SQL statements such SELECT, INSERT, etc. Data Control Language includes commands such as GRANT, and mostly concerns with rights, permissions and other controls of the database system.

563 Q.How do you get the number of rows affected by query?

ANS :SELECT COUNT (user_id) FROM users would only return the number of user_id's.

564 Q.If the value in the column is repeatable, how do you find out the unique values?

Ans: Use DISTINCT in the query, such as SELECT DISTINCT user_firstname FROM users; You can also ask for a number of distinct values by saying SELECT COUNT (DISTINCT user_firstname) FROM users;

565 Q. How do you return the a hundred books starting from 25th?

Ans: SELECT book_title FROM books LIMIT 25, 100. The first number in LIMIT is the offset, the second is the number.

566 Q. What does this query mean: SELECT user_name, user_isp FROM users LEFT JOIN isps USING (user_id) ?

Ans: It's equivalent to saying SELECT user_name, user_isp FROM users LEFT JOIN isps WHERE users.user_id=isps.user_id

567 Q.How do you find out which auto increment was assigned on the last insert?

Ans:SELECT LAST_INSERT_ID() will return the last value assigned by the auto_increment function. Note that you don't have to specify the table name.

568 Q.What does –i-am-a-dummy flag to do when starting MySQL?

Ans: Makes the MySQL engine refuse UPDATE and DELETE commands where the WHERE clause is not present.

569 Q.How would you delete a column?

Ans:ALTER TABLE techpreparation_answers DROP answer_user_id.

570 Q. What is difference between mysql_connect and mysql_pconnect?

Ans:Mysql_connect() opens a new connection to the database while mysql_pconnect() opens a persistent connection to the database. This means that each time the page is loaded mysql_pconnect() does not open the database.

Mysql_close() cannot be used to close the persistent connection. Though it can be used to close mysql_connect().

Q.What is MySQL data directory? How to determine the location of the data directory?

Ans: MySQL stores its data on the disk on the data dictionary. Each subdirectory under this data dictionary represents a MySQL database, inside those directories. By default the information managed my MySQL = server mysqld is stored in data directory.A default location of data directory in windows is C:\mysql\data or C:\Program Files\MySQL\MySQL Server 5.0 \data..

571 Q.Explain advantages of MyISAM over InnoDB?

Ans: Much more conservative approach to disk space management - each MyISAM table is stored in a separate file, which could be compressed then with myisamchk if needed. With InnoDB the tables are stored in tablespace, and not much further optimization is possible. All data except for TEXT and BLOB can occupy 8,000 bytes at most. No full text indexing is available for InnoDB. TRhe COUNT(*)s execute slower than in MyISAM due to tablespace complexity.

572 Q.What are HEAP tables in MySQL?

Ans: HEAP tables are in-memory. They are usually used for high-speed temporary storage. No TEXT or BLOB fields are allowed within HEAP tables. You can only use the comparison operators = and <=>. HEAP tables do not support AUTO_INCREMENT. Indexes must be NOT NULL.

573 Q.How do you control the max size of a HEAP table?

Ans: MySQL config variable max_heap_table_size.

574 Q.Database Basics

Ans: Databases are managed by a relational database management system (RDBMS). An RDBMS supports a database language to create and delete databases and to manage and search data. The database language used in almost all DBMSs is SQL, a set of statements that define and manipulate data. After creating a

database, the most common SQL statements used are INSERT, UPDATE, DELETE, and SELECT, which add, change, remove, and search data in a database, respectively.

Database
A repository to store data.

Table
The part of a database that stores the data. A table has columns or attributes, and the data stored in rows.

Attributes
The columns in a table. All rows in table entities have the same attributes. For example, a customer table might have the attributes name, address, and city. Each attribute has a data type such as string, integer, or date.

Rows
The data entries in a table. Rows contain values for each attribute. For example, a row in a customer table might contain the values "Matthew Richardson," "Punt Road," and "Richmond." Rows are also known as records.

Relational model
A model that uses tables to store data and manage the relationship between tables.

Relational database management system
A software system that manages data in a database and is based on the relational model. DBMSs have several components described in detail in Chapter 1.

SQL
A query language that interacts with a DBMS. SQL is a set of statements to manage databases, tables, and data.

Constraints
Restrictions or limitations on tables and attributes. For example, a wine can be produced only by one winery, an order for wine can't exist if it isn't associated with a customer, having a name attribute could be mandatory for a customer.

Primary key
One or more attributes that contain values that uniquely identify each row. For example, a customer table might have the primary key of cust ID. The cust ID attribute is then assigned a unique value for each customer. A primary key is a constraint of most tables.

Index

A data structure used for fast access to rows in a table. An index is usually built for the primary key of each table and can then be used to quickly find a particular row. Indexes are also defined and built for other attributes when those attributes are frequently used in queries.

Entity-relationship modeling
A technique used to describe the real-world data in terms of entities, attributes, and relationships.

Normalized database
A correctly designed database that is created from an ER model. There are different types or levels of normalization, and a third-normal form database is generally regarded as being an acceptably designed relational database.

MySQL Command Interpreter
The MySQL command interpreter is commonly used to create databases and tables in web database applications and to test queries. Throughout the remainder of this chapter we discuss the SQL statements for managing a database. All these statements can be directly entered into the command interpreter and executed. The statements can also be included in server-side PHP scripts, as discussed in later chapters.

575 Q. How to take the Dump of a MySQL Database ?

Ans : Mysqldump databasename > dumpname

576 Q.What is Kernel? Explain the task it performs.

Ans: Kernel is used in UNIX like systems and is considered to be the heart of the operating system. It is responsible for communication between hardware and software components. It is primarily used for managing the systems resources as well.
Kernel Activities:
The Kernel task manager allows tasks to run concurrently.
Managing the computer resources: Kernel allows the other programs to run and use the resources.Resources include i/o devices, CPU, memory. Kernel is responsible for Process management. It allows multiple processes to run simultaneously allowing user to multitask. Kernel has an access to the

systems memory and allows the processes to access the memory when required. Processes may also need to access the devices attached to the system. Kernel assists the processes in doing so.For the processes to access and make use of these services, system calls are used.

577 Q. Different types of Kernel.

Ans . Monolithic Vs Modular Kernel
Monolithic kernel is one single program that contains all of the code necessary to perform every kernel related task. Most UNIX and BSD kernels are monolithic by default.
Some advantages hinge on these points:
Since there is less software involved it is faster. As it is one single piece of software it should be smaller both in source and compiled forms.
Less code generally means less bugs which can translate to fewer security problems.

Modular kernel allows an administrator to add functionality only when required. Keeping only what's necessary in kernel memory reduces the kernel's memory footprint and increases its overall performance. A few advantages to the modular kernel are: Faster development time for drivers that can operate from within modules. No reboot required for testing (provided the kernel is not destabilized). On demand capability versus spending time recompiling a whole kernel for things like new drivers or subsystems. Faster integration of third party technology (related to development but pertinent unto itself nonetheless).

578 Q.What is Initial Ram disk?

Ans: An initial ram disk is a temporary file system used in the boot process of the Linux kernel. **initrd** and **initramfs** refer to slightly different schemes for loading this file system into memory. Both are commonly used to make preparations before the real root file system can be mounted.

579 Q. Explain Different Option for updating kernel?

Ans:
A. install new kernel from rmp file
a. Get the Newest kernel and put in /tmp
b. rpm -i /tmp/kernal-2.4.19-3.i386.rpm (Not ugrading.can be switched to old keranl if need)
c. Now Kernal ,RamDisk etc..are installed in /boot
d. New stanza added to grub.com (you need alter "default")

B. customizes, install and configure from kernel source code.

Required RMP's tool for customizing kernel source code.

1. Kernel-source-*
2. glibc-kernheaders-*
3. glibc-deve;-*
4. cpp-*
5. binutils-*
6. gcc-*
7. tcl-*
8. tk-*

580 Q. Explain the Steps to compile and install kernel?

Ans: make mrproper -- clean up the directory tree
make xconfig -- tools for customixing the kernal
make dep --- makes and build the dependecies
make clean --- cleans any unecessary files used to build the previous steps
make bzimage -- actually builds the kernel
make modules -- Builds the modules
make modules_install -- Install kernel modules
make install -- copy the new kernel files to associated directory.

581 Q.What are ISO image files?

Ans: An ISO image is an archive file (also known as a disk image) of an optical disc in a format defined by the International Organization for Standardization (ISO). ISO image files typically have a file extension of .iso. The name ISO is taken from the ISO 9660 file system used with CD-ROM media according towikipedia.org.

582 Q. How to create an ISO image?

Ans: As of now I know two methods to create an ISO file. One is through dd command (disk dump) and other is mkisofs command

583 Q. Creating ISO files from cd-rom/dvd by using dd command

Ans: #dd if=/dev/cdrom of=/temp/songs.iso

dd if=/dev/dvd of=/temp/songs1.iso # for dvd

dd if=/dev/scd0 of=/temp/songs2.iso # if cdrom is scsi

Let me explain what actually the above command will do. dd is a disk dump command which will present in most of the Linux/nix systems and "if"specifies input file and "of" specifies output file. So we are specifying to dd command what is our input file and where is our output file.

584 Q. Creating ISO file using mkisofs command.

Ans: Method1: Creating an iso image from cd-rom

#mkisofs /dev/cdrom -o /temp/songs3.iso

Method2: Creating an iso image from a folder

Suppose i am having a folder /opt/data which contains some data to be converted in to .iso file, follow below step

#mkisofs -o /temp/songs4.iso /opt/data

585 Q. How to mount an ISO file?

Ans: #mount -o loop /temp/songs4.iso /opt/mountpoint

586 Q. How to unmount an ISO file?

Ans: #umount -lf /opt/mountpoint

587 Q: - Is there Graphical editors for LDAP ?

Ans:-
Yes, Following are some GUI based tools for LDAP
- GQ
- Java LDAP Browser/Editor
- Softerra LDAP Browser

588 Q: - What can i do if my application doesn't speak to LDAP ?

Ans:-
Gateway that translate one directory access protocol into another.

589 Q: - How can i join information contained in different directories ?

Ans:-
Distributed, Multivendor directories glued together by referrals and references.

590 Q: - What is "LDIF" ?

Ans:-

The LDAP Interchange Format (LDIF) is a standard text file format for storing LDAP

configuration information and directory contents. LDIF files are often used to import new data into your directory or make changes to existing data.

591 Q: - Name the object class types ?

- Structural Object class
- Auxiliary Object class
- Abstract object classes

592 Q: - What is the name of main configuration file name for LDAP server ?

Ans:-

slapd.conf

593 Q: - What is LDAP ?

Ans:-

LDAP stands for Lightweight Directory Access Protocol. In plain and simple terms, its a database whereby it has all the details of all of organizations, individuals, and other resources such as files and devices in a network, whether on the Internet or on corporate intranetand whether or not you know the domain name, IP address, or geographic whereabouts. An LDAP directory can be distributed among many servers on a network, then replicated and synchronized regularly. An LDAP server is also known as a Directory System Agent (DSA). Its a not a relational database. Outlook and other email programs uses LDAP to search for a recipient in an organization.

594 Q: - Whats the relation ship between LDAP and JNDI?

Ans:-

JNDI has classes provided by SUN that will help ur appln interact with and LDAP server. JNDI appln work similarly to JDBC applns once and be free to use 'drivers' from different vendors. SUN provides the "driver" that will help interact with the LDAP server. Sun also provides "drivers" for other naming services (like CORBA).

595 Q: - Why LDAP is called light weight?

Ans:-

LDAP (Lightweight Directory Access Protocol) is a protocol for communications between LDAP servers and LDAP clients.
LDAP servers store "directories" which are access by LDAP clients.
LDAP is called lightweight because it is a smaller and easier protocol which was derived from the X.500 DAP
(Directory Access Protocol) defined in the OSI network protocol stack.

596 Q: - what is SLAPD?

Ans:-
SLAPD stands for Stand-Alone LDAP.Clients connect to the server over the LDAP protocol, usually using a network-based connection (though SLAPD provides a UNIX socket listener).

597 Q: - Which daemons are required for LDAP server?

Ans:-
slapd and slurpd

598 Q: - Tell me the name of three LDAP Client utilities or Applications

Ans:-

ldapsearch
ldapadd
ldapmodify

599 Q: - Define Schemas?

Ans:-
Schemas provide definitions of the different object classes and attribute types that OpenLDAP should support. Using these, OpenLDAP can determine what entries it is allowed to store, whether any given entry is valid, and how entries should optimally be stored.

600 Q: - Explain modulepath directive?

Ans:-

The modulepath directive provides the full path to the directory where the modules (the compiled libraries) are stored.

601 Q: - Explain moduleload directive?

Ans:-
The moduleload directive instructs OpenLDAP to load a particular module.

602 Q: - What is HDB?

Ans:

HDB is the new generation storage mechanism for OpenLDAP. Like its predecessor, the BDB backend, HDB uses the Oracle BerkeleyDB database for storage, but HDB stores entries hierarchically, a perfect fit for LDAP's tree structure. The old BDB backend is still supported, and you can use it by specifying bdb instead of hdb in the database directive.

603 Q: - Which utility is used to Encrypt the password?

Ans:-
ldappasswd

604 Q: - How you will verify LDAP configuration file?
Ans:-

Use "slaptest" utility.
slaptest -v -f /etc/ldap/slapd.conf

605 Q: - Which configuration file is required for LDAP clients?
Ans:-
ldap.conf

606 Q: - Explain "SIZELIMIT" directive?
Ans:-

This directive indicates the upper limits on the number of records returned.

607 Q: - Explain "TIMELIMIT" directive?

Ans:-

This directive will give the information about the amount of time the client will wait for the server to respond.

608 Q: - Tell me the name of logical operators which are used in ldap filters.
Ans:-

AND (&), OR (|), and NOT (!)

609 Q: - What Does slapadd Do?
Ans:-

The slapadd utility reads the slapd.conf file, loads the appropriate backend databases, and then reads LDIF data.

610 Q: - Which web based tool you have used for LDAP?
Ans:-

phpLDAPadmin

611 Q: – What is FTP ?

FTP stands for File Transfer Protocol. An FTP server allows clients to connect to it either anonymously or with a username and password combination. After successful authentication, files can be transferred back and forth between the server and client. The files are neither encrypted nor compressed.

612 Q: – How to deny specific users access to the FTP server ?

To deny specific users access to the FTP server, add their usernames to the /etc/vsftpd/ ftpusers file. By default, system users such as root and nobody are included in this list.

613 Q: – **Can we create logs for ftp authenticated sessions ?**

Yes, If the xferlog_enable directive in vsftpd.conf is set to YES, file transfers using the FTP protocol are logged to /var/log/xferlog. Information such as a time stamp, IP address of the client, the file being transferred, and the username of the person who authenticated the connection is included in the log entry.

614 Q: – **What is meaning of max_clients parameter ?**

Maximum number of clients that can connect at one time. If set to 0, the number of clients is unlimited.

615 Q: – Is there any way to monitor clients connected to vsftpd?

Yes. We actually have two slightly different methods to monitor vsftpd clients. First, make sure you have enabled the config option, "setproctitle_enable=YES" like in our example above and restart your vsftpd server. Then run the command "watch ps -Cvsftpd -o user, pid, stime, cmd" to watch the processes including ip, username and actions like idle or data retrieval.

616 Q: – **I want to copy multiple files with out prompting for any info, how can I do that one?**

ftp -i ftpserver

617 Q: – **Local users cannot log in. How to resolve this issue?**

Check "local_enable=YES" in your /etc/vsftpd/vsftpd. conf to allow local users to log in.

618 Q: – **For Redhat Linux or Fedora which package is required for FTP service ?**

Red Hat Enterprise Linux 5 or FEDORA includes the vsftpd FTP service.
vsftpd-2.0.5-12.el5 (For Redhat)

619 Q: How to deny specific users access to the FTP server ?
To deny specific users access to the FTP server, add their usernames to the /etc/vsftpd/
ftpusers file. By default, system users such as root and nobody are included in this list.

620 Q: Can we create logs for ftp authenticated sessions ?
Yes, If the xferlog_enable directive in vsftpd.conf is set to YES, file transfers using the FTP protocol are logged to /var/log/xferlog. Information such as a time stamp, IP address of the client, the file being transferred, and the username of the person who authenticated the connection is included in the log entry.

621 Q: What is meaning of max_clients parameter ?
Maximum number of clients that can connect at one time.
If set to 0, the number of clients is unlimited.

622 Q: On which port VSFTP server works ?
FTP uses two ports, 20 and 21. By default, the FTP server listens for

requests on port 21.
After a connection is established, the client sends commands to the server on port 21.
However, port 20 is used when the server sends data back to the client.

623 Q: How to restart VSFTP server ?

service vsftpd restart

624 Q: How to allow Anonymous FTP ?

Anonymous FTP is enabled by default by setting the anonymous_enable directive in
/etc/vsftpd/vsftpd.conf to YES.

625 Q: - For Redhat Linux or Fedora which package is required for FTP service ?

Red Hat Enterprise Linux 6 or FEDORA includes the vsftpd FTP service.
Rpm -qa | grep vsftpd to chceck before installing
yum install vsftpd* -y to install vsftpd package

626 Q: - Important Configuration file for vsftp server ?

The FTP server uses the /etc/vsftpd/vsftpd.conf configuration file. Using this file, you
can set options for displaying a custom banner message after users log in, setting the
default file permissions for uploaded files, and setting the port on which to listen for
incoming connections.

627 Q: - What is FTP ?

FTP client connects to the FTP server by establishing an FTP control connection to port 21 of the server. Your commands such as 'ls' and 'get' are sent over this connection. Whenever the client requests data over the control connection, the server initiates data transfer connections back to the client. The source port of these data transfer connections is always port 20 on the server, and the destination port is a high port (greater than 1024) on the client.

628 Q: - What is Passive mode?

Passive mode, like active mode, is initiated by the FTP client application. When requesting data from the server, the FTP client indicates it wants to access the data in passive mode and the server provides the IP address and

a random, unprivileged port (greater than 1024) on the server. The client then connects to that port on the server to download the requested information.

629 Q: - Explain directive "session_support”?

When enabled, vsftpd attempts to maintain login sessions for each user through Pluggable Authentication Modules (PAM).

630 Q: - Is there any way to monitor clients connected to vsftpd?

Yes. We actually have two slightly different methods to monitor vsftpd clients. First, make sure you have enabled the config option, "setproctitle_enable=YES" like in our example above and restart your vsftpd server. Then run the command "watch ps -Cvsftpd -o user, pid, stime, cmd" to watch the processes including ip, username and actions like idle or data retrieval.

631 Q: - I want to copy multiple files with out prompting for any info, how can I do that one?

ftp -i ftpserver

632 Q: - Local users cannot log in. How to resolve this issue?

Check "local_enable=YES" in your /etc/vsftpd/vsftpd. conf to allow local users to log in.

633 Q: - How to change vsftpd default port?

Set "listen_port" option in "vsftpd.conf"

634 Q: - How to restrict some IP's not use my FTP server?

Use TCP_WRAPPERS,
go to /etc/hosts.deny
write,
vsftpd: .example.com (or) ip address
save it and do the following
service vsftpd restart
chkconfig vsftpd on

635 Q: - Does vsftpd support IPv6?

Yes

636 Q . What is the difference between service and process?

A process is any piece of software that is running on a computer. For example, your anti-virus software runs in the background as a process, which was automatically started when the computer booted. Some processes start when your computer boots, others are started manually when needed.

Some processes are services that publish methods to access them, so other programs can call them as needed. Printing services would be an example of a service type of process, where your email program can just call the print services process to say it wants to print, and the service does the actual work.

638 Q . How to view crond status? If it's show service is not found.

Service crond restart

639 Q . My clients are getting services from servers but how to know which client is using which service. is there any files to keep information about these? Clients used ftp, nis, samba, apache, squid, nfs and mail services how to know how many users got service from server side with date, time and client system ip?

Mail server – /var/log/mail/maillog [RedHat,centos]
ssh – /var/log/secure
Apache – /var/log/http/access.log
nfs – /var/lib/nfs/rmtab

640 Q . How to FTP user access other directory except his own home directory?

vim /etc/vsftpd/vsftpd.conf
Chroot_list_enable=yes

641 Q . What are the Linux-based security tools?

Selinux
Firewall
iptables
Tcp-wrappers

642 Q . What are the basic elements of firewall?

A firewall should be able to filter packets (drop/pass them) based on certain rules specified by the user. The rules may be used to identify an incoming packet to the computer or outgoing packet from the computer, it can be based on target port number/ip add , traffic from a particular Network card etc…

The firewall rules can be in a tabular form (saved on the disk) from where the firewall software can read them and implement it. iptables firewall on Linux is a great example

643 Q . What is a command to display top 10 users who are using huge space?

du -sh /home/* | sort -r | head -10

644 Q . How do find all failed login attempts via ssh?

tail -f /var/log/secure | grep Failed

645 Q . How do you configure Linux system as a router?

vim /etc/sysctl.conf
net.ipv4.ip_forward=1
system-config-network
eth0 192.168.1.120 eth0:1 172.24.0.1
255.255.255.0 255.255.0.0
172.24.0.1 192.168.1.120

646 Q . What is the UID and GID of root user? Can a normal user can change the ownership of a file? What is the command to change ownership of a file?

The root UID/GID is 0 (zero). Which is why he can able to intervene in all normal users files even though he don't had permission. A normal user will don't have the permission to change ownership of file. The command to change ownership is < chown user.user file >

647 Q . What is the diff b/w ext2 and ext3?

Ext3 is a tiny bit slower than ext2 is, but it holds tremendous advantages. There is really only one difference between ext2 and ext3, and that is that ext3 uses a journal to prevent filesystem corruption in the case of an unclean shutdown (ie. before the filesystem is synced to disk). That makes ext3 a bit slower than ext2 since all metadata changes are written to the journal, and then flushed to disk, but on the other hand you don't risk having the entire filesystem destroyed at power failure or if an unwitted person turns the computer off uncleanly. You don't have to check the filesystem after an unclean shutdown either. Ext3 has three levels of journalling. Metadata (ie. internal filesystem structures) are always journalled, so that the filesystem itself is never corrupted. How ordinary data is written to the file system is controllable, though. The default option is the "ordered" mode, which causes file contents to be written to the filesystem before metadata is even committed to the journal. The highest reliable mode is called the "journal" mode, which causes file data to be committed to the journal before it is flushed to its final place, like the metadata. The least reliable mode, but rumoured to be the fastest, is called the "writeback" mode, which makes no promises at all regarding the consistency of file data. Only metadata is output reliably in writeback mode. So as for anything else, it's mainly a matter of priority. If you don't want ultimate speed, go with ext3. If you need the highest speed that is

theoratically aquirable though, then go with ext2. For that to be effective you'll probably need a really advanced hard drive controller, though.

648 Q . As the system administrator you need to review Bob's cronjobs. What command would you use?

crontab –lu Bob

649 Q . What command is used to remove the password assigned to a group?

gpasswd –r groupname

650 Q . What are the different RAID levels?

♣ RAID level 0
♣ RAID level RAID level 1
♣ RAID level 2
♣ RAID level 3
♣ RAID level 4
♣ RAID level 5
♣ RAID level 6
♣ RAID level 10
♣ RAID level 50

651 Q . How do you create a swapfile?

dd if=/dev/zero of=/swapfile bs=1024 count=200M
mkswap /swapfile
swapon /swapfile

652 Q . What does nslookup do?

Nslookup is a program used to find information about internet Domain Name server.
The two modes of nslookup are: Interactive and non-interactive.
Using 'interactive mode' user can query the name servers for the information pertaining to hosts and domains.
Using 'non-interactive mode' the user can just print the name and requested information of a host.

653 Q . What is the difference between UDP and TCP?

TCP is a Transmission Control Protocol.
UDP is a User Datagram Protocol.
There are four major differences between UDP and TCP:

1. TCP can establish a Connection and UDP cannot.

2. TCP provides a stream of unlimited length, UDP sends Small packets.

3.TCP gurantees that as long as you have a connection data sent will arrive at the destination, UDP provides not guarantee delivery.
4.UDP is faster for sending small amounts of data since no connection setup is required, the data can be sent in less time then it takes for TCP to establish a connection.

654 Q . What command do you run to check file system consistency?

Need to run fsck [file system consistency check] command to check file system consistency and repair a Linux / UNIX file system.

fsck

655 Q . What is the command to remove Lvm ,Pv and vg

1st remove the entry on /etc/fstab file & save – quit.
2nd remove LVM – lvremove lvname
3rd remove VG – vgremove vgname
4th remove PV – pvremove pvname

656 Q . How to schedule cron backup to run on 4th Saturday of month?

* * * * 6 weekdaynum 4 && sh /backup/test.sh

657 Q . What is an inode?

ext2 and ext3 file systems keep a list of the files they contain in a table called an inode table. The inode is referenced by its number. This is unique within a file system.

The inode contains the metadata about files. Among the data stored in the inode is

File type

File permissions

Link count

User ID number of the file owner and the group ID number of the associated group

Last modification time

Location of the data on the hard disk

Other metadata about the file

ls -li – view inode number only

stat /etc/passwd – view inode details

658 Q . How to see unallocated hard disk space on linux

df -h

659 Q . How do u find remote machine operating system and version?

nmap -A –v 192.168.1.100

660 Q . How do you port scanning with netstat command?

netstat –an

661 Q . What is mean by system calls?

A system call is the mechanism used by an application program to request service from the operating system.

On Unix-based and POSIX-based systems, popular system calls are open, read, write, close, wait, exec, fork, exit, and kill. Many of today's operating systems have hundreds of system calls. For example, Linux has 319 different system calls. FreeBSD has about the same (almost 330). Tools such as strace and truss report the system calls made by a running process.

662 Q . Important port no:

NFS	–	2049
FTP	–	21
SAMBA	–	445
SSH	–	22
DNS	–	53
POP3	–	110
IMAP	–	143
HTTPS	–	443
HTTP – 80		

663 Q . How do u extract files from iso cd images in linux?

mount –o loop disk1.iso /mnt/iso

664 Q . What is the use of KVM switch?

KVM switch allows user to take control of two or more computers using the resources that are shared like mouse, keyboard, monitor, etc. This allows the administrator to access multiple servers from a single place. Administrator can access fileserver or internal server or proxy server and can separate out these servers from each other. This gives the freedom to administrator to use the components anywhere and make it efficient.

665 Q . What is the requirement of KVM switch?

KVM switch is required to access more than one server from one place. The requirements of different switches differ from one institute to another. There are switches that give no protection but provide higher efficiency. Some switch offer the password to protect the switch. The password helps in locking the switch so that more security can be given. The switches provide protection to the user so that the data can be kept safe.

666 Q . What is the difference between passive and active switches?

- Passive switch doesn't have any active components and it is a hard wired connection whereas, active switches are used to retain the actions performed by hardware devices that are connected during switching process.

- Passive switches are wired internally between Input/Output ports whereas, active switches uses the computers connection for their connections they don't rely on external or internal wires.

- Passive switches are simple and cheap whereas, active switches are a bit expensive in terms of its application and it requires separate power supplies.

667 Q . What are the various uses of KVM?

KVM acts as a virtualization platform that is required by machines to run other operating system in synchronization with host operating system. It is an inbuilt virtualization platform that comes with Linux kernel. It is used for non-graphic servers and supports wide range of operating system to run with. It runs along with QEMU to provide strong virtualization platform.

668 Q. What are the different formats supported by KVM?

KVM supports various formats in synchronization with QEMU as it supports many raw images, native QEMU format of qcow2, VMWare format like .vmx, .vmdk, etc. VirtualBox format like .vbx, etc. and many more. QEMU and KVM together give the advanced functionality to run virtualized servers and platforms.

669 Q . How can I use KVM with a non-privileged user?

To use KVM as a non-privileged user firs the creation of a group has to take place in which addition of the users takes places and then making their group. The file that has to be modified for the groups and by which the permissions can be set resides in /dev/kvm that has to be changed to give permissions to non-privileged users. To set the permission the following line has to be added or replaced in the file

mentioned above.

KERNEL=="kvm", GROUP= "kvm"

670 Q . How to check the hardware acceleration provided by KVM?

KVM module provides hardware acceleration for better performance and to provide better efficiency in between the virtual machines.
To check the hardware acceleration checks the system configuration settings by using the following commands:

qemu-system-x86_64 -hda myvm.qcow2

If it gives any error message or module not find message then the modules are not loaded correctly.

671 Q . What is the procedure to set up network for guest to make it accessible from host machine?

KVM known as kernel virtual machine uses QEMU for device emulation. In this there are two networking modes that are defined. First mode is root networking mode and another mode is network bridge mode. If the IP address to the virtual machines has to be assigned then the kvm need to be up and running.

To setup the network interface across the host machine the command that is required to be followed is as follows:

/sbin/ip
/usr/sbin/brctl
/usr/sbin/tunctl

672 Q . What is the difference between KVM and Xen?

- KVM is an internal part of the linux kernel module and uses regular memory and scheduler like linux, whereas Xen is an external hypervisor that takes the control and divides the resources between the guest machines.
- KVM is easy to use and provide more features, whereas Xen is powerful but it requires good amount of knowledge to operate.
- KVM runs on processors that support hvm, whereas Xen runs non-hvm compatible processors.
- KVM doesn't have any support for para-virtualization, whereas Xen supports para-virtualization that is used for device drivers to improve the performance of input/output.

673 Q . How KVM does support dynamic memory management?

KVM allows the resources to be shared to the guest operating systems. KVM allocates the memory dynamically between many processes. KVM allows the host to allocate the memory for the guest and beyond it guest can't use memory. The guest system that has a balloon driver keeps the guest from taking full memory. This ballooning is controlled by balloon monitor. KVM allows the pages that are similar to be grouped or collapse together in one to save the space this way the dynamic memory management can be performed by KVM.

674 Q: What is Virtualization ?

Ans: Virtualization is a technique for creating virtual resources (rather than the actual) such as server, storage device, network and Operating system. Virtualization is dis-associating the tight bond between software and hardware.

675 Q: What are the different types of Virtualization ?

Ans: Virtualization can be used in different ways and can take many different forms. Some of them are listed below :

- Server Virtualization
- Network Virtualization
- Hardware virtualization
- Application virtualization
- Desktop virtualization
- User virtualization

676 Q: What is the difference between full virtualization & para virtualization ?

Ans: Full virtualization & para virtualization both comes under the Hardware virtualization. Some of the difference between them are listed below :

Full Virtualization : It is a virtualization in which guest machine(virtual machines) is unware that it is in virtualized environment therefore hardware is virtualized by the host operating system so that the guest can issue commands to what it thinks is actual hardware but really are just simulated hardware devices created by the host

Para Virtualization : It is a virtualization in which guest machine is aware that it is in virtualized environment . If guest machine require resources like memory & cpu , it issues command to guest operating system instead of directly communication with actual hardware.

677 Q: What is hypervisor ?

Ans: Hypervisor is a peace of a software that is being install on the physical machine , which then further creates and run virtual machines. Virtual machine are known as guest machines and host machine is the hypervisor on which different virtual machines are created.

678 Q: What are different hypervisors available in Linux ?

Ans: Xen & KVM are two hypervisor available in linux.

679 Q: What is the difference between Xen & KVM ?

Ans: For Xen hypervisor first we have to install Xen kernel and have to boot the machine with Xen kernel where as KVM is kernel based Virtualization , we don’t need any extra kernel for

KVM. KVM is a module in Kernel. Xen hypervisor by default doesn't support full virtualization whereas KVM supports Full virtualization.

680 Q: What is Type-1 and Type-2 hypervisor ?

Ans: Type-1 hypervisor is bare metal hypervisor runs on bare metal of hardware. Hyper-V and ESXI Server are the examples of type-1 hypervisor. Type-2 hypervisor is hosted by operating system. Examples of type-2 hypervisor are Microsoft Virtual Server & VMware Server.

681 Q: What is Dom0 in Xen ?

Ans: Dom0 or Domain0 is the initial domain started by xen hypervisor. It has the special rights like to start new domain and access the hardware directly. Dom0 is responsible for running all of the device drivers for the hardware.

682 Q: How to verify Virtualization Technology (VT) is enabled in your server's BIOS or not ?

Ans : grep -E 'svm|vmx' /proc/cpuinfo

683 Q: What is the use of virsh command ?

Ans: virsh is the interface or command for managing the virtual machines based on KVM & Xen hypervisor. On virsh interface virtual machines are identified by their domain names , so virsh is generally used to list current domains , to create , pause & shutdown domains.

684 Q: How to identify the KVM version ?

Ans: To find the KVM version use the command 'virsh version'

685 Q: Which command is used to list all virtual machine running on the KVM hypervisor ?

Ans: Using the command **'virsh list –all**' we can list all virtual machines irrespective of their states.

686 Q: How to forcefully shutdown the KVM based virtual machine from the command line ?

Ans: We can forcefully shutdown the VM using the command '**virsh destroy machine_name**'.This command should only be used in a case where VM is in Hung state because forcefully shutdowm may cause filesystem corruption.

687 Q: What are the basic requirements of VM live migration in KVM ?

Ans: Some of the basic requirements are listed below :

- The guest image or virtual machine image must be located on a shared storage and it must be accessible using **iSCSI, NFS, GFS2** or **Fibre Channel**.

- The shared storage must be mounted on the same path on both the hypervisors / hosts.
- Both hypervisors / hosts must run the same version of **KVM**.
- Both guests or VMs must have the same network configuration & bridging configuration (their IPs must be different)

688 Q: Which command is used in KVM for VMs live migration ?

Ans: 'virsh migrate –live machine_name qemu+ssh://destination_server/system'

689 Q: What are the different states of a VM in Xen hypervisor ?

Ans: A VM can have different states like

- r – Running
- b – Blocked
- c – crashed
- s – Shutdown
- p – Paused

690 Q: How to get the console of guest or virtual machine in Xen ?

Ans: xm console <domain-id>

691 Q: How to shutdown,reboot & start VMs (domain-ids) in Xen ?

Ans: Use xm command :

xm shutdown [domain-id]
xm reboot [domain-id]
xm start [domain-id]

692 Q: How to get hardware information of KVM guest machine ?

Ans: Use the command 'virsh dominfo <domain-name / VM Name>'

693 Q: How to connect a particular VM using virt-viewer ?

Ans: virt-viewer -c qemu:///system <VM_Name>